Constrained particle on a group: from propagators to correlators

Guanda Lin

Leinweber Institute for Theoretical Physics and Department of Physics,
University of California, Berkeley, California 94720, U.S.A.

geoff_guanda_lin@berkeley.edu

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Abstract

We develop a particle-on-a-group formulation of super-JT gravity aimed at computing supersymmetric correlators. We show that the (super)JT gravity can be described by a particle moving on the isometry group satisfying constraints from boundary conditions of (super)JT gravity. In this language the $\mathcal{N} = 2$ and $\mathcal{N} = 4$ theories are described by constrained particles on $SU(1, 1|1)$ and $PSU(1, 1|2)$. Solving the constraints gives the super-Schwarzian actions. We also quantize the reduced superparticle, with careful treatment of the fermionic constraints. We then derive the physical worldline supercharges from the requirement that the transformations preserve the constraints. These charges allow us to construct supersymmetric interval propagators in invariant variables and to formulate boundary-anchored Wilson-line operators for both superconformal primary and descendant insertions. Finally, we use these ingredients to build an algorithm for correlators. We obtain the $\mathcal{N} = 2$ and $\mathcal{N} = 4$ three-point composition kernels and zero-energy scalar three-point functions. For four-point functions, the same method reproduces the standard bosonic JT OTOC, gives an explicit zero-energy OTOC in $\mathcal{N} = 2$ and $\mathcal{N} = 4$ SJT, which is potentially useful for studying Berry curvature and BPS chaos.

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1 Introduction

Recent years have witnessed significant progress in understanding the universal low-energy dynamics of near-extremal and near-BPS black holes. For a large class of black holes, the near-horizon region develops a long nearly-AdS_2 throat. After dimensional reduction, the universal gravitational dynamics near the edge of this throat is described by JT gravity. In the simplest case the only boundary mode is the reparametrization of the throat boundary, whose dynamics is the Schwarzian theory, equivalently the disk-level description of Jackiw–Teitelboim gravity [1–6].

For supersymmetric black holes the corresponding low-energy theory is super-JT gravity. The dimensional reduction now keeps the reparametrization mode as well as the boundary supersymmetries and R-symmetry modes fixed by the near-horizon superisometry. Different amounts of supersymmetry for black holes give $\mathcal{N} \leq 4$ super-JT. These structures control the low-energy dynamics of near-BPS black holes, including the BPS ground-state sector and the continuum of near-BPS excitations [7–9].

A useful heuristic in these discussions is to follow the relevant fields and operators from the asymptotic region to the edge of the near-horizon region, and then map the low-energy problem to JT or super-JT. This UV-to-throat map lets one study the black hole as a quantum mechanical system: its spectrum and density of states are obtained from the JT or super-JT partition function [7–15], while its response to perturbations is probed by reducing higher-dimensional fields to matter insertions dressed by the same near-horizon dynamics [16–26]. In these discussions the properties of the low-temperature black hole are encoded in correlators, mostly two-point functions, receiving quantum corrections from the Schwarzian fluctuations of the near-horizon geometry.

Beyond the spectrum and response, one also wants observables related to chaotic behavior, and they often involve more complicated higher-point correlators. The standard example is the out-of-time-order correlator (OTOC) [27–34], which measures how a simple perturbation is scrambled by black hole dynamics. In nearly-AdS_2 systems it gives the familiar maximal chaos signal of the Schwarzian/JT mode [4–6, 35–38]. Interestingly, even in a protected BPS sector there can still be a useful notion of chaos [39–41]. This is special because the relevant states are exactly degenerate ground states, so there is neither scrambling-type chaotic time evolution nor level-statisics notion of chaoswithin the protected subspace. The chaos is instead in the eigenvectors: it describes how the degenerate ground states mix with each other under perturbations. One specific probe is the non-Abelian Berry curvature, which measures the geometric phase acquired by the degenerate ground states as the parameters of the system are adiabatically varied [42]. Gravitional prediction should give statistical properties of these Berry curvatures, in particular, moments of Berry curvatures can be captured by (higher-point) correlators in gravity: for example, the second moment involves time-ordered and out-of-time-ordered four-point correlators.

With this motivation for computing correlators, we next ask how they can be determined in practice. From the mathematical point of view, these correlators are standard representation-theoretic objects.

For example, the ordinary JT OTOC can be written in terms of the $SL(2,\mathbb{R})$ $6j$ symbol. This means that the problem can be separated into a few pieces: the representations which can appear in an intermediate channel, the dressed three-point functions, and the crossing data which relates different pairings of the operators. For the supergroups relevant to $\mathcal{N} = 2, 4$ super-JT, the analogous channel data, three-point structures and crossing kernels have not been worked out, at least to our knowledge. One goal of this paper is to derive some physicscally relevant parts of these data.

Among the different formulations of JT gravity, the particle-on-a-group description is especially convenient for this purpose. It gives a physicist's route to the same representation-theoretic answers. The reason is that the particle-on-a-group formulation keeps the symmetry explicit: gravity and matter multiplets are organized by their transformation properties, so the correlators have a clear representation-theoretic meaning. Conversely, the same symmetry also organizes the final answers, since the natural building blocks are group invariants and covariants. This is the approach we take in this paper.

Summary of results. In this paper, we set up and clarify the particle-on-a-group description of (super-)JT gravity, and study applications of this formalism to supersymmetric correlators. We first review the bosonic prototype in Section 2 as a warm-up. Ordinary JT can be written as an $SL(2,\mathbb{R})$ BF theory whose boundary dynamics is a constrained particle on $SL(2,\mathbb{R})$. We explain how the JT boundary condition becomes the particle constraint, why the constrained particle describes the same dynamics, and how the constraint should be imposed quantum mechanically. The same logic will be supersymmetrized in the rest of the paper. For reference, we provide a self-contained derivation in Appendix B.

We then perform the supersymmetric generalization in Section 3, with emphasis on the $\mathcal{N} = 2$ and $\mathcal{N} = 4$ theories. The starting point is to identify how the supergravity boundary conditions become constraints for the particle on $SU(1,1|1)$ or $PSU(1,1|2)$. We translate the fields among the supergravity, super-BF, and particle-on-a-group descriptions, identify the constraint equations, and derive the super-Schwarzian action from the constrained particle on a supergroup. Therefore, at least classically, the particle-on-a-group formalism gives a group-theoretic realization of the same (super-)Schwarzian boundary dynamics.

We then quantize the reduced theory and address the subtleties associated with the constraints. In the supersymmetric case, the positive fermionic constraints satisfy an algebra that closes onto the bosonic parabolic charge. Once this bosonic charge is fixed, these fermionic constraints become second class and cannot all be imposed directly on ordinary scalar wavefunctions. Our particle-on-a-group derivation explains the auxiliary fermion χ in the LMRS Hilbert-space construction [40]: it is the degree of freedom associated with the positive-fermion sector of the constrained particle, and the reduced wavefunctions are schematically $F(x, \rho, a, \theta_-, \bar{\theta}_-, \chi)$. The same discussion on constraints also fixes the particle measure used when wavefunctions are glued.

In Section 4, we clarify the distinction and relation between two symmetries: the (super)symmetry

of the target space, which is the group, and the symmetry of the worldline. We often refer to them as bulk and boundary symmetries. The bulk symmetry is the $SU(1,1|1)$ or $PSU(1,1|2)$ isometry of the super-AdS$_2$ spacetime, while the boundary symmetry is the $\mathcal{N}=2$ or $\mathcal{N}=4$ super-Poincare symmetry of the boundary superline. The first result is a first-principles derivation of the relation between bulk and boundary supercharges. Although closely related, a bare bulk supercharge does not preserve the super-JT boundary condition in general; rather, it must be supplemented by a compensating large diffeomorphism. In the particle-on-a-group language, a bare $SU(1,1|1)$ or $PSU(1,1|2)$ translation would cause the trajectory to fail to satisfy the constraints. We find an equivalent constraint-preserving supercharge, which is precisely the $\mathcal{N}=2$ or $\mathcal{N}=4$ super-Poincare supercharge [25, 40, 43–45]. Using these charges, we construct supersymmetric propagators organized by bulk-supergroup invariants and explain the reduction to Liouville quantum mechanics [25, 40].

Moreover, the same distinction between bulk and boundary supersymmetry controls Wilson-line operators. A bulk field is a representation of the bulk isometry group, but its boundary-anchored two-point function has a more complicated transformation law than a bare $SU(1,1|1)$ or $PSU(1,1|2)$ transformation, because of the compensating diffeomorphisms required by the constraint. We show that for superconformal primaries, the bulk matter two-point function becomes a simple open Wilson line. For superdescendants, however, the construction is more subtle and must use the physical super-Poincare supercharge. This extends the familiar bosonic BF Wilson-line/Schwarzian bilocal correspondence [37]. For practical purposes, we also express the Wilson-line operators in terms of bulk-supergroup invariants. As an example, we discuss the marginal operator used for the D1–D5–P moduli deformation in Section 6, whose bilocal is obtained by acting with $\mathcal{N}=4$ supercharges on the short-multiplet primary bilocal.

In Section 5, we turn these ingredients into an algorithm for computing correlation functions. Starting from the reduced propagators and Wilson-line operators, we glue boundary intervals with the reduced particle measure and quotient by the diagonal global symmetry. The resulting integral is written in invariant length variables. The basic building block is the three-length composition kernel I_3, obtained by composing two interval propagators at a common endpoint. For ordinary JT this is the familiar bosonic I_3 kernel [46–48]. In the supersymmetric theories the same construction gives $\mathcal{N}=2$ and $\mathcal{N}=4$ kernels with a finite component structure reflecting the fermions and the R-symmetry data of the propagator. As a first application, we compute zero-energy scalar three-point functions in the $\mathcal{N}=2$ [49] and $\mathcal{N}=4$ theories. A technical point in this calculation is that the kernels are involved and the length-space integrands are complicated. We develop an integration-by-parts method in Appendix C to simplify them.

We finally apply the same length-space algorithm to four-point functions. The time-ordered channel is obtained from a product of two-point functions, while the crossed channel gives the OTOC. As a check on the method, Appendix D applies the same length-integral technology to the bosonic theory and reproduces the standard $\mathcal{N}=0$ JT OTOC answer [5, 36, 37, 50–52]. In the $\mathcal{N}=2$ zero-energy case

the length integrals can be reduced by integration by parts to an explicit finite answer, and we also analyze the corresponding large-Δ approximation. In the $\mathcal{N}=4$ BPS case the exact integral structure is more involved, but the same method gives a controlled large-Δ saddle approximation.

We end the paper with Section 6, where we discuss concrete protocols for two future calculations: the representation-theoretic data needed for a general closed-form super-JT OTOC, and the application to Berry curvature for D1–D5–P black hole microstates.

2 Warm-up: Bosonic-JT from constrained particle on a group

In this section we review ordinary JT gravity in the form used as the bosonic template for the rest of the paper. The detailed derivations are collected in Appendix B; here we only keep the ingredients needed in the main text. We start from the metric dilaton-gravity description, pass to the Schwarzian boundary mode, rewrite the same boundary problem in the $SL(2,\mathbb{R})$ BF language, and finally obtain the constrained particle on $SL(2,\mathbb{R})$. The metric and BF descriptions give the same boundary dynamics, but they emphasize different structures. The metric derivation makes the boundary curve geometrical, while the BF/group-particle description makes the $SL(2,\mathbb{R})$ representation theory and boundary Wilson lines manifest. This also fixes the notation for later use.

The starting point is the metric formulation of Euclidean JT gravity. The theory contains a two-dimensional metric $G_{\mu\nu}$ and a dilaton Φ [1, 2], with action

$$I_{\rm JT} = -\frac{1}{2}\int_M d^2X\,\sqrt{G}\,\Phi(R+2) - \int_{\partial M} ds\,\Phi_\partial(K-1). \tag{2.1}$$

The dilaton equation sets the curvature to $R=-2$, and hence the bulk metric is locally AdS_2. The nontrivial mode is therefore the regulated boundary curve. This is the standard nearly-AdS_2 derivation of the Schwarzian boundary action [5, 6].

The nearly-AdS_2 boundary condition fixes the leading proper length of the boundary together with the leading dilaton profile,

$$h_{uu}\big|_{\partial M} = \left(\frac{q}{\epsilon}\right)^2, \qquad \Phi\big|_{\partial M} = \frac{\phi_r}{\epsilon}, \qquad \epsilon \to 0. \tag{2.2}$$

Many Schwarzian conventions set the boundary einbein to one and absorb the physical length into the range of u. We instead keep the constant q explicit; we will see why later. The boundary condition fixes the asymptotic scale, but it still allows the boundary curve to fluctuate through the map $u \mapsto x(u)$. We call this reparametrization $x(u)$; common alternatives are $t(u)$ and $f(u)$.

Solving the constant-curvature constraint and evaluating the boundary extrinsic-curvature term

gives the Schwarzian action

$$I_{\rm Sch} = -\phi_r \int du\, \{x,u\}, \qquad \{x,u\} = \frac{x'''}{x'} - \frac{3}{2}\left(\frac{x''}{x'}\right)^2 . \tag{2.3}$$

Thus the metric derivation describes JT by the boundary reparametrization $x(u)$, modulo the global $SL(2,\mathbb{R})$ redundancy of AdS_2.

One convenient way to encode the Schwarzian in the metric and bridge to the first-order description is to use Fefferman–Graham gauge. Writing $ds^2 = dr^2 + (qe^r + e^{-r}\mathcal{L}(u) + \cdots)^2\, du^2$ and fixing r to be a large cutoff value, one can solve the boundary condition (2.2). In the AdS/CFT language, the growing piece is the source and is held fixed, while the decaying piece is the normalizable data, which we call the vev data in this paper. The extrinsic-curvature condition then determines $\mathcal{L}(u) = -(2q)^{-1}\{x,u\}$. We now translate this source/vev organization to the first-order $SL(2,\mathbb{R})$ BF description.

Now we turn to the first-order formulation. This is simply the passage to first-order variables, "taking the square root of the metric," as with the vielbein in ordinary gravity; see Appendix B for details. A convenient way to package the data is to introduce an $SL(2,\mathbb{R})$ gauge connection $\mathcal{A}$ and an adjoint scalar $\mathcal{X}$, whose components include the dilaton. This is the BF description [37, 38, 53, 54]. In this language the metric variables $(G_{\mu\nu}, \Phi)$ are replaced by $(\mathcal{A}, \mathcal{X})$, and the topological nature of the bulk theory becomes manifest. The bulk action is

$$I_{\rm BF,bulk} = -\int_M \langle \mathcal{X}, \mathcal{F}[\mathcal{A}] \rangle\,, \qquad \mathcal{F}[\mathcal{A}] = d\mathcal{A} + \mathcal{A}\wedge\mathcal{A}. \tag{2.4}$$

Here $\langle\,,\,\rangle$ denotes the invariant bilinear form on $\mathfrak{sl}(2,\mathbb{R})$, with generator conventions fixed in Appendix A.1. The equations of motion impose $\mathcal{F}[\mathcal{A}] = 0$, i.e. $\mathcal{A}$ is a flat connection, and make $\mathcal{X}$ covariantly constant. Together these equations reproduce the metric JT equations through the first-order formalism.

The boundary condition must also be translated into BF variables. In the BF description, the Dirichlet boundary condition in the metric formalism becomes a mixed boundary condition [37, 55]. We implement it by adding the boundary term

$$I_{\rm BF} = -\int_M \langle \mathcal{X}, \mathcal{F}[\mathcal{A}] \rangle + \int_{\partial M} du \left[\langle \mathcal{X}, \mathcal{A}_u \rangle - \frac{1}{2\phi_r} \langle \mathcal{X}, \mathcal{X} \rangle \right]. \tag{2.5}$$

Varying the full action gives

$$\mathcal{X}\big|_{\partial M} = \phi_r\, \mathcal{A}_u\big|_{\partial M}. \tag{2.6}$$

The source/vev split described above is encoded in the BF connection: the Fefferman–Graham gauge above translates to the following radial-stripped boundary connection,

$$\mathcal{A}_u = q\, \mathsf{L}_- + \mathcal{L}(u)\, \mathsf{L}_+. \tag{2.7}$$

The coefficient $\mathcal{L}(u)$ is the same vev mode that appeared in the Fefferman–Graham expansion, now written as the L_+ component of the radial-stripped connection.

We now extract the boundary particle from the flat BF connection. Since $\mathcal{F}[\mathcal{A}] = 0$, we can write $\mathcal{A} = \mathcal{G}^{-1}d\mathcal{G}$, with $\mathcal{G}(r,u) \in SL(2,\mathbb{R})$. Restricting to the boundary gives $\mathcal{A}_u = \mathcal{G}^{-1}\dot{\mathcal{G}}$. This $\mathcal{G}^{-1}\dot{\mathcal{G}}$ can be interpreted as a group current, which we denote by $\mathscr{J}(u)$. The BF boundary action then gives the first-order action for a particle on the group,

$$I_{\rm pG}[\mathcal{G},\mathcal{P}] = \phi_r \int du \left[\left\langle \mathcal{P}, \mathcal{G}^{-1}\dot{\mathcal{G}}\right\rangle - \frac{1}{2}\langle \mathcal{P},\mathcal{P}\rangle\right]. \tag{2.8}$$

Here $\mathcal{P} = \mathcal{X}/\phi_r$ is the normalized particle momentum. If we integrate out $\mathcal{P}$, we obtain the second-order form

$$I_{\rm pG}[\mathcal{G}] = \frac{\phi_r}{2}\int du \left\langle \mathcal{G}^{-1}\dot{\mathcal{G}}, \mathcal{G}^{-1}\dot{\mathcal{G}}\right\rangle. \tag{2.9}$$

This is still the unconstrained particle on $SL(2,\mathbb{R})$.

Ordinary JT is obtained only after imposing the condition inherited from the fixed boundary source. We highlight the following simple calculation. Using the Gauss parameterization $\mathcal{G}(u) = e^{x(u)\mathsf{L}_-}e^{\rho(u)\mathsf{L}_0}e^{\gamma(u)\mathsf{L}_+}$, the corresponding current is

$$\mathcal{G}^{-1}\dot{\mathcal{G}} = e^{\rho}\dot{x}\,\mathsf{L}_- + (\dot{\rho} - 2\gamma e^{\rho}\dot{x})\,\mathsf{L}_0 + \left(\dot{\gamma} + \gamma\dot{\rho} - \gamma^2 e^{\rho}\dot{x}\right)\mathsf{L}_+. \tag{2.10}$$

Imposing

$$\mathcal{G}^{-1}\dot{\mathcal{G}} = q\,\mathsf{L}_- + \mathcal{L}(u)\,\mathsf{L}_+ \tag{2.11}$$

gives

$$e^{\rho} = \frac{q}{\dot{x}}, \qquad \gamma = \frac{\dot{\rho}}{2q} = -\frac{\ddot{x}}{2q\dot{x}}, \qquad \mathcal{L}(u) = -\frac{1}{2q}\{x,u\}. \tag{2.12}$$

We mention that closely related derivations of this BF and group-particle route to the Schwarzian can be found in [55,56]. In our normalization $\langle q\mathsf{L}_- + \mathcal{L}\mathsf{L}_+, q\mathsf{L}_- + \mathcal{L}\mathsf{L}_+\rangle = 4q\mathcal{L}$. Hence the quadratic particle action reduces to

$$I_{\rm pG} = -\phi_r \int du\,\{x,u\}. \tag{2.13}$$

This reproduces the Schwarzian action from the constrained group particle. This also gives the explicit dictionary between the boundary curve and the group particle: the Schwarzian reparametrization $x(u)$ is the lower-triangular coordinate of $\mathcal{G}(u)$, while the Schwarzian derivative is encoded in the L_+ component $\mathcal{L}(u)$ of the reduced current.

Having established the particle-on-a-group formalism at the action level, we now turn to its quantization. The problem is a quantum-mechanical system with a constraint. Here we only summarize the result needed later; the more detailed Hamiltonian and path-integral derivation is in Appendix B. The source condition is imposed as a right parabolic charge constraint, $\langle \mathcal{P}, \mathsf{L}_+\rangle = q$. The appearance of L_+ follows from the invariant pairing: it is the momentum conjugate to the L_- source in (2.7). A

first-class form of the constrained particle is obtained by introducing a Lagrange multiplier, $\lambda(u)$,

$$I[\mathcal{G},\mathcal{P},\lambda]=\phi_r\int du\left[\left\langle\mathcal{P},\mathcal{G}^{-1}\dot{\mathcal{G}}-\lambda\mathsf{L}_+\right\rangle-\frac{1}{2}\langle\mathcal{P},\mathcal{P}\rangle+q\lambda\right]. \tag{2.14}$$

This action has a local right N_+ redundancy,

$$\mathcal{G}(u)\to\mathcal{G}(u)e^{\alpha(u)\mathsf{L}_+},\qquad \lambda(u)\to\lambda(u)+\dot{\alpha}(u),\qquad N_+=\exp(\mathbb{R}\mathsf{L}_+), \tag{2.15}$$

with the corresponding adjoint action on $\mathcal{P}$. Thus the constrained particle involves both a fixed parabolic charge and a quotient by the gauge orbit generated by this constraint.

The effect of this gauge redundancy is clearest in Hamiltonian language. Physical wavefunctions are equivariant under the right N_+ action: $\Psi_q(x,\rho,\gamma)=e^{-q\gamma}F(x,\rho)$, or infinitesimally $L_+^R\Psi_q=-q\Psi_q$. Acting on these wavefunctions, the $SL(2,\mathbb{R})$ Casimir becomes the reduced Hamiltonian, while the Haar measure descends to the quotient measure,

$$\widehat{H}_{\mathrm{JT},q}=\frac{1}{2\phi_r}\left(\partial_\rho^2+\partial_\rho+qe^{-\rho}\partial_x\right),\qquad d\mu_{\mathrm{phys}}=\frac{e^\rho\,d\rho\,dx\,d\gamma}{d\gamma}=e^\rho\,dx\,d\rho. \tag{2.16}$$

The eigenfunctions of this reduced Hamiltonian give the Schwarzian propagator; in the length variable used below they become the familiar Bessel profiles $\Psi_s(L)\propto K_{2is}(2qL)$, up to normalization. The measure in (2.16) is the one used whenever interval wavefunctions are glued.

Let us end with one more reason for using the particle-on-a-group approach: it organizes operator insertions well by symmetry. A matter field transforms in a representation $\mathcal{R}$ of the isometry group, and on a flat connection its Wilson line reduces to the endpoint matrix element $\mathcal{U}_\Delta(i,j)=\mathcal{R}_\Delta(\mathcal{G}_i^{-1}\mathcal{G}_j)$. Choosing scalar-primary endpoint polarizations and using the source constraint $e^\rho\dot{x}=q$ at each endpoint gives the usual Schwarzian bilocal as

$$\begin{aligned}\mathcal{O}_\Delta(i,j)&=q^{2\Delta}{}_\Delta\langle\text{primary}|\mathcal{R}_\Delta(\mathcal{G}_i^{-1}\mathcal{G}_j)|\text{primary}\rangle_\Delta\\&=\left[\frac{qe^{-\frac{1}{2}(\rho_i+\rho_j)}}{x_j-x_i}\right]^{2\Delta}=\left[\frac{\dot{x}_i\dot{x}_j}{(x_i-x_j)^2}\right]^{\Delta}.\end{aligned} \tag{2.17}$$

Thus, already in bosonic JT, the particle-on-a-group language gives a compact way to organize correlators by reduced wavefunctions, the physical measure, and Wilson-line bilocals. Appendix D uses precisely these ingredients to reproduce the standard OTOC kernel in length space.

3 Super-JT from the particle on a group

Section 2 described bosonic JT as a constrained particle on $SL(2,\mathbb{R})$. We now apply the same logic for super-JT. We begin in Section 3.1 by setting up notation and explaining the bridge from dilaton supergravity to first-order BF theory, which is the natural bridge to the particle on a group. The

standard reference for this (super)gravity-to-BF map in the $\mathcal{N}=4$ case is [7].[1] We follow the same source/vev organization, but rewrite it in a form adapted to the particle on a group. Section 3.2 then gives the main classical check of the particle on a supergroup formalism. After imposing the constraints from super-JT boundary conditions, we show that the particle action reduces to the $\mathcal{N}=2$ or $\mathcal{N}=4$ super-Schwarzian action, meaning that the constrained particle is a group-theoretic manifestation of super-Schwarzian dynamics. Finally, Section 3.3 addresses the quantization of this constrained system. The fixed parabolic charge condition is standard, while the new fermionic constraints require a more careful treatment because they become second class after the parabolic charge is fixed. This is the natural origin of the auxiliary fermions in the LMRS construction [40], and it also fixes the reduced Hilbert space and the path-integral or wavefunction gluing measure used in later sections.

3.1 Boundary data and radial weights

As in Section 2, we treat super-JT gravity in the usual AdS/CFT way: the gravitational theory is a two-dimensional dilaton gravity with a prescribed boundary condition, and that boundary condition is most transparent after choosing a holographic radial gauge near infinity. We then explain the passage to first-order variables in a little more detail, introducing the frame fields and packaging the supergravity fields into a gauge theory connection. This gives a direct route from (super)BF theory to the particle on a group.

Coordinates and fields. We use the superspace coordinates $\mathbf{Z}^M = (X^\mu = (r,u); \Theta^{p\alpha}, \bar{\Theta}_p{}^\alpha)$ and $z^m = (u, \vartheta^p, \bar{\vartheta}_p)$. Here $\mathbf{Z}^M$ are bulk supercoordinates and z^m are the boundary superline coordinates. In the main text we will only use component fields; the flat superline basis and the corresponding derivatives are recorded in Appendix A.2.[2] The index p has one value for $\mathcal{N}=2$ and two values for $\mathcal{N}=4$. We write the R-symmetry group as $\mathfrak{R} = U(1)_\mathfrak{R}$ for $\mathcal{N}=2$, and $\mathfrak{R} = SU(2)_\mathfrak{R}$ for $\mathcal{N}=4$. The index I denotes an $\mathfrak{R}$-algebra index: $I=\mathfrak{R}$ for $U(1)_\mathfrak{R}$, and $I=i=1,2,3$ for $SU(2)_\mathfrak{R}$. The component supergravity fields are

$$E_\mu{}^a, \qquad \omega_\mu, \qquad A_\mu{}^I, \qquad \Psi^p_\mu, \qquad \bar{\Psi}_{\mu,p}. \tag{3.1}$$

They are the frame, spin connection, R-symmetry gauge field, and gravitini.

Asymptotic modes and boundary conditions in radial gauge. As in the bosonic discussion, radial weights distinguish fixed sources from vev data. We now apply this separation to the frame, gravitini, and $\mathfrak{R}$-symmetry gauge field. The growing modes are held fixed at the boundary, while the

[1] In [7] the boundary super-reparametrization is parametrized by $f(\tau)$, $g(\tau) \in SU(2)$, and fermions $\eta^p(\tau), \bar{\eta}_p(\tau)$. These become our particle variables $x(u)$, $U(u) \in SU(2)_\mathfrak{R}$, and $\theta^p_-(u), \bar{\theta}_{-,p}(u)$. Their component super-Schwarzian multiplet $(S_b, S^p_f, \bar{S}_{f,p}, S^i_T)$, or equivalently the BF boundary fields $(\tilde{L}, \psi^p, \bar{\psi}_p, B^i_\tau)$, is represented here by the reduced current components $(\mathcal{L}, \mathcal{S}^p, \bar{\mathcal{S}}_p, \mathcal{R}^i)$.

[2] In particular, Appendix A.2 gives Δu, $d\vartheta^p$, $d\bar{\vartheta}_p$, and the derivatives $D_p, \bar{D}^p$.

decaying modes are the vev data[3] integrated over in the gravitational path integral.

It is convenient to make this separation in radial gauge. We impose radial gauge on the component fields,

$$E^{\hat{r}} = dr, \qquad E_u{}^{\hat{r}} = 0, \qquad \Psi^p_r = \bar{\Psi}_{r,p} = 0, \qquad A_r{}^I = 0. \tag{3.2}$$

The boundary data are carried by the transverse components

$$E_u{}^{\hat{u}}, \qquad \Psi^p_u, \qquad \bar{\Psi}_{u,p}, \qquad A_u{}^I. \tag{3.3}$$

We now solve the radial equations mode by mode. For the frame field radial gauge leaves

$$E^{\hat{u}} = E_u{}^{\hat{u}}(r,u)\, du, \qquad ds^2 = dr^2 + \Big(E_u{}^{\hat{u}}(r,u)\Big)^2\, du^2. \tag{3.4}$$

The asymptotic solution is

$$E_u{}^{\hat{u}}(r,u) = e^r e^{(0)}_u(u) + e^{-r}\mathcal{L}(u) + \cdots . \tag{3.5}$$

The leading coefficient $e^{(0)}_u$ is the metric source, while $\mathcal{L}$ is the stress-tensor vev. The spin connection follows from the torsion constraint,

$$\omega^{\hat{r}\hat{u}} = \partial_r E_u{}^{\hat{u}}(r,u)\, du = \Big(e^r e^{(0)}_u - e^{-r}\mathcal{L} + \cdots\Big)\, du. \tag{3.6}$$

The gravitino equation is the fermionic curvature equation. In radial gauge the leading radial part and its two eigensolutions are

$$\begin{aligned}
0 &= \left(\partial_r + \frac{1}{2}\gamma_{\hat{r}}\right)\Psi^p_u + \cdots , \\
\frac{1}{2}(1-\gamma_{\hat{r}})\Psi^p_u = e^{r/2} v^{(0)p}_u + \cdots , \qquad & \frac{1}{2}(1+\gamma_{\hat{r}})\Psi^p_u \quad = e^{-r/2}\mathcal{S}^p + \cdots , \\
0 &= \left(\partial_r + \frac{1}{2}\gamma_{\hat{r}}\right)\bar{\Psi}_{u,p} + \cdots , \\
\frac{1}{2}(1-\gamma_{\hat{r}})\bar{\Psi}_{u,p} = e^{r/2}\bar{v}^{(0)}_{u,p} + \cdots , \qquad & \frac{1}{2}(1+\gamma_{\hat{r}})\bar{\Psi}_{u,p} \quad = e^{-r/2}\bar{\mathcal{S}}_p + \cdots .
\end{aligned} \tag{3.7}$$

The $e^{r/2}$ coefficients $v^{(0)p}_u, \bar{v}^{(0)}_{u,p}$ are the gravitino sources. The $e^{-r/2}$ coefficients $\mathcal{S}^p, \bar{\mathcal{S}}_p$ are the supercurrent vevs.

The R-symmetry gauge field equation is the $\mathfrak{R}$-curvature equation. Its radial component is

$$F_{\mathfrak{R},ru}{}^I = \partial_r A_u{}^I - \partial_u A_r{}^I + f^I{}_{JK} A_r{}^J A_u{}^K = 0. \tag{3.8}$$

[3] We use "vev" as shorthand for vacuum expectation value, or more generally the holographic one-point-function data conjugate to a source.

Using $A_r{}^I = 0$, this becomes

$$\partial_r A_u{}^I = 0. \tag{3.9}$$

Thus the R-symmetry gauge field has zero radial weight. Since it has zero weight, the radial equation does not by itself split $A_u{}^I$ into a growing source mode and a decaying vev mode. The boundary condition depends on the ensemble: one may fix the boundary holonomy, or one may fix the conjugate $\mathfrak{R}$-flux/charge. In the rest of the paper we use the fixed $\mathfrak{R}$-flux, equivalently fixed $\mathfrak{R}$-charge, ensemble. The $SU(2)_{\mathfrak{R}}$ group element $U(u)$ is therefore retained as a particle variable in the $\mathcal{N} = 4$ theory, and its conjugate current $\mathcal{R}^i(u)$ appears as part of the super-Schwarzian vev multiplet.

The Dirichlet boundary condition fixes the source coefficients and lets the vev coefficients fluctuate. For the zero-weight R-symmetry field this statement is understood with the fixed-flux ensemble just specified. The bosonic metric source and vanishing gravitino source condition is

$$e_u^{(0)} = \mathfrak{q}, \qquad v_u^{(0)p} = \bar{v}_{u,p}^{(0)} = 0. \tag{3.10}$$

This condition does not constrain the vev data $\mathcal{L}, \mathcal{S}^p, \bar{\mathcal{S}}_p$, which are allowed to fluctuate. In the supersymmetric discussion we denote this fixed boundary source by $\mathfrak{q}$, following the LMRS notation [40].

First-order form as a bridge. We now package the same boundary data into the super-BF variables. This is the supergravity analogue of the first-order map reviewed in Section 2: the frame, spin connection, $\mathfrak{R}$-symmetry gauge field, and gravitini become components of a single superconnection $\mathcal{A}$, while the dilaton multiplet becomes the adjoint scalar $\mathcal{X}$. We use $\mathsf{L}_\pm, \mathsf{L}_0$ for the $SL(2)$ generators, R_I for the $\mathfrak{R}$-symmetry generators, and $\mathsf{F}_{\pm,p}, \bar{\mathsf{F}}^p_\pm$ for the fermionic generators. With the radial weights used below, the $\mathsf{F}_{-,p}, \bar{\mathsf{F}}^p_-$ coefficients are growing source modes, while the $\mathsf{F}_{+,p}, \bar{\mathsf{F}}^p_+$ coefficients are decaying vev modes; see Appendix A.1 for the algebra and pairing. We write this connection schematically as

$$\mathcal{A} = \mathcal{A}^- \mathsf{L}_- + \mathcal{A}^0 \mathsf{L}_0 + \mathcal{A}^+ \mathsf{L}_+ + \mathsf{A}^I \mathsf{R}_I + \Psi^p_- \mathsf{F}_{-,p} + \bar{\Psi}_{-,p} \bar{\mathsf{F}}^p_- + \Psi^p_+ \mathsf{F}_{+,p} + \bar{\Psi}_{+,p} \bar{\mathsf{F}}^p_+. \tag{3.11}$$

The first-order equations of motion are the BF equations

$$\mathcal{F}[\mathcal{A}] \equiv d\mathcal{A} + \mathcal{A} \wedge \mathcal{A} = 0, \qquad D_{\mathcal{A}}\mathcal{X} \equiv d\mathcal{X} + [\mathcal{A}, \mathcal{X}] = 0. \tag{3.12}$$

The flatness equation packages the component equations for the frame, spin connection, $\mathfrak{R}$-symmetry gauge field, and gravitini; the second equation is the dilaton-multiplet equation. The first-order radial gauge is

$$\mathcal{A}_r = \mathsf{L}_0. \tag{3.13}$$

The radial part of the flatness equation is

$$\partial_r \mathcal{A}_u + [\mathsf{L}_0, \mathcal{A}_u] = 0. \tag{3.14}$$

If $[\mathsf{L}_0, \mathsf{T}_w] = w\,\mathsf{T}_w$, the coefficient of T_w is e^{-wr} times a boundary field. Therefore

$$\begin{aligned}\mathcal{A}(r,u) &= \mathsf{L}_0\,dr \\ &\quad + du\Big[e^r e_u^{(0)}\,\mathsf{L}_- + e^{r/2}v_u^{(0)p}\,\mathsf{F}_{-,p} + e^{r/2}\bar{v}_{u,p}^{(0)}\,\bar{\mathsf{F}}_-^p + \mathsf{A}_u{}^I\,\mathsf{R}_I + e^{-r/2}\mathcal{S}^p\,\mathsf{F}_{+,p} + e^{-r/2}\bar{\mathcal{S}}_p\,\bar{\mathsf{F}}_+^p + e^{-r}\mathcal{L}\,\mathsf{L}_+\Big].\end{aligned} \tag{3.15}$$

The minus-components are the sources and the plus-components are the vev data. The zero-weight coefficient $\mathsf{A}_u{}^I$ is treated according to the chosen $\mathfrak{R}$-ensemble; in a fixed-flux ensemble its conjugate charge is the R-current $\mathcal{R}^I$. For $\mathcal{N} = 4$, the vev fields $(\mathcal{L}, \mathcal{S}^p, \bar{\mathcal{S}}_p, \mathcal{R}^i)$ form the component super-Schwarzian multiplet, and the $\mathcal{N} = 2$ case is the $U(1)_{\mathfrak{R}}$ truncation [7].

The adjoint scalar $\mathcal{X}$ is the first-order dilaton multiplet. Besides the ordinary JT dilaton and auxiliary torsion fields, it contains the dilatini and the $\mathfrak{R}$-symmetry multipliers. Its radial equation gives

$$\mathcal{X}(r,u) = e^r X_-\,\mathsf{L}_- + e^{r/2}\zeta_-^p\,\mathsf{F}_{-,p} + e^{r/2}\bar{\zeta}_{-,p}\,\bar{\mathsf{F}}_-^p + X_{\mathcal{R}}{}^I\,\mathsf{R}_I + e^{-r/2}\zeta_+^p\,\mathsf{F}_{+,p} + e^{-r/2}\bar{\zeta}_{+,p}\,\bar{\mathsf{F}}_+^p + e^{-r}X_+\,\mathsf{L}_+. \tag{3.16}$$

The minus-components are fixed by the renormalized dilaton data, while the plus-components are conjugate vev data. We only use the radial part of $D_{\mathcal{A}}\mathcal{X} = 0$ here; the transverse equation will reappear below.

The BF action with the JT boundary term is

$$I[\mathcal{A}, \mathcal{X}] = -\int_M \langle \mathcal{X}, \mathcal{F}[\mathcal{A}]\rangle + \int_{\partial M} du\,\left[\langle \mathcal{X}, \mathcal{A}_u\rangle - \frac{1}{2\phi_r}\langle \mathcal{X}, \mathcal{X}\rangle\right]. \tag{3.17}$$

Here $\langle\,,\,\rangle$ is the invariant supertrace pairing, with the gravity normalization fixed in Appendix A.1. The boundary term imposes the JT mixed boundary condition

$$\mathcal{X}|_{\partial M} = \phi_r\,\mathcal{A}_u|_{\partial M}\,, \qquad \delta\mathcal{X}|_{\partial M} = \phi_r\,\delta\mathcal{A}_u|_{\partial M}\,, \tag{3.18}$$

as reviewed in Section 2, with details in Appendix B.2. For related references on this BF/boundary-particle viewpoint in the bosonic and $\mathcal{N} = 4$ cases, see also [7, 37, 57].

For the bosonic metric source and vanishing gravitino sources, the boundary value of the connection is

$$\mathcal{A}_u|_{\partial M} = e^r\mathfrak{q}\,\mathsf{L}_- + 0\times\mathsf{F}_{-,p} + 0\times\bar{\mathsf{F}}_-^p + 0\times\mathsf{L}_0 + \mathsf{A}_u{}^I\,\mathsf{R}_I + e^{-r/2}\mathcal{S}^p\,\mathsf{F}_{+,p} + e^{-r/2}\bar{\mathcal{S}}_p\,\bar{\mathsf{F}}_+^p + e^{-r}\mathcal{L}\,\mathsf{L}_+. \tag{3.19}$$

The displayed zero components, together with the fixed L_- source, will be important later: after stripping the radial weights, they become the constraint equations in (3.26). The plus-components

$\mathcal{L}, \mathcal{S}^p, \bar{\mathcal{S}}_p$ are still dynamical vev data.[4]

Flatness, $\mathcal{F}[\mathcal{A}] = 0$, lets us write the connection locally as $\mathcal{A} = \mathcal{G}^{-1} d\mathcal{G}$, where $\mathcal{G}(r, u)$ is a group-valued field on AdS_2. After stripping the radial weights, the boundary condition on $\mathcal{A}_u$ becomes a condition on the boundary current $\mathcal{G}^{-1}(u)\dot{\mathcal{G}}(u)$, with $\mathcal{G}(u) = \mathcal{G}(r, u)|_{\partial M}$. The detailed boundary reduction is reviewed in Section 2 and Appendix B.2.

3.2 Super-Schwarzian from the particle on a group

We now perform the supersymmetric analogue of the current calculation in Section 2. The goal is to show that imposing the super-JT source constraints on the group particle produces the super-Schwarzian multiplet. For the explicit calculation we use the $\mathcal{N} = 2$ theory with $U(1)_{\mathfrak{R}}$ generator $\mathsf{J}_{\mathfrak{R}}$, and we comment on the $\mathcal{N} = 4$ case at the end. The particle-on-a-group action follows from the BF boundary action (3.17) by replacing the boundary gauge connection $\mathcal{A}_u$ with $\mathcal{G}^{-1}(u)\dot{\mathcal{G}}(u)$, with normalized particle momentum $\mathcal{P} = \mathcal{X}/\phi_r$. In this language the same $\mathfrak{q}$ is the fixed parabolic charge of the group particle, identified with the boundary source above. The first-order particle action is

$$I[\mathcal{G}, \mathcal{P}] = \phi_r \int du \left[\left\langle \mathcal{P}, \mathcal{G}^{-1}\dot{\mathcal{G}} \right\rangle - \frac{1}{2} \langle \mathcal{P}, \mathcal{P} \rangle \right]. \tag{3.20}$$

One can integrate out $\mathcal{P}$ to get

$$I[\mathcal{G}] = \frac{\phi_r}{2} \int du \left\langle \mathcal{G}^{-1}\dot{\mathcal{G}}, \mathcal{G}^{-1}\dot{\mathcal{G}} \right\rangle. \tag{3.21}$$

Variables and constraints. We use the ordered parametrization

$$\mathcal{G} = e^{x\mathsf{L}_-} e^{\theta_- \mathsf{F}_- + \bar{\theta}_- \bar{\mathsf{F}}_-} e^{\rho \mathsf{L}_0} e^{a \mathsf{J}_{\mathfrak{R}}} e^{\theta_+ \mathsf{F}_+ + \bar{\theta}_+ \bar{\mathsf{F}}_+} e^{\gamma \mathsf{L}_+}. \tag{3.22}$$

The variables $x, \theta_-, \bar{\theta}_-, a$ are the surviving boundary variables after the source conditions are imposed. The variables $\rho, \theta_+, \bar{\theta}_+, \gamma$ are solved for by the minus-component constraints. The $U(1)_{\mathfrak{R}}$ variable $a(u)$ is kept because the zero-weight R-symmetry boundary condition is an ensemble choice. We use a as a Lie-algebra coordinate; the unitary convention e^{iaJ} is recovered by $a \to ia$.

Define

$$\mathscr{J} = \mathcal{G}^{-1}\dot{\mathcal{G}}. \tag{3.23}$$

and we denote by $\mathscr{J}_{\mathsf{T}}$ its coefficient along the generator T. The component which replaces the bosonic $\dot{x}$ is

$$\Pi = \dot{x} + \bar{\theta}_- \dot{\theta}_- + \theta_- \dot{\bar{\theta}}_-. \tag{3.24}$$

[4] If one later imposes the remaining transverse equation of motion, namely the u-component of $D_{\mathcal{A}}\mathcal{X} = 0$, this gives $\partial_u \mathcal{L} = 0$, $\partial_u \mathcal{S}^p = 0$, and $\partial_u \bar{\mathcal{S}}_p = 0$. Equivalently, these constant vev data solve the Schwarzian equations of motion.

This is the component form of the invariant superline one-form. A direct multiplication gives the components of $\mathscr{J}$ needed for the source constraint:

$$\begin{aligned}
\mathscr{J}_{\mathsf{L}_-} &= e^{\rho}\Pi, \\
\mathscr{J}_{\mathsf{F}_-} &= e^{a+\rho/2}\dot{\theta}_- + e^{\rho}\Pi\,\theta_+ + O(\theta_-^3), \\
\mathscr{J}_{\bar{\mathsf{F}}_-} &= e^{-a+\rho/2}\dot{\bar{\theta}}_- - e^{\rho}\Pi\,\bar{\theta}_+ + O(\theta_-^3), \\
\mathscr{J}_{\mathsf{L}_0} &= \dot{\rho} - 2\gamma e^{\rho}\Pi + O(\theta_-^4).
\end{aligned} \tag{3.25}$$

The absence of a quadratic correction in $\mathscr{J}_{\mathsf{L}_0}$ gives a non-trivial check: the two terms which would come from solving $\mathscr{J}_{\mathsf{F}_-} = 0$ and $\mathscr{J}_{\bar{\mathsf{F}}_-} = 0$ cancel.

We now impose the source conditions obtained by stripping the radial weights from (3.19),

$$\mathscr{J}_{\mathsf{L}_-} = \mathfrak{q}, \qquad \mathscr{J}_{\mathsf{F}_-} = 0, \qquad \mathscr{J}_{\bar{\mathsf{F}}_-} = 0, \qquad \mathscr{J}_{\mathsf{L}_0} = 0. \tag{3.26}$$

These equations fix the metric source, turn off the gravitino sources, and choose the simple L_0 representative. They do not set the plus-component vev data to zero. Solving them gives

$$e^{\rho} = \frac{\mathfrak{q}}{\Pi}, \qquad \theta_+ = -\frac{e^{a+\rho/2}}{\mathfrak{q}}\dot{\theta}_- + O(\theta_-^3), \qquad \bar{\theta}_+ = \frac{e^{-a+\rho/2}}{\mathfrak{q}}\dot{\bar{\theta}}_- + O(\theta_-^3), \qquad \gamma = \frac{\dot{\rho}}{2\mathfrak{q}} + O(\theta_-^4). \tag{3.27}$$

This is the supersymmetric version of the bosonic solution (2.12), with $\dot{x}$ replaced by Π.

Reduced current and super-Schwarzian. After the auxiliary variables are eliminated, the reduced current has the same form as in Appendix A.2,

$$\mathscr{J}^{(2)}_{\text{red}} = \mathfrak{q}\,\mathsf{L}_- + \mathcal{R}\,\mathsf{J}_{\mathfrak{R}} + \mathcal{S}\,\mathsf{F}_+ + \bar{\mathcal{S}}\,\bar{\mathsf{F}}_+ + \mathcal{L}\,\mathsf{L}_+. \tag{3.28}$$

To quadratic order in the boundary fermions,

$$\mathcal{R} = \dot{a} - \frac{\dot{\theta}_-\dot{\bar{\theta}}_-}{\Pi} + O(\theta_-^4), \tag{3.29}$$

and the plus fermionic components begin as

$$\begin{aligned}
\mathcal{S} &= -\frac{e^{a+\rho/2}}{\mathfrak{q}}\left(\ddot{\theta}_- - \frac{\dot{\Pi}}{\Pi}\dot{\theta}_-\right) + O(\theta_-^3), \\
\bar{\mathcal{S}} &= \frac{e^{-a+\rho/2}}{\mathfrak{q}}\left(\ddot{\bar{\theta}}_- - \frac{\dot{\Pi}}{\Pi}\dot{\bar{\theta}}_-\right) + O(\theta_-^3).
\end{aligned} \tag{3.30}$$

The stress-tensor component is

$$\mathcal{L} = -\frac{1}{2\mathfrak{q}}\left[\frac{\ddot{\Pi}}{\Pi} - \frac{3}{2}\left(\frac{\dot{\Pi}}{\Pi}\right)^2\right] + \frac{1}{\mathfrak{q}\Pi}\left(\ddot{\theta}_-\dot{\bar{\theta}}_- - \dot{\theta}_-\ddot{\bar{\theta}}_-\right) + O(\theta_-^4). \tag{3.31}$$

These vev fields are the plus-components of the same reduced current after the source components have been fixed.

Substituting (3.28) into the quadratic action gives

$$\frac{1}{2}\left\langle \mathscr{J}^{(2)}_{\rm red}, \mathscr{J}^{(2)}_{\rm red}\right\rangle = 2\mathfrak{q}\,\mathcal{L} - 2\mathcal{R}^2 + 2\mathcal{S}\bar{\mathcal{S}} \text{ terms}. \tag{3.32}$$

The resulting particle action has the form

$$I_{\rm red} = 2\phi_r\mathfrak{q}\int du\,\mathcal{L}_{\mathcal{N}=2}, \tag{3.33}$$

up to total derivatives and the normalization convention for the $U(1)_{\mathfrak{R}}$ generator. Equations (3.32) and (3.33) are therefore the component form of the $\mathcal{N}=2$ super-Schwarzian action written in superspace in Appendix A.2; see also [7, 58].

$\mathcal{N}=4$ case. For $\mathcal{N}=4$, we perform the same calculation in $PSU(1,1|2)$. The single minus pair $\theta_-, \bar{\theta}_-$ becomes $\theta^p_-, \bar{\theta}_{-,p}$, and the $U(1)_{\mathfrak{R}}$ factor $e^{a\mathsf{J}_{\mathfrak{R}}}$ is replaced by $U(u) \in SU(2)_{\mathfrak{R}}$. The invariant superline one-form becomes

$$\Pi = \dot{x} + \bar{\theta}_{-,p}\dot{\theta}^p_- + \theta^p_-\dot{\bar{\theta}}_{-,p}. \tag{3.34}$$

Solving the $\mathcal{N}=4$ source constraints gives

$$e^\rho = \frac{\mathfrak{q}}{\Pi}, \qquad \theta^p_+ = -\frac{e^{\rho/2}}{\mathfrak{q}}(U^{-1})^p{}_q\dot{\theta}^q_- + O(\theta^3), \qquad \bar{\theta}_{+,p} = \frac{e^{\rho/2}}{\mathfrak{q}}\dot{\bar{\theta}}_{-,q}U^q{}_p + O(\theta^3), \qquad \gamma = \frac{\dot{\rho}}{2\mathfrak{q}} + O(\theta^4). \tag{3.35}$$

The reduced current is most cleanly written in terms of the group-current coefficients $(\mathcal{R}^i, \mathcal{S}^p, \bar{\mathcal{S}}_p, \mathcal{L})$.

$$\mathscr{J}^{(4)}_{\rm red} = \mathfrak{q}\,\mathsf{L}_- + \mathcal{R}^i\,\mathsf{R}_i + \mathcal{S}^p\,\mathsf{F}_{+,p} + \bar{\mathcal{S}}_p\,\bar{\mathsf{F}}^p_+ + \mathcal{L}\,\mathsf{L}_+. \tag{3.36}$$

The analog of (3.29)–(3.31) is

$$\begin{aligned}
\mathcal{R}^i &= \frac{i}{2}\,\mathrm{Tr}\Big(\sigma^i U^{-1}\dot{U}\Big) - \frac{i}{2\Pi}(\sigma^i)^q{}_p(U^{-1})^p{}_r\dot{\theta}^r_-\,\dot{\bar{\theta}}_{-,s}U^s{}_q + O(\theta^4),\\
\mathcal{S}^p &= -\frac{e^{\rho/2}}{\mathfrak{q}}\left[(U^{-1}\ddot{\theta}_-)^p - \frac{\dot{\Pi}}{\Pi}(U^{-1}\dot{\theta}_-)^p\right] + O(\theta^3),\\
\bar{\mathcal{S}}_p &= \frac{e^{\rho/2}}{\mathfrak{q}}\left[(\ddot{\bar{\theta}}_-U)_p - \frac{\dot{\Pi}}{\Pi}(\dot{\bar{\theta}}_-U)_p\right] + O(\theta^3),
\end{aligned}$$

$$\mathcal{L} = -\frac{1}{2\mathfrak{q}}\left[\frac{\ddot{\Pi}}{\Pi} - \frac{3}{2}\left(\frac{\dot{\Pi}}{\Pi}\right)^2\right] + \frac{1}{\mathfrak{q}\Pi}\left[(U^{-1}\ddot{\theta}_-)^p(\dot{\bar{\theta}}_- U)_p - (U^{-1}\dot{\theta}_-)^p(\ddot{\bar{\theta}}_- U)_p\right] + O(\theta^4). \tag{3.37}$$

These are the direct coefficients of $\mathcal{G}^{-1}\dot{\mathcal{G}}$. The reduced action is

$$\frac{1}{2}\left\langle \mathscr{J}^{(4)}_{\text{red}}, \mathscr{J}^{(4)}_{\text{red}} \right\rangle = 2\mathfrak{q}\,\mathcal{L} - \mathcal{R}^i\mathcal{R}^i + \mathcal{S}^p\bar{\mathcal{S}}_p \text{ terms}. \tag{3.38}$$

Using the pairings in (A.8), this gives the reduced action $I^{(4)}_{\text{red}} = 2\phi_r\mathfrak{q}\int du\,\mathcal{L}_{\mathcal{N}=4}$, where $\mathcal{L}_{\mathcal{N}=4}$ is the component Lagrangian. Thus the particle-on-a-group calculation reproduces the $\mathcal{N}=4$ super-Schwarzian action [7, 59, 60].

3.3 Hamiltonian reduction and the fermionic polarization

As explained in Section 2, to quantize the constrained particle on a group, we not only need to impose the constraint equation but also need to quotient out a (right) subgroup N_+. In super-JT we impose the same bosonic charge together with the positive fermionic charges. The new point is that, after $L^R_+ = -\mathfrak{q}$ is fixed, the positive fermionic constraints satisfy some non-trivial algebra.. We will show how to deal with this and how the LMRS auxiliary fermions χ^p appears.

How should the fermionic constraints be imposed? The super-FG boundary condition fixes the minus-component sources in (3.26). Using the invariant pairing, this becomes a fixed right-charge condition for the generators L_+, F_+, and $\bar{\mathsf{F}}_+$:

$$\mathfrak{n}^{\text{susy}}_+ = \text{span}\left\{\mathsf{L}_+,\, \mathsf{F}_{+,p},\, \bar{\mathsf{F}}^p_+\right\}, \qquad p = 1, 2, \tag{3.39}$$

with one value of p in the $\mathcal{N}=2$ truncation. We impose

$$\mu^R_{\mathsf{L}_+} = -\mathfrak{q}, \qquad \mu^R_{\mathsf{F}_{+,p}} = 0, \qquad \mu^R_{\bar{\mathsf{F}}^p_+} = 0, \tag{3.40}$$

or, quantum mechanically, $L^R_+\Psi = -\mathfrak{q}\Psi$. The bosonic equation fixes the dependence of the wavefunction along the right N_+ coordinate. The fermionic equations require more care, because after the L_+ charge is fixed the F_+ and $\bar{\mathsf{F}}_+$ constraints no longer close into a quantity that vanishes on the constrained surface. Indeed,

$$\left\{\mathsf{F}^R_{+,p}, \bar{\mathsf{F}}^{R,q}_+\right\} = \delta_p{}^q\,\mathsf{L}^R_+. \tag{3.41}$$

If we tried to impose both fermionic constraints directly as scalar annihilation conditions,

$$\mathsf{F}^R_{+,p}\Psi = 0, \qquad \bar{\mathsf{F}}^{R,q}_+\Psi = 0, \tag{3.42}$$

then their anticommutator would give

$$0 = \left\{ \mathsf{F}^R_{+,p}, \bar{\mathsf{F}}^{R,q}_{+} \right\} \Psi = -\mathfrak{q}\, \delta_p{}^q \Psi. \tag{3.43}$$

Thus the obstruction is precise: the bracket of two constraints which are both set to zero becomes the nonzero number $-\mathfrak{q}$.

This is the standard distinction between first-class and second-class constraints. For a first-class set, the brackets close into the constraints themselves, and therefore vanish after the constraints are imposed. Here the $\mathsf{F}_+, \bar{\mathsf{F}}_+$ pair is second class once the L_+ charge is fixed. The fixed-value equations are still legitimate equations on phase space, but they should not be imposed simultaneously as annihilation equations on an ordinary scalar wavefunction. One can either solve the second-class system with the Dirac bracket, or enlarge the phase space so that an equivalent first-class system is obtained before quantization.

We first illustrate the two equivalent treatments in a minimal model. Consider one pair of independent Grassmann variables $\theta, \bar{\theta}$, with conjugate momenta $\pi_\theta, \pi_{\bar{\theta}}$, and impose the two constraints

$$C = \pi_\theta + \mathfrak{q}\bar{\theta}, \qquad \bar{C} = \pi_{\bar{\theta}} + \mathfrak{q}\theta. \tag{3.44}$$

With the standard brackets one obtains $\{C, \bar{C}\} \sim 2\mathfrak{q}$. For $\mathfrak{q} \neq 0$, C and $\bar{C}$ are therefore a second-class pair.

The direct treatment is to use the Dirac bracket. For any second-class constraint set $\phi_A = 0$, define the Dirac bracket by

$$\{f, g\}_D = \{f, g\} - \{f, \phi_A\}(\Delta^{-1})^{AB}\{\phi_B, g\}, \tag{3.45}$$

if the constraint-bracket matrix $\Delta_{AB} = \{\phi_A, \phi_B\}$ is invertible on the constraint surface. It is constructed so that, for all functions f,

$$\{f, \phi_A\}_D = 0. \tag{3.46}$$

In the present example, this gives $\{\theta, \bar{\theta}\}_D = 2\mathfrak{q}^{-1}$ and $\{\theta, \theta\}_D = \{\bar{\theta}, \bar{\theta}\}_D = 0$. Thus θ and $\bar{\theta}$ become conjugate fermionic variables on the reduced phase space. The original coordinate-momentum bracket therefore becomes a bracket between θ and $\bar{\theta}$ themselves.[5]

For the supergroup quotient it is more convenient to use an equivalent first-class construction. Instead of replacing the Poisson bracket by the Dirac bracket, we enlarge the phase space by an auxiliary fermionic pair (χ, π_χ), with $\{\chi, \pi_\chi\} = 1$. The old second-class equations $C = \bar{C} = 0$ are then

[5] Equivalently, the Dirac bracket is the inverse of the reduced fermionic symplectic form $\Omega_{\text{red}} \sim 2\mathfrak{q}\, d\bar{\theta}\, d\theta$. This form is nondegenerate on the remaining $(\theta, \bar{\theta})$ directions, so those directions are not null directions of the reduced phase space. The reduced Berezin measure is the one associated with this fermionic symplectic form, up to the same convention-dependent signs and factors.

replaced by the shifted constraints

$$\widehat{C} = C + 2\mathfrak{q}\chi, \qquad \widehat{\bar{C}} = \bar{C} - \pi_\chi. \tag{3.47}$$

The coefficient of χ is fixed by the same number that appears in the original bracket $\{C, \bar{C}\}$. With this choice, the bracket of $\mathfrak{q}\chi$ with π_χ cancels the nonzero term in $\{C, \bar{C}\}$, and $\widehat{C}, \widehat{\bar{C}}$ become first class. Upon quantization, χ acts by multiplication and its conjugate acts by differentiation with respect to χ. This construction is equivalent to the Dirac-bracket reduction, but keeps the reduction in a form adapted to imposing constraints by a quotient.

We now apply the same construction to the supergroup particle. The role of $C, \bar{C}$ is played by the two fixed fermionic right charges $\mu^R_{\mathsf{F}_{+,p}}$ and $\mu^R_{\bar{\mathsf{F}}^p_+}$. Their bracket is nonzero because $\mu^R_{\mathsf{L}_+}$ has already been fixed to $-\mathfrak{q}$. For each p, we therefore introduce a new Grassmann variable χ^p and its conjugate momentum $\pi_{\chi,p}$, with $\{\chi^p, \pi_{\chi,q}\} = \delta^p{}_q$. The variable χ^p is therefore introduced only to supply the missing bracket: the terms $\mathfrak{q}\chi^p$ and $\pi_{\chi,p}$ cancel the obstruction in (3.41). This gives a first-class formulation of the fixed fermionic right charges, which can then be imposed before quantization in the same way as the bosonic right charge.

Constrained action and reduced wavefunctions. We now implement the first-class constraints in the worldline action. Let λ denote the even Lagrange multiplier for the L_+ condition, and let $\epsilon^p, \bar{\epsilon}_p$ denote the odd Lagrange multipliers for the two fermionic conditions. With the conventions used below, the shifted constraints are

$$\mu^R_{\mathsf{L}_+} + \mathfrak{q} = 0, \qquad \mu^R_{\mathsf{F}_{+,p}} + \mathfrak{q}\,\chi^p = 0, \qquad \mu^R_{\bar{\mathsf{F}}^p_+} - \pi_{\chi,p} = 0. \tag{3.48}$$

The auxiliary first-order term is

$$I_\chi = \int du\, \pi_{\chi,p}\dot{\chi}^p, \tag{3.49}$$

and the constrained first-order particle action is

$$I_{\mathrm{red}} = \phi_r \int du \left[\left\langle \mathcal{P}, \mathcal{G}^{-1}\dot{\mathcal{G}} \right\rangle - \frac{1}{2}\langle \mathcal{P}, \mathcal{P}\rangle - \lambda\left(\mu^R_{\mathsf{L}_+} + \mathfrak{q}\right) - \epsilon^p\left(\mu^R_{\mathsf{F}_{+,p}} + \mathfrak{q}\,\chi^p\right) - \bar{\epsilon}_p\left(\mu^R_{\bar{\mathsf{F}}^p_+} - \pi_{\chi,p}\right)\right] + I_\chi. \tag{3.50}$$

The constraints follow directly from this action.

It is useful to spell out the local form of these equations. We isolate the $\mathsf{F}_+, \bar{\mathsf{F}}_+, \mathsf{L}_+$ coordinates by writing their block as $e^{\Theta_+}e^{y\mathsf{L}_+}$. The right charges are represented by momenta conjugate to $y, \theta^p_+, \bar{\theta}_{+,p}$, and after using the bosonic equation $p_y = -\mathfrak{q}$ in the fermionic charges one obtains

$$\mu^R_{\mathsf{L}_+} = p_y, \qquad \mu^R_{\mathsf{F}_{+,p}} = \pi_{\theta^p_+} - \frac{\mathfrak{q}}{2}\bar{\theta}_{+,p}, \qquad \mu^R_{\bar{\mathsf{F}}^p_+} = \pi_{\bar{\theta}_{+,p}} - \frac{\mathfrak{q}}{2}\theta^p_+. \tag{3.51}$$

Substituting (3.51) into (3.48) gives

$$p_y = -\mathfrak{q}, \qquad \pi_{\theta^p_+} - \frac{\mathfrak{q}}{2}\bar{\theta}_{+,p} = -\mathfrak{q}\chi^p, \qquad \pi_{\bar{\theta}_{+,p}} - \frac{\mathfrak{q}}{2}\theta^p_+ = \pi_{\chi,p}. \tag{3.52}$$

The last two equations are the local first-class version of the second-class fermionic pair discussed above.[6] Thus, after introducing χ^p, one can safely eliminate θ^p_+ and $\bar{\theta}_{+,p}$; the constrained dependence on θ^p_+ and $\bar{\theta}_{+,p}$ is represented by the single variable χ^p. The first equation fixes the y-dependence, so the reduced wavefunction can be written as

$$\Psi = e^{-\mathfrak{q}y} F(x, \rho, U, \theta^p_-, \bar{\theta}_{-,p}, \chi^p), \qquad U \in SU(2)_{\mathfrak{R}}. \tag{3.53}$$

Reduced measure. We finally describe the reduced measure. At the path-integral level, the momentum integrations over $\mathcal{P}$ and $\pi_{\chi,p}$ impose the usual conjugate-momentum relations. After these variables are integrated out, the remaining path integral is over the group variable $\mathcal{G}$, the multipliers $\lambda, \epsilon^p, \bar{\epsilon}_p$, and the auxiliary fermions χ^p. Here λ imposes the L_+ constraint, while ϵ^p and $\bar{\epsilon}_p$ impose the two fermionic constraints in (3.48). Since the total constraints are first class, they generate local right shifts by $\mathbf{N}^{\text{susy}}_+ = \exp \mathfrak{n}^{\text{susy}}_+$. The path integral must therefore divide by the formal volume of this redundant gauge direction. Schematically,

$$Z_{\text{quot}} = \int \frac{D\mathcal{G}\, D\lambda\, D\epsilon\, D\bar{\epsilon}\, D\chi}{\text{Vol}(\mathbf{N}^{\text{susy}}_+)}\, e^{-I_{\text{red}}}. \tag{3.54}$$

This is the same gauge-fixing and measure argument reviewed in Section 2 for the bosonic particle, with the fermionic right shifts included. In current variables, the quotient is what permits the simple source representative. For the bosonic particle this is the choice $s(u) = 0$ in $\mathscr{J} = \mathcal{G}^{-1}\dot{\mathcal{G}} = \mathfrak{q}\mathsf{L}_- + s(u)\mathsf{L}_0 + \mathcal{L}(u)\mathsf{L}_+$; in the supersymmetric case it is the corresponding source condition (3.26).

In the Hamiltonian description, the same fixed-source condition is imposed by the right constraints (3.48). Those constraints give equivariance under right multiplication by the $\mathsf{L}_+, \mathsf{F}_+, \bar{\mathsf{F}}_+$ subgroup. The quotient just described is the corresponding statement for the physical inner product. The wavefunction

[6] Keeping a single value of p gives the $\mathcal{N} = 2$ truncation. In [40], the operators $G^R_+, \bar{G}^R_+$ are the analogues of our constrained right charges $\mu^R_{\mathsf{F}_+}, \mu^R_{\bar{\mathsf{F}}_+}$. The related covariant derivatives $D_+ = \partial_{\theta_+} + \bar{\theta}_+\partial_\gamma$ and $\bar{D}_+ = \partial_{\bar{\theta}_+} + \theta_+\partial_\gamma$ enter the remaining supergroup generators. Compared with $G^R_+, \bar{G}^R_+$, they have the opposite sign in the $\theta_+\partial_\gamma$ or $\bar{\theta}_+\partial_\gamma$ term, with the $e^{\pm ia}$ phases stripped off. On the sector with the $G^R_+, \bar{G}^R_+$ constraints and $L^R_+ = -\mathfrak{q}$, the construction of [40] replaces the constrained $(\theta_+, \bar{\theta}_+)$ dependence by one fermion, $D_+ \to 2e^{ia}\chi\,\partial_\gamma$ and $\bar{D}_+ \to e^{-ia}\partial_\chi$, which is the quantized version of our shifted constraints: $\mathfrak{q}\chi^p$ acts by multiplication and $\pi_{\chi,p}$ by ∂_{χ^p}. Here y, χ correspond to the γ, χ variables of [40] up to normalization, while our $(\theta^p_+, \bar{\theta}_{+,p})$ variables differ from the $(\theta_+, \bar{\theta}_+)$ variables of [40] by the $e^{\pm ia}$ rotation, so the coordinate equations look slightly different.

gluing measure is therefore read off from the same gauge-fixed path-integral measure.[7]

$$d\mu_{\rm phys} \sim d\mu_{\rm BH}^{\rm red}\, d\chi^1 d\chi^2 = e^{-\rho} d\rho\, dx\, d\mu_{SU(2)} d\theta_-^1 d\theta_-^2 d\bar{\theta}_{-,1} d\bar{\theta}_{-,2} d\chi^1 d\chi^2, \tag{3.55}$$

so inserting complete reduced states between two interval propagators uses this quotient measure, not the unreduced Berezin–Haar measure on the full supergroup. The auxiliary χ^p variables remain in the physical measure because they are the fermionic variables left by the first-class implementation of the constraints.

4 Worldline supercharges, supersymmetric propagator and Wilson-line operators

In this section we turn from the construction of the constrained particle to the objects that it computes: supercharges, propagators, and boundary-anchored Wilson-line operators. The central point is that the boundary theory does not realize the full bulk supergroup in a naive way. In the bulk, matter fields and Wilson lines transform under the full superisometry group, such as $PSU(1,1|2)$ for the $\mathcal{N}=4$ theory, while the boundary particle realizes the $\mathcal{N}=4$ super-Poincare symmetry preserved by the super-JT boundary conditions. Section 4.1 derives this relation: the boundary supercharges arise from bulk supergroup generators only after supplementing them by the compensating transformations required to stay in the fixed-source representative. This gives the particle-on-a-group derivation of the worldline supercharges used in [40,43]. Section 4.2 then constructs the corresponding supersymmetric propagator. We organize the two-endpoint dependence in terms of invariants of the diagonal bulk supergroup action, and then impose the boundary supercharge conditions; the resulting invariant functions are the building blocks that will later be glued into length kernels. Finally, Section 4.3 applies the same logic to matter insertions. A boundary-anchored two-point function is represented by a bulk Wilson line, but the boundary endpoints are not arbitrary bulk points: they are restricted to the same fixed-source embedding used in the Hamiltonian reduction. As a result, primary components give the familiar invariant bilocals, while supercharge descendants must be obtained using the compensated boundary supercharges rather than bare bulk generators. This distinction is essential for the moduli-operator bilocal used in the Berry-curvature discussion of Section 6.

4.1 The physical worldline supercharges

From boundary supertranslations to reduced worldline supercharges. We begin by identifying the charge to be constructed. The physical worldline supercharge is the Noether charge for

[7] For comparison, in the $\mathcal{N}=2$ truncation the convention of [40] uses $d\mu_{\rm phys}^{\mathcal{N}=2} = (2\pi\mathfrak{q})^{-1} d\rho\, dx\, d\theta_-\, d\bar{\theta}_-\, da\, d\chi$. With $\Psi_0^{\mathcal{N}=2}(L) = \frac{2\mathfrak{q}L}{\sqrt{\pi}} K_{1/2}(2\mathfrak{q}L)$ and $L = e^{-\ell/2}$, their $j=0$ norm is $\int d\ell\, 2(\Psi_0^{\mathcal{N}=2}(L))^2 = 1$, where the factor of 2 counts the two fermionic structures. We also reproduce this measure with our method.

a rigid supertranslation of the boundary superline with coordinates $z = (u, \vartheta^p, \bar{\vartheta}_p)$ and flat one-form $\Delta u = du + i\, d\vartheta^p\, \bar{\vartheta}_p + i\, d\bar{\vartheta}_p\, \vartheta^p$. A constant supertranslation acts schematically as

$$\delta_\epsilon \vartheta^p = \epsilon^p, \qquad \delta_\epsilon \bar{\vartheta}_p = 0, \qquad \delta_\epsilon u = -\epsilon^p \bar{\vartheta}_p, \tag{4.1}$$

with the barred transformation obtained by conjugation.

We next translate this statement into the image variables used by the particle action. The coordinate x is the reparametrized boundary time u. We restrict to the minus fermions and regard $\theta^p_-, \bar{\theta}_{-,p}$ as the reparametrized versions of $\vartheta^p, \bar{\vartheta}_p$, because the reduction above has eliminated the independent positive variables $\theta^p_+, \bar{\theta}_{+,p}$.[8] Thus the bare charge is supplied by the minus part of the boundary representative, $\mathcal{G}_- = e^{x\mathsf{L}_-} e^{\theta^p_- \mathsf{F}_{-,p} + \bar{\theta}_{-,p} \bar{\mathsf{F}}^p_-}$. The needed minus supertranslation algebra is

$$\{\mathsf{F}_{-,p}, \bar{\mathsf{F}}^q_-\} \sim \delta_p{}^q \mathsf{L}_-, \qquad [\mathsf{L}_-, \mathsf{F}_{-,p}] = [\mathsf{L}_-, \bar{\mathsf{F}}^p_-] = 0. \tag{4.2}$$

Acting from the left by $\epsilon^p \mathsf{F}_{-,p}$ then gives

$$\delta_\epsilon \theta^p_- = \epsilon^p, \qquad \delta_\epsilon \bar{\theta}_{-,p} = 0, \qquad \delta_\epsilon x = -\epsilon^p \bar{\theta}_{-,p}. \tag{4.3}$$

Equation (4.3) preserves $\Pi = \dot{x} + \bar{\theta}_{-,p} \dot{\theta}^p_- + \theta^p_- \dot{\bar{\theta}}_{-,p}$, the component form of Δu, and is therefore the target-space version of (4.1). Thus, before any positive data are fixed as in (3.26), the algebraic bare charges are $q_{\mathrm{bare},p} = \mathsf{F}_{-,p}$ and $\bar{q}^p_{\mathrm{bare}} = \bar{\mathsf{F}}^p_-$.

The q_{bare} charges have nontrivial commutators with the positive generators appearing in the constraints, and therefore cannot be the physical supercharges of the superline:

$$\{\mathsf{L}_+ - \mathfrak{q}, \mathsf{F}_{-,p}\} \sim \mathsf{F}_{+,p}, \qquad \{\bar{\mathsf{F}}^q_+, \mathsf{F}_{-,p}\} \sim \delta_p{}^q\, \mathsf{L}_0 + (\mathfrak{R}\text{-generators}). \tag{4.4}$$

The first bracket vanishes on the constrained surface, while the second one does not. Equivalently, the bare minus supertranslation moves the particle trajectory out of the chosen fixed-source slice in phase space. In the gravitational description, a bare supertranslation generated by $\mathsf{F}_{-,p}$ would break FG gauge.

To fix this, we perform the procedure of "fixing constraints with constraints", which is possible if the constraints satisfy non-trivial algebra. If ϕ_A denotes the positive equations $(\mathsf{L}_+ - \mathfrak{q}, \mathsf{F}_+, \bar{\mathsf{F}}_+)$, a physical worldline supercharge must obey the tangency condition

$$\{\phi_A, q_{\mathrm{phys}}\}\big|_{\phi=0} = 0. \tag{4.5}$$

[8] More precisely, the group coordinates are target-space variables, while $(u, \vartheta^p, \bar{\vartheta}_p)$ are worldline variables. Eliminating the independent positive variables is not the same as setting the target $\theta^p_+, \bar{\theta}_{+,p}$ to zero; already in (3.27), they are functions of the minus variables. Solving this embedding directly is expected to reproduce the supercharge expressions of [7]; here we instead improve the leading minus charge so that it preserves the positive-data constraints.

This condition requires that, after the constraints are imposed, the Hamiltonian flow generated by $q_{\rm phys}$ remains tangent to the constrained surface. We may therefore improve the bare charge by constrained generators,

$$q_{\rm phys} = q_{\rm bare} + \lambda^A \phi_A + \lambda^{AB}\phi_A\phi_B + \cdots, \qquad q_{\rm phys}\big|_{\phi=0} = q_{\rm bare}\big|_{\phi=0}, \tag{4.6}$$

so the value of the charge on the reduced surface is unchanged. Higher powers are allowed because the constrained generators need not commute. The compensator is fixed by imposing[9]

$$\{\phi_A, q_{\rm bare}\}\big|_{\phi=0} + \lambda^B \{\phi_A, \phi_B\}\big|_{\phi=0} = 0. \tag{4.7}$$

The nonzero bracket of the positive fermionic pair, discussed in (3.41), is precisely what makes this cancellation possible at fixed non-zero $\mathfrak{q}$.

The same construction has a direct group-theoretic interpretation. Let $\mathbf{Z}(z)$ denote the embedding of the boundary superline into super-AdS$_2$ superspace, using the same boldface superspace notation as in Section 3. The group representative is then $\mathcal{G}(\mathbf{Z}(z))$, namely the boundary representative evaluated at the target superspace point. The embedding $\mathbf{Z}(z)$ is not arbitrary, because $\mathcal{G}(\mathbf{Z})$ must obey the fixed-source constraints on $\mathcal{G}^{-1}d\mathcal{G}$. A physical supertranslation must map one allowed embedding to another, $\mathbf{Z}(z) \mapsto \mathbf{Z}(z) + \epsilon\,\delta\mathbf{Z}(z)$.[10] A naive left action by $\mathsf{F}_{-,p}$ fails to do this by itself, because it generally refactorizes as

$$e^{\epsilon \mathsf{F}_{-,p}}\mathcal{G}(\mathbf{Z}) = \mathcal{G}(\mathbf{Z} + \epsilon\,\delta\mathbf{Z})\, e^{\epsilon\alpha_+(\mathbf{Z})} + O(\epsilon^2), \qquad \alpha_+(\mathbf{Z}) \in \mathfrak{n}_+^{\rm susy}. \tag{4.8}$$

The extra positive factor is precisely the piece which moves the configuration away from the fixed-source representative. The compensator in $q_{\rm phys}$ absorbs $e^{\epsilon\alpha_+(\mathbf{Z})}$ and restores the fixed-source representative. The $\mathcal{N}=2$ and $\mathcal{N}=4$ examples below provide explicit checks of this construction.

The $\mathcal{N}=2$ case. In the $\mathcal{N}=2$ truncation the compensating construction takes a particularly simple form. The physical representative whose bare piece is F_- is

$$q_{\rm phys} = \mathsf{F}_- - \frac{\mathsf{L}_0 + \mathsf{R}}{\mathsf{L}_+}\mathsf{F}_+. \tag{4.9}$$

The second term is the positive compensator. It vanishes on the fixed positive surface, but its Hamiltonian flow cancels the component of the bare F_- flow which would leave the fixed-source slice. The barred charge is

$$\bar{q}_{\rm phys} = \bar{\mathsf{F}}_- + \frac{\mathsf{L}_0 - \mathsf{R}}{\mathsf{L}_+}\bar{\mathsf{F}}_+. \tag{4.10}$$

[9] For higher powers in (4.6), the full condition is $\{\phi_A, q_{\rm phys}\}$ lying in the ideal generated by the ϕ_A's. Expanding order by order in ϕ gives the higher-order Poisson-bracket conditions; the displayed equation is the leading one.

[10] Here the embedding is varied in a fixed super-AdS$_2$ background with fixed simple boundary metric, as in the standard Schwarzian derivation from a fluctuating cutoff curve; see [38].

In the one-complex-fermion case, no higher positive-fermion correction can be constructed.

For later use, we introduce numerator-cleared versions of these rational representatives. With the sign convention $Q = -q_{\rm phys}\mathsf{L}_+$ and $\bar{Q} = -\bar{q}_{\rm phys}\mathsf{L}_+$, one obtains

$$\begin{aligned} Q &= (\mathsf{L}_0 + \mathsf{R})\,\mathsf{F}_+ - \mathsf{F}_-\mathsf{L}_+, \\ \bar{Q} &= -(\mathsf{L}_0 - \mathsf{R})\,\bar{\mathsf{F}}_+ - \bar{\mathsf{F}}_-\mathsf{L}_+. \end{aligned} \tag{4.11}$$

These charges obey the fixed-parabolic normalization

$$Q^2 = \bar{Q}^2 = 0, \qquad \{Q, \bar{Q}\} = \mathcal{C}_{\mathcal{N}=2}\mathsf{L}_+, \tag{4.12}$$

where the ordered quadratic $\mathfrak{psu}(1,1|1)$ Casimir is $\mathcal{C}_{\mathcal{N}=2} = \mathsf{L}_0^2 + \mathsf{L}_-\mathsf{L}_+ - \mathsf{R}^2 + \mathsf{F}_-\bar{\mathsf{F}}_+ - \bar{\mathsf{F}}_-\mathsf{F}_+$.

The $\mathcal{N} = 4$ case. In the $\mathcal{N} = 4$ theory the same correction contains one additional feature. The positive fermion constraints form a doublet, which allows the nilpotent bilinear $\mathsf{F}_{+,1}\mathsf{F}_{+,2}$. We use $a = 1, 2$ for the $SU(2)_{\mathfrak{R}}$ doublet index and $\mathsf{R}_\pm = \mathsf{R}_1 \pm i\mathsf{R}_2$. For the charge whose bare piece is $\mathsf{F}_{-,1}$, the physical representative is

$$q^1_{\rm phys} = \mathsf{F}_{-,1} - \frac{\mathsf{L}_0 + \mathsf{R}_3}{\mathsf{L}_+}\mathsf{F}_{+,1} - \frac{\mathsf{R}_+}{\mathsf{L}_+}\mathsf{F}_{+,2} + \frac{1}{(\mathsf{L}_+)^2}\mathsf{F}_{+,1}\bar{\mathsf{F}}^2_+\mathsf{F}_{+,2}. \tag{4.13}$$

The first two compensating terms are the direct analogues of the $\mathcal{N} = 2$ correction. The cubic term is allowed because of the nilpotent positive bilinear. In the normalization of Appendix A.1, nilpotency and closure fix its coefficient to be 1.

The corresponding numerator-cleared charge is $Q^1 = -(\mathsf{L}_+)^2 q^1_{\rm phys}$, namely

$$Q^1 = (\mathsf{L}_0 + \mathsf{R}_3)\mathsf{F}_{+,1}\mathsf{L}_+ + \mathsf{R}_+\mathsf{F}_{+,2}\mathsf{L}_+ - \mathsf{F}_{-,1}(\mathsf{L}_+)^2 - \mathsf{F}_{+,1}\bar{\mathsf{F}}^2_+\mathsf{F}_{+,2}. \tag{4.14}$$

The remaining three numerator-cleared charges are obtained by the same compensation, together with conjugation and $SU(2)_{\mathfrak{R}}$:

$$\begin{aligned} Q^2 &= (\mathsf{L}_0 - \mathsf{R}_3)\mathsf{F}_{+,2}\mathsf{L}_+ + \mathsf{R}_-\mathsf{F}_{+,1}\mathsf{L}_+ - \mathsf{F}_{-,2}(\mathsf{L}_+)^2 - \mathsf{F}_{+,2}\bar{\mathsf{F}}^1_+\mathsf{F}_{+,1}, \\ \bar{Q}_1 &= -(\mathsf{L}_0 - \mathsf{R}_3)\bar{\mathsf{F}}^1_+\mathsf{L}_+ + \mathsf{R}_-\bar{\mathsf{F}}^2_+\mathsf{L}_+ - \bar{\mathsf{F}}^1_-(\mathsf{L}_+)^2 + \bar{\mathsf{F}}^1_+\mathsf{F}_{+,2}\bar{\mathsf{F}}^2_+, \\ \bar{Q}_2 &= -(\mathsf{L}_0 + \mathsf{R}_3)\bar{\mathsf{F}}^2_+\mathsf{L}_+ + \mathsf{R}_+\bar{\mathsf{F}}^1_+\mathsf{L}_+ - \bar{\mathsf{F}}^2_-(\mathsf{L}_+)^2 + \bar{\mathsf{F}}^2_+\mathsf{F}_{+,1}\bar{\mathsf{F}}^1_+. \end{aligned} \tag{4.15}$$

The terms proportional to $\mathsf{F}_{-,a}$ or $\bar{\mathsf{F}}^a_-$ are the original minus supertranslations. The remaining terms are precisely the positive compensators required to stay inside the fixed-source slice. The numerator-cleared $\mathcal{N} = 4$ charges satisfy

$$\{Q^a, \bar{Q}_b\} = \delta^a{}_b\mathcal{C}_{\mathcal{N}=4}(\mathsf{L}_+)^3, \qquad \{Q^a, Q^b\} = \{\bar{Q}_a, \bar{Q}_b\} = 0. \tag{4.16}$$

Here the quadratic $\mathfrak{psu}(1,1|2)$ Casimir is

$$\mathcal{C}_{\mathcal{N}=4} = \mathsf{L}_0^2 + \mathsf{L}_-\mathsf{L}_+ - \mathsf{L}_0 - \mathsf{R}_3^2 - \mathsf{R}_-\mathsf{R}_+ - \mathsf{R}_3 + \mathsf{F}_{-,a}\bar{\mathsf{F}}_+^a - \bar{\mathsf{F}}_-^a\mathsf{F}_{+,a}. \tag{4.17}$$

and this Casimir becomes the reduced Hamiltonian.

4.2 The supersymmetric propagator

We next construct the propagator for the particle on a group. The supercharges obtained above generate the boundary super-Poincare algebra, and the particle on a group must furnish a representation of this algebra. Consequently the propagator is fixed by charge conditions: it carries the appropriate $SU(2)_{\mathfrak{R}}$ quantum numbers and satisfies the differential equations generated by the supercharges.

The two-endpoint propagator is a function of the group variables at both endpoints. Following [40], we first use the symmetry under diagonal left multiplication to organize this dependence into invariant variables. Once these variables are chosen, the construction proceeds in three steps. We determine the action of the supercharges on the invariants, write the most general invariant ansatz, and impose the resulting charge equations.

Construction of invariants. We begin with the group coordinates before eliminating the positive fermions. An endpoint of the particle on a group carries $x, \rho, U, y, \theta_-^p, \bar{\theta}_{-,p}, \theta_+^p, \bar{\theta}_{+,p}$. These are the coordinates of the Gauss representative of the particle on a group. At this stage the $\theta_+^p, \bar{\theta}_{+,p}$ variables are still present; their elimination and the introduction of the χ^p polarization occur only after the fixed positive data are imposed. The invariant variables are defined as in [40]. Before any positive variables are eliminated, a two-endpoint function is an invariant if it is annihilated by the diagonal left action of the global supergroup.[11] Writing $\mathcal{D}^L_{\mathsf{T},i}$ and $\mathcal{D}^L_{\mathsf{T},f}$ for the left differential operator generated by $\mathsf{T} \in \mathfrak{psu}(1,1|2)$ at the initial and final endpoints, this condition is

$$\left(\mathcal{D}^L_{\mathsf{T},i} + \mathcal{D}^L_{\mathsf{T},f}\right) I = 0, \qquad \mathsf{T} \in \mathfrak{psu}(1,1|2). \tag{4.18}$$

Thus an invariant is unchanged when both endpoints are moved by the same global superisometry. Before eliminating the positive fermions, we introduce the auxiliary quantities

$$\widetilde{X}_{if} = x_f - x_i - \theta^a_{-,i}\bar{\theta}_{-,f,a} - \bar{\theta}_{-,i,a}\theta^a_{-,f}, \qquad \theta^a_{if,-} = \theta^a_{-,f} - \theta^a_{-,i}, \qquad \bar{\theta}_{if,-,a} = \bar{\theta}_{-,f,a} - \bar{\theta}_{-,i,a}.$$

[11] This follows from the definition of the endpoint states and propagator. The group-labeled states are defined by $|\mathcal{G}, i\rangle = U(\mathcal{G})|i\rangle$, so a physical left action sends $|\mathcal{G}, i\rangle$ to $|a\mathcal{G}, i\rangle$. Since the Hamiltonian is the Casimir, the propagator is invariant under simultaneous left multiplication, as in (B.49). Right multiplication is different: it gives equivariance, not invariance. See Section 2 and Appendix B.2, especially (B.47)–(B.50), for details.

The first terms in the invariant variables are

$$
\begin{aligned}
L &= \frac{e^{-\frac{1}{2}(\rho_i+\rho_f)}}{\widetilde{X}_{if}} + O(\theta^4),\\
W^a{}_b &= \left[U_i \begin{pmatrix} 1 - \dfrac{\varepsilon_{cd}\theta^c_{if,-}\bar\theta_{if,-,d}}{\widetilde{X}_{if}} & \dfrac{(\bar\theta_{if,-})^2}{\widetilde{X}_{if}} \\ \dfrac{(\theta_{if,-})^2}{\widetilde{X}_{if}} & 1 + \dfrac{\varepsilon_{cd}\theta^c_{if,-}\bar\theta_{if,-,d}}{\widetilde{X}_{if}} \end{pmatrix} U_f^{-1} \right]^a{}_b + O(\theta^4),\\
Y_i - Y_f &= y_i - y_f + \frac{e^{-\rho_i}+e^{-\rho_f}}{\widetilde{X}_{if}} + \left(\theta^a_{+,i}\bar f_{i,a} - f^a_i\bar\theta_{+,i,a}\right) - \left(\theta^a_{+,f}\bar f_{f,a} - f^a_f\bar\theta_{+,f,a}\right) + O(\theta^4).\\
f^a_\alpha &= \theta^a_{+,\alpha} + \frac{e^{-\rho_\alpha/2}(U_\alpha)^a{}_b\,\bar\theta^b_{if,-}}{\widetilde{X}_{if}} + O(\theta^3),\\
\bar f_{\alpha,a} &= \bar\theta_{+,\alpha,a} + \frac{e^{-\rho_\alpha/2}(U_\alpha^{-1})_a{}^b\,\theta^b_{if,-}}{\widetilde{X}_{if}} + O(\theta^3).
\end{aligned}
\tag{4.19}
$$

Here $\alpha = i, f$ and $\varepsilon_{12} = 1$. The complete all-fermion expressions are recorded in the Mathematica supplement.

We now impose the final positive constraints of Section 3.3. The fermionic constraints eliminate the independent $\theta^a_+, \bar\theta_{+,a}$ dependence and replace it by the χ^a polarization. This step also fixes the treatment of the positive-fermion terms in $Y_i - Y_f$ in (4.19). Using the local constraints (3.52), the $\theta_+, \bar\theta_+$ dependence is part of the positive fiber; after passing to the χ-polarization we choose the reduced representative $\theta^a_{+,\alpha} = \bar\theta_{+,\alpha,a} = 0$, so $(Y_i - Y_f)_{\rm red} = y_i - y_f + (e^{-\rho_i} + e^{-\rho_f})/\widetilde{X}_{if} + O(\theta^4)$. The fermionic data which originally sat in the positive variables are not discarded. They are carried by the η^a_α variables below and by the χ^a_α-derivative representation of the conjugate variables. In this polarization the reduced fermionic invariants are

$$
\eta^a_\alpha \equiv f^a_\alpha\big|_{\rm red} = \chi^a_\alpha + \frac{e^{-\rho_\alpha/2}(U_\alpha)^a{}_b\,\bar\theta^b_{if,-}}{\widetilde{X}_{if}} + O(\theta^3), \qquad \alpha = i, f. \tag{4.20}
$$

In this polarization, the conjugate $\bar f_{\alpha,a}$ variables are represented by derivatives with respect to χ^a_α. The bosonic right constraint fixes the Y-dependence, and therefore the two-endpoint wavefunction has the fixed-charge form

$$
P_{if} = e^{-\mathfrak{q}(Y_i - Y_f)}\Psi_{\rm WdW}(L, W, \eta_i, \eta_f). \tag{4.21}
$$

The reduced factor $\Psi_{\rm WdW}$ will be identified below with the Wheeler–DeWitt wavefunction.

Action of the supercharges. The differential operators used below are obtained directly from the supercharges in (4.14)–(4.15). No additional ansatz is required. Since a left action on an endpoint group element is represented in coordinates by a right-invariant vector field, each algebra generator T in the polynomial charge is replaced by the corresponding endpoint operator $\mathcal{D}^R_{\mathsf{T}}$. For example, (4.14)

becomes

$$Q^1 = \left(\mathcal{D}^R_{\mathsf{L}_0} + \mathcal{D}^R_{\mathsf{R}_3}\right)\mathcal{D}^R_{\mathsf{F}_{+,1}}\mathcal{D}^R_{\mathsf{L}_+} + \mathcal{D}^R_{\mathsf{R}_+}\mathcal{D}^R_{\mathsf{F}_{+,2}}\mathcal{D}^R_{\mathsf{L}_+} - \mathcal{D}^R_{\mathsf{F}_{-,1}}\left(\mathcal{D}^R_{\mathsf{L}_+}\right)^2 - \mathcal{D}^R_{\mathsf{F}_{+,1}}\mathcal{D}^R_{\bar{\mathsf{F}}^2_+}\mathcal{D}^R_{\mathsf{F}_{+,2}}. \tag{4.22}$$

This operator acts on a function of the original endpoint variables through the invariant variables L, Y, W, η_i, η_f. Equivalently, if $F_{\text{inv}} = F(L, Y, W, \eta_i, \eta_f)$, then

$$Q^a F_{\text{inv}} = Q^a F\big(L(x,\rho,U,\theta_-,\bar\theta_-,\chi),\, Y(x,\rho,U,\theta_-,\bar\theta_-,\chi),\, W(x,\rho,U,\theta_-,\bar\theta_-,\chi),\, \eta_i, \eta_f\big)\big|_{\text{inv}}\,, \tag{4.23}$$

where the last step means that the result is rewritten again in the invariant variables. This procedure gives the Q charge as a differential operator on the invariant variables. Let $\mathsf{R}_{f,3}, \mathsf{R}_{f,\pm}$ denote the $SU(2)_{\mathfrak{R}}$ differential operators on W generated from the final endpoint, and let $\mathsf{R}_{i,3}, \mathsf{R}_{i,\pm}$ denote the corresponding initial-endpoint operators. We also write the opposite-orientation Wilson line as $\bar W$, defined by $W^a{}_c \bar W^c{}_b = \bar W^a{}_c W^c{}_b = \delta^a{}_b$. The endpoint label on ∂_{Y_f} or ∂_{Y_i} records which endpoint positive coordinate is differentiated before the fixed-charge phase is imposed. For the final endpoint, we obtain

$$\begin{aligned}
Q^1_f &= \left[\frac{1}{2}(L\partial_L - 2\mathsf{R}_{f,3})\partial_{\eta^1_f} + \partial_{\eta^2_f}\mathsf{R}_{f,+} - L\left(W^1{}_1\eta^1_i + W^1{}_2\eta^2_i\right)\partial^2_{Y_f} + \frac{1}{2}\partial_{\eta^1_f}\partial_{\eta^2_f}\eta^2_f\right]\partial_{Y_f},\\
\bar Q_{f,1} &= \left[-\frac{1}{2}(L\partial_L + 2\mathsf{R}_{f,3})\eta^1_f + \eta^2_f\mathsf{R}_{f,-} - L\left(W^2{}_2\partial_{\eta^1_i} - W^2{}_1\partial_{\eta^2_i}\right) - \frac{1}{2}\eta^1_f\eta^2_f\partial_{\eta^2_f}\right]\partial^2_{Y_f},\\
Q^2_f &= \left[-\frac{1}{2}(L\partial_L + 2\mathsf{R}_{f,3})\partial_{\eta^2_f} - \partial_{\eta^1_f}\mathsf{R}_{f,-} - L\left(W^2{}_1\eta^1_i + W^2{}_2\eta^2_i\right)\partial^2_{Y_f} - \frac{1}{2}\partial_{\eta^2_f}\partial_{\eta^1_f}\eta^1_f\right]\partial_{Y_f},\\
\bar Q_{f,2} &= \left[\frac{1}{2}(L\partial_L - 2\mathsf{R}_{f,3})\eta^2_f - \eta^1_f\mathsf{R}_{f,+} - L\left(W^1{}_1\partial_{\eta^2_i} - W^1{}_2\partial_{\eta^1_i}\right) + \frac{1}{2}\eta^2_f\eta^1_f\partial_{\eta^1_f}\right]\partial^2_{Y_f}.
\end{aligned} \tag{4.24}$$

Repeating the same calculation at the initial endpoint, with $Y_i - Y_f$ and W kept fixed, gives

$$\begin{aligned}
Q^1_i &= \left[\frac{1}{2}(L\partial_L - 2\mathsf{R}_{i,3})\partial_{\eta^1_i} + \partial_{\eta^2_i}\mathsf{R}_{i,+} - L\left(\bar W^1{}_1\eta^1_f + \bar W^1{}_2\eta^2_f\right)\partial^2_{Y_i} + \frac{1}{2}\partial_{\eta^1_i}\partial_{\eta^2_i}\eta^2_i\right]\partial_{Y_i},\\
\bar Q_{i,1} &= \left[-\frac{1}{2}(L\partial_L + 2\mathsf{R}_{i,3})\eta^1_i + \eta^2_i\mathsf{R}_{i,-} - L\left(\bar W^2{}_2\partial_{\eta^1_f} - \bar W^2{}_1\partial_{\eta^2_f}\right) - \frac{1}{2}\eta^1_i\eta^2_i\partial_{\eta^2_i}\right]\partial^2_{Y_i},\\
Q^2_i &= \left[-\frac{1}{2}(L\partial_L + 2\mathsf{R}_{i,3})\partial_{\eta^2_i} - \partial_{\eta^1_i}\mathsf{R}_{i,-} - L\left(\bar W^2{}_1\eta^1_f + \bar W^2{}_2\eta^2_f\right)\partial^2_{Y_i} - \frac{1}{2}\partial_{\eta^2_i}\partial_{\eta^1_i}\eta^1_i\right]\partial_{Y_i},\\
\bar Q_{i,2} &= \left[\frac{1}{2}(L\partial_L - 2\mathsf{R}_{i,3})\eta^2_i - \eta^1_i\mathsf{R}_{i,+} - L\left(\bar W^1{}_1\partial_{\eta^2_f} - \bar W^1{}_2\partial_{\eta^1_f}\right) + \frac{1}{2}\eta^2_i\eta^1_i\partial_{\eta^1_i}\right]\partial^2_{Y_i}.
\end{aligned} \tag{4.25}$$

On the fixed-charge sector (4.21), these expressions reproduce the Liouville supercharges of [25] after replacing the endpoint derivatives by $\partial_{Y_i} \to -\mathfrak{q}$ and $\partial_{Y_f} \to +\mathfrak{q}$.

BPS propagator. The BPS propagator is annihilated by the supercharges at both endpoints and obeys the $SU(2)_{\mathfrak{R}}$-singlet condition at each endpoint,

$$Q_i^a P_{if} = \bar{Q}_{i,a} P_{if} = 0, \qquad Q_f^a P_{if} = \bar{Q}_{f,a} P_{if} = 0, \qquad \mathcal{J}_i P_{if} = \mathcal{J}_f P_{if} = 0, \tag{4.26}$$

where $\mathcal{J}_i$ and $\mathcal{J}_f$ denote the $SU(2)_{\mathfrak{R}}$-singlet conditions at the two endpoints.

In the reduced η-variable convention, the two $SU(2)_{\mathfrak{R}}$-singlet fermion structures are

$$\eta_f^a W_a{}^b \eta_{i,b}, \qquad \eta_i^1 \eta_i^2 + \eta_f^1 \eta_f^2.$$

Therefore the invariant ansatz is

$$P^{\mathrm{BPS}} = e^{-\mathfrak{q}(Y_i - Y_f)} \left[A(L)\, \eta_f^a W_a{}^b \eta_{i,b} + B(L) \left(\eta_i^1 \eta_i^2 + \eta_f^1 \eta_f^2 \right) \right]. \tag{4.27}$$

The first structure contracts the two endpoint fermions through the $SU(2)_{\mathfrak{R}}$ Wilson line, while the second is the endpoint singlet in the χ-polarization. Acting with the reduced supercharges on (4.27) gives two coupled first-order radial equations for $A(L)$ and $B(L)$. In the convention used here they can be written as

$$(L\partial_L - 1)\, A(L) + 2\mathfrak{q} L\, B(L) = 0, \qquad L\partial_L B(L) + 2\mathfrak{q} L\, A(L) = 0. \tag{4.28}$$

The solution is

$$A(L) = L K_0(2\mathfrak{q}L), \qquad B(L) = L K_1(2\mathfrak{q}L). \tag{4.29}$$

The resulting BPS interval propagator is

$$P^{\mathrm{BPS}} = e^{-\mathfrak{q}(Y_i - Y_f)} \left[L K_0(2\mathfrak{q}L)\, \eta_f^a W_a{}^b \eta_{i,b} + L K_1(2\mathfrak{q}L) \left(\eta_i^1 \eta_i^2 + \eta_f^1 \eta_f^2 \right) \right]. \tag{4.30}$$

Non-BPS propagator. Away from the BPS sector, the supermultiplet contains several components, and the propagator becomes correspondingly matrix-valued. For fixed energy E, equivalently fixed Casimir eigenvalue, and spin j, we use the non-BPS multiplet. For a more detailed discussion of these multiplets and their wavefunctions, see [7, 25].

$$\mathcal{M}_{E,j} = |\mathrm{H}; E, j\rangle \oplus |\Psi; E, j - \tfrac{1}{2}\rangle \oplus |\chi; E, j - \tfrac{1}{2}\rangle \oplus |\mathrm{L}; E, j - 1\rangle. \tag{4.31}$$

Here $\mathrm{H}, \Psi, \chi, \mathrm{L}$ are component labels, and L should not be confused with the invariant L. Schematically, the multiplet has the diamond structure

$$\begin{array}{ccc} & |\mathrm{H}; E, j\rangle & \\ \swarrow Q & & Q \searrow \\ |\Psi; E, j-\tfrac{1}{2}\rangle & & |\chi; E, j-\tfrac{1}{2}\rangle \\ Q \searrow & & \swarrow Q \\ & |\mathrm{L}; E, j-1\rangle & \end{array} \tag{4.32}$$

where each arrow denotes an appropriate component of the supercharge. The bottom state is the spin-$j-1$ component obtained after two supercharge actions.[12] For the non-BPS multiplet with highest spin $j = \frac{1}{2}$, the L component is absent. The continuous part of the spectrum is labelled by $s > 0$. In the $\mathcal{N} = 4$ convention of [25], a non-BPS multiplet of highest spin j has $E = E_{\mathrm{gap}}(j) + s^2$, with $E_{\mathrm{gap}}(j) = j^2$ up to the overall Schwarzian energy normalization. It is useful to package a boundary-segment state as

$$\mathbf{s} = (s, j, \Phi), \tag{4.33}$$

where s is the continuous energy label, j is the highest spin of the worldline supermultiplet, and Φ labels the component inside the multiplet, such as $\mathrm{H}, \Psi, \chi, \mathrm{L}$.

Let us display the H-component for the highest-spin $j = \frac{1}{2}$ multiplet, which is produced most directly by the invariant calculation. In the endpoint ordering used here, introduce

$$\eta_i \cdot W \cdot \eta_f \equiv \varepsilon_{ab} \eta_i^a W^b{}_c \eta_f^c, \qquad \mathcal{U}_1 = W^1{}_1 \, \eta_i \cdot W \cdot \eta_f, \qquad \mathcal{U}_2 = W^1{}_1 \left(\eta_i^1 \eta_i^2 + \eta_f^1 \eta_f^2\right), \qquad \mathcal{U}_3 = \eta_i^1 \eta_f^2. \tag{4.34}$$

The endpoint-orientation sign is fixed to be $+1$ throughout this paragraph. Before imposing the supercharge equations, the invariant ansatz is

$$P_{HH}^{1/2} = e^{-\mathfrak{q}(Y_i - Y_f)} \left[\mathfrak{q} A(L) \mathcal{U}_1 + B(L) \mathcal{U}_2 + C(L) \mathcal{U}_3\right]. \tag{4.35}$$

The factor multiplying $A(L)$ is part of the normalization convention for this component. The functions A, B, C are arbitrary at this stage. The first-order charge equations, for example the pair generated by Q_f^1 and $\bar{Q}_{i,1}$, fix B and C in terms of A:

$$B(L) = \frac{A(L) - \frac{1}{2} L A'(L)}{L}, \qquad C(L) = \frac{2\left(\mathfrak{q}^2 L^2 - 2\right) A(L) + L\left(3A'(L) - L A''(L)\right)}{4\mathfrak{q} L^2}. \tag{4.36}$$

[12] The naive second supercharge action also contains pieces in the higher-spin channel, so obtaining the L state includes the usual orthogonalization.

On the fixed-charge sector, the endpoint supercharge algebra acting on the propagator is

$$\left\{Q^a_\alpha, \bar{Q}_{\alpha,b}\right\} P = \delta^a{}_b\, \mathfrak{q}^3\, \widehat{H}\, P, \qquad \alpha = i, f. \tag{4.37}$$

Using this equation, the Hamiltonian equation on this component becomes the radial equation

$$\left[(L\partial_L - 1)^2 - (2\mathfrak{q}L)^2 + 4s^2\right] A(L) = 0. \tag{4.38}$$

The normalizable solution is

$$A_s(L) = LK_{2is}(2\mathfrak{q}L). \tag{4.39}$$

Therefore the H-component propagator is

$$P^{1/2}_{HH} = e^{-\mathfrak{q}(Y_i - Y_f)} \left[\mathfrak{q}A_s(L)\mathcal{U}_1 + \frac{A_s(L) - \frac{1}{2}LA'_s(L)}{L}\mathcal{U}_2 + \frac{2\left(\mathfrak{q}^2L^2 - 2\right)A_s(L) + L\left(3A'_s(L) - LA''_s(L)\right)}{4\mathfrak{q}L^2}\mathcal{U}_3\right]. \tag{4.40}$$

In this multiplet, Ψ and χ are the two scalar components. Their diagonal propagators are obtained by applying endpoint supercharges to $P^{1/2}_{HH}$:

$$P^{1/2}_{\Psi\Psi} \propto Q^2_i \bar{Q}_{f,2} P^{1/2}_{HH}, \qquad P^{1/2}_{\chi\chi} \propto \bar{Q}_{i,1} Q^1_f P^{1/2}_{HH}, \tag{4.41}$$

with the proportionality constants fixed by the normalization of the component states.

Measure and relation to Liouville quantum mechanics. The gluing measure is the quotient Berezin–Haar measure from the Hamiltonian reduction. In the coordinate convention of Section 3.3, the $\mathcal{N} = 4$ measure contains

$$d\mu_{\text{phys}} = e^{-\rho} d\rho\, dx\, d\mu_{SU(2)} d\theta^1_- d\theta^2_- d\bar{\theta}_{-,1} d\bar{\theta}_{-,2} d\chi^1 d\chi^2, \tag{4.42}$$

up to an overall normalization constant. What needs to be noticed in the $\mathcal{N} = 4$ discussion is the factor $e^{-\rho}$: this factor plays an important role when translating the group propagator to Liouville quantum mechanics. There one inserts $P(1,2)P(2,1)$, divides by the simultaneous global G action on the two endpoints, and uses the symmetry to fix one representative of the two-point configuration. In the present $\mathcal{N} = 4$ theory $G = \mathrm{PSU}(1,1|2)$, while in the bosonic detour it is $SL(2,\mathbb{R})$. The invariant variables then reduce to the length, the Euler angle, and the two remaining χ variables, and the final answer is written with the flat $d\ell$ Liouville measure. In our variables, after the same kind of gauge fixing, only one radial integration remains. Suppressing the $SU(2)_{\mathfrak{R}}$ and fermionic contractions, the radial part has the schematic form

$$\int \frac{d\mu_{\text{phys}}(A) d\mu_{\text{phys}}(B)}{\mathrm{Vol}(G)} P_{AB} P_{BA} \longrightarrow \int d\rho_B\, e^{-\rho_B}\, \mathcal{R}(\rho_B)^2. \tag{4.43}$$

The arrow means that a representative of the diagonal orbit has been chosen. The same quotient-measure mechanism is the one used below in the endpoint formulas for three- and four-point functions. In the same gauge the invariant length is $L = e^{-\ell/2} \propto e^{-\rho_B/2}$. Thus

$$\int d\rho_B\, e^{-\rho_B}\, \mathcal{R}(\rho_B)^2 = \int d\ell\, \left[e^{-\rho_B/2}\mathcal{R}(\rho_B)\right]^2 \sim \int d\ell\, \left[L\,\mathcal{R}(L)\right]^2 . \tag{4.44}$$

This is the useful way to remember the relation: the leftover measure factor is shared by the two propagators in the norm. Each Liouville wavefunction therefore receives one extra factor $e^{-\rho_B/2} = L$ relative to the corresponding group propagator component. For example, the K_0 component of the Liouville BPS wavefunction is $L^2 K_0(2\mathfrak{q}L)$, with $L = e^{-\ell/2}$, rather than the $LK_0(2\mathfrak{q}L)$ appearing in the BPS propagator above. With this measure dictionary, any diagonal propagator in the same supermultiplet can be obtained from the corresponding Liouville quantum mechanics wavefunction in [25] by the replacements

$$e^{-\ell/2} \longrightarrow L, \qquad g^a{}_b \longrightarrow W^a{}_b, \qquad (\psi_l, \bar{\psi}_l; \psi_r, \bar{\psi}_r) \longrightarrow (\eta_i, \partial_{\eta_i}; \partial_{\eta_f}, \eta_f), \tag{4.45}$$

with the fixed charge restored in the Bessel function argument as $2e^{-\ell/2} \to 2\mathfrak{q}L$, and with the same fermion polarization as in (4.24)–(4.25). The composition of propagators will be discussed more structurally in Section 5.

4.3 Bulk representations and Wilson-line operators

The final ingredient needed for correlators is the operator inserted between two endpoints of the particle on a group. We will not repeat the bosonic derivation in Section 2 here. The point of this subsection is instead to isolate the new feature in the supersymmetric case. A matter field carries a bulk $PSU(1,1|2)$ representation, while the boundary dynamics is described by a reduced $\mathcal{N} = 4$ super-Poincare representation. The Wilson line is the bridge between these two languages, but this difference leads to a few subtleties that we will try to resolve in this subsection.

Symmetries and representations. There are two representation spaces which should be kept separate. The first one is the representation of $PSU(1,1|2)$ in which the bulk matter operator transforms. We denote it by $\mathbf{R}$, and write its vectors as $\mathbf{v} \in \mathbf{R}$; we sometimes also call these vectors polarizations. Concretely, the matter operator can be packaged as a superfield in this representation, $\mathcal{O}_{\mathbf{R}}(X, \theta_\pm, \bar{\theta}_\pm)$. Expanding it as a polynomial in the fermionic variables defines the component fields, which are ordinary fields of X. The odd generators $\mathsf{F}_{\pm,a}, \bar{\mathsf{F}}^a_\pm$ act on the same representation space and move one component field into another. These component fields should still be distinguished from the vectors $\mathbf{v}$. Each component field contains a tower of modes transforming under the bosonic subalgebra $SL(2,\mathbb{R}) \times SU(2)_{\mathfrak{R}}$. Choosing a component and a mode in this tower is what gives a vector $\mathbf{v} \in \mathbf{R}$. All such modes of all

component fields together form the $PSU(1,1|2)$ representation $\mathbf{R}$.[13]

The second representation is the one carried by the particle on a group, whose propagator was discussed above. This Hilbert space carries a $\mathcal{N}=4$ super-Poincare representation. We write a state label schematically as $\mathbf{s}=(s,j,\Phi)$. Although this representation is closely related to $PSU(1,1|2)$, the particle on a group must also obey the fixed positive-charge constraints, so not every group-element configuration is allowed. Therefore $\mathbf{s}$ is acted on by the physical worldline supercharges of Section 4.1, rather than by the bare $\mathsf{F}_{\pm,a}, \bar{\mathsf{F}}^a_{\pm}$ generators.

Component bilocals: primaries, compensators, and descendants. We now spell out what the preceding covariance statement means for actual component two-point functions. In fixed Euclidean AdS_2, before the boundary mode is made dynamical, a component matter two-point function is fixed by the isometries. If the component carries $SL(2)$ weight h and transforms in an $SU(2)_{\mathfrak{R}}$ representation of spin j, its boundary limit has the schematic form

$$G^{(h,j)}{}_m{}^n(i,j) = C_{h,j}\, L_{ij}^{2h} \left[D^{(j)}(U_{ij})\right]_m{}^n, \tag{4.46}$$

where L_{ij} is the invariant boundary length, $U_{ij}\in SU(2)_{\mathfrak{R}}$ is the R-symmetry Wilson-line group element between the two endpoints, and $D^{(j)}(U_{ij})$ is its matrix in the spin-j representation. For $j=0$ this reduces to a single power of L_{ij}. For the fundamental representation, $j=\frac{1}{2}$, it is proportional to $L_{ij}(U_{ij})^a{}_b$. The corresponding full matter Wilson-line matrix element is

$$W^{\mathbf{R}}_{ij}(\mathbf{v}_i,\mathbf{v}_j) = \langle \mathbf{v}_i|\mathbf{R}(\mathcal{G}_i^{-1}\mathcal{G}_j)|\mathbf{v}_j\rangle. \tag{4.47}$$

Here $\mathcal{G}_i$ and $\mathcal{G}_j$ are the reduced endpoint group elements, and $\mathbf{v}_i,\mathbf{v}_j\in\mathbf{R}$ are matter polarizations. They select the component field and the mode of that component at the two endpoints. After the Hamiltonian reduction this fixed-background statement is not automatic, so let us first focus on the simple cases where it survives in the reduced variables. The simple one-term form in (4.46) is guaranteed only for polarizations on which the positive generators used in the Hamiltonian reduction act trivially. More explicitly, suppose the endpoint polarization is annihilated by the positive subgroup,

$$\mathsf{L}_+\mathbf{v}=0, \qquad \mathsf{F}_{+,p}\mathbf{v}=0, \qquad \bar{\mathsf{F}}^p_+\mathbf{v}=0, \tag{4.48}$$

with the corresponding dual condition at the bra endpoint. Then the compensating factor $h_A(X;\epsilon)\in \mathbf{N}^{\mathrm{susy}}_+$ in

$$e^{\epsilon A}\mathcal{G}(X) = \mathcal{G}(X+\epsilon\delta_A X)\, h_A(X;\epsilon) + O(\epsilon^2) \tag{4.49}$$

[13] We also comment on terminology related to the fermionic generators. When we refer to a superconformal primary, we mean the vector killed by the positive generators, in particular by $\mathsf{F}_{+,a}, \bar{\mathsf{F}}^a_+$. This is a mode of the leading component field. Other component fields in the same multiplet often contain L_+-primary modes, but they need not contain an F_+-primary mode.

acts trivially on the endpoint polarizations. In that case the component bilocal is obtained by simply evaluating the Wilson line on the reduced invariants. For the bottom $SU(2)_{\mathfrak{R}}$-singlet matter component, this is the direct supersymmetric analogue of (2.17):

$$\begin{aligned}\mathcal{O}_{\Delta}^{\text{scalar}}(i,j) &= \left[\frac{\Pi_i\Pi_j}{\widetilde{X}_{ij}^2}\right]^{\Delta} = L_{ij}^{2\Delta},\\ \widetilde{X}_{ij} &= x_j - x_i - \theta^a_{-,i}\bar{\theta}_{-,j,a} - \bar{\theta}_{-,i,a}\theta^a_{-,j}, \qquad L_{ij} = \frac{e^{-\frac{1}{2}(\rho_i+\rho_j)}}{\widetilde{X}_{ij}}.\end{aligned} \tag{4.50}$$

Here the source condition has been used to replace the endpoint super-Jacobians by the reduced length L_{ij}. The scalar insertion used in Section 5 is this simplest case.

This simplification should not be extended to arbitrary components of a matter multiplet. A component obtained from another one by a fermionic generator is still computed from the same Wilson line, but the generator has to be represented on the reduced variables. The point is that the bare action of $\mathsf{F}_{-,p}$, or of its conjugate, does not preserve the fixed-source representative. It must be completed by the same positive compensator which appeared in the construction of the physical worldline supercharges. Thus, at the level of reduced bilocals, one should not write "act with the bare F_- on the old answer." The descendant bilocal is therefore the result of the same compensated worldline action used in Section 4.1.

A simple short-multiplet example makes the distinction explicit. Let the reduced throat multiplet contain a fermionic component whose primary two-point function is

$$\mathcal{O}_{\psi}^{\text{modulia}}{}_{b}(i,f) = L_{if}\, W^a{}_b(i,f). \tag{4.51}$$

This is the one-term answer for a primary polarization with $h=\frac{1}{2}$ and $j=\frac{1}{2}$. The bosonic supercharge descendant in the same short multiplet is obtained by applying the corresponding reduced supercharges to the Wilson line. In the normalization where the purely bosonic term has coefficient one, the useful model operator is

$$\mathcal{O}_{\text{marg}}^{\text{moduli}}(i,f) = L_{if}^2 + i\, L_{if}\, \eta_f^a W_a{}^b(i,f)\eta_{i,b}. \tag{4.52}$$

Equation (4.52) is not a new primary formula. It is the descendant of (4.51). The second term is the finite effect of the compensating positive transformation in the physical worldline supercharge.

Interestingly, this also matches the Liouville quantum mechanics derivation. In the $\mathcal{N}=4$ Liouville variables of [25], the spin-$\frac{1}{2}$ Wilson-line component is

$$\mathcal{O}_{1/2} = e^{-\frac{1}{2}\ell}\, D^{1/2}_{\frac{1}{2},\frac{1}{2}}(g). \tag{4.53}$$

Acting with the supercharges which give the bosonic descendant gives

$$-i\{\bar{Q}_{l,1},[Q_r^2,\mathcal{O}_{1/2}]\} = e^{-\ell} + ie^{-\frac{1}{2}\ell}\,\psi_l^a D_a^{1/2b}(g)\psi_{r,b}. \tag{4.54}$$

This is the Liouville version of (4.52).[14] This example will be useful in the Berry-phase discussion of Section 6.

5 Correlators from the supersymmetric length kernel

We now turn the particle-on-a-group formalism into an algorithm for computing correlators. The basic strategy is to use the propagators and Wilson-line operators constructed in Section 4, glue them using the reduced particle measure reviewed in Section 2, and rewrite the result in terms of invariant length variables. In this language the fundamental object is the three-length composition kernel I_3. Section 5.1 explains how this kernel arises from gluing two propagators at a common endpoint. The bosonic kernel is the familiar JT I_3 kernel, while its $\mathcal{N}=2$ and $\mathcal{N}=4$ analogues are obtained by applying the same gluing procedure to the supersymmetric propagators. Once this kernel is known, scalar three-point functions follows almost directly from length integrals: we compute the zero-energy $\mathcal{N}=2$ scalar three-point function in closed Gamma-function form, and the $\mathcal{N}=4$ BPS scalar three-point function as a finite sum of balanced ${}_4F_3(1)$'s.

Section 5.2 applies the same length-space algorithm to four-point functions. The time-ordered channel is obtained by iterating the three-point kernel, while the crossed channel gives the out-of-time-ordered correlator. We derive the zero-energy scalar OTOC for the crossed bilocal pairing $\mathcal{O}_\Delta(1,3)\mathcal{O}_\Delta(2,4)$ and then study its large-Δ approximation. In the $\mathcal{N}=2$ case the resulting length integral can be evaluated explicitly as a finite hypergeometric sum, while in the $\mathcal{N}=4$ BPS case the same method gives a controlled saddle approximation. The integral manipulations used in this section are collected in Appendix C, and Appendix D shows that the identical length-integral method reproduces the classic $\mathcal{N}=0$ JT OTOC.

5.1 The composition kernel and scalar three-point functions

The composition kernel. It is useful to start from the bosonic $\mathcal{N}=0$ problem. Let $P_{ij}^{\mathcal{N}=0}$ be the propagator for an interval with endpoints i,j. In the fixed-charge sector it has the form $P_{ij}^{\mathcal{N}=0} = e^{-qY_{ij}}\Psi_{\rm WdW}^{\mathcal{N}=0}(L_{ij})$. The composition law is

$$\int d\mu_{\rm phys}(j)\, P_{ij}^{\mathcal{N}=0} P_{jk}^{\mathcal{N}=0} = P_{ik}^{\mathcal{N}=0}. \tag{5.1}$$

[14] The $\mathcal{N}=2$ submultiplet gives the same one-line check: for $\mathcal{O}_{1/2} = e^{-\frac{1}{2}\ell - ia}$, the reduced supercharges give $-i\{\bar{Q}_l,[Q_r,\mathcal{O}_{1/2}]\} = e^{-\ell} + ie^{-\frac{1}{2}\ell - ia}\bar{\psi}_l\psi_r$.

To extract the length kernel, we quotient the common global symmetry and use L_{ij}, L_{jk} as the two integration variables, with L_{ik} fixed by the external endpoints. In these variables the product of phases can be written as

$$e^{-qY_{ij}}e^{-qY_{jk}} = e^{-qY_{ik}}\,\exp\left[-q\left(\frac{L_{ij}L_{jk}}{L_{ik}}+\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ik}L_{ij}}{L_{jk}}\right)\right], \tag{5.2}$$

The left-hand side is the phase part of the two propagators being composed. On the right-hand side, the endpoint phase $e^{-qY_{ik}}$ combines with $\Psi^{\mathcal{N}=0}_{\mathrm{WdW}}(L_{ik}) = K_{2is}(2qL_{ik})$ to give the bosonic JT propagator $P^{\mathcal{N}=0}_{ik}$. The remaining exponential on the right-hand side is the phase contribution that stays inside the length integral; together with the reduced length measure, it defines the three-length composition kernel. In other words, after the quotient the measure and phase part takes the form[15]

$$d\mu_{\mathrm{phys}}(j)\,e^{-qY_{ij}}e^{-qY_{jk}} = \frac{dL_{ij}}{L_{ij}}\frac{dL_{jk}}{L_{jk}}\,e^{-qY_{ik}}\,I_3^{\mathcal{N}=0}(L_{ij},L_{jk},L_{ik}), \tag{5.3}$$

The bosonic kernel is

$$I_3^{\mathcal{N}=0}(L_{ij},L_{jk},L_{ik}) = \frac{1}{\pi}\,\exp\left[-q\left(\frac{L_{ij}L_{jk}}{L_{ik}}+\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ik}L_{ij}}{L_{jk}}\right)\right]. \tag{5.4}$$

This is the bosonic Schwarzian I_3 kernel [36, 37].

The supersymmetric calculation follows the same logic, but the factor multiplying the WdW wavefunction contains fermions and R-symmetry data. The simpler $\mathcal{N}=2$ zero-energy calculation makes this point explicit. Translated to the present length convention, the $j=0$ propagator of [40] is

$$P^{\mathcal{N}=2}_{0,ij} = e^{-\mathfrak{q}Y_{ij}}\left(\eta_j e^{-\frac{i}{2}\Sigma_{ij}}+\eta_i e^{\frac{i}{2}\Sigma_{ij}}\right)\Psi_0^{\mathcal{N}=2}(L_{ij}), \tag{5.5}$$

where the $j=0$ normalization is fixed by the norm convention described above, and $\Psi_0^{\mathcal{N}=2}(L) = \frac{2\mathfrak{q}L}{\sqrt{\pi}}K_{1/2}(2\mathfrak{q}L) = \sqrt{\mathfrak{q}L}\,e^{-2\mathfrak{q}L}$. The analogue of (5.3) is obtained by stripping off the two radial factors $\Psi_0^{\mathcal{N}=2}$ and doing the $U(1)_{\mathfrak{R}}$ and Grassmann part of the intermediate-endpoint integral:

$$\begin{aligned}\int_{U(1)_{\mathfrak{R}},\mathrm{Gr.}} & d\mu^{\mathcal{N}=2}_{\mathrm{phys}}(j)\,e^{-\mathfrak{q}Y_{ij}}\left(\eta_j e^{-\frac{i}{2}\Sigma_{ij}}+\eta_i e^{\frac{i}{2}\Sigma_{ij}}\right)e^{-\mathfrak{q}Y_{jk}}\left(\eta_k e^{-\frac{i}{2}\Sigma_{jk}}+\eta_j e^{\frac{i}{2}\Sigma_{jk}}\right)\\ &= \frac{dL_{ij}}{L_{ij}}\frac{dL_{jk}}{L_{jk}}\,e^{-\mathfrak{q}Y_{ik}}\left(\eta_k e^{-\frac{i}{2}\Sigma_{ik}}+\eta_i e^{\frac{i}{2}\Sigma_{ik}}\right)I_3^{\mathcal{N}=2}(L_{ij},L_{jk},L_{ik}).\end{aligned} \tag{5.6}$$

[15] Equation (5.3) should be read as the result of the two-fold integral over the intermediate endpoint. Passing to the two length variables L_{ij}, L_{jk} also brings in the Jacobian of this change of variables, which is included in the displayed I_3 normalization.

This defines the corresponding three-length kernel. Its value is

$$I_3^{\mathcal{N}=2}(L_{ij},L_{jk},L_{ik}) = \exp\left[-\mathfrak{q}\left(\frac{L_{ij}L_{jk}}{L_{ik}}+\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ik}L_{ij}}{L_{jk}}\right)\right] \times\left[-1+\frac{2\mathfrak{q}}{L_{ij}L_{jk}L_{ik}}\left(L_{ij}L_{jk}+L_{jk}L_{ik}+L_{ik}L_{ij}\right)^2\right]. \tag{5.7}$$

This uses the same length convention as the $\mathcal{N}=4$ formula below: the Bessel function argument is $2\mathfrak{q}L$.

For $\mathcal{N}=4$, the kernel contains two component matrices,

$$\begin{aligned}
\left(\mathbb{I}_{ijk}^{(0)}\right)_{00} &= -\frac{\mathfrak{q}^2}{4}\left(\frac{L_{ij}L_{jk}}{L_{ik}}+\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ik}L_{ij}}{L_{jk}}\right)^2+\frac{5\mathfrak{q}}{4}\left(\frac{L_{ij}L_{jk}}{L_{ik}}+\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ik}L_{ij}}{L_{jk}}\right)-1 \\
&\quad -\mathfrak{q}^2\left(L_{ij}^2+L_{jk}^2+L_{ik}^2\right), \\
\left(\mathbb{I}_{ijk}^{(0)}\right)_{01} &= \mathfrak{q}L_{jk}\left[2-\mathfrak{q}\left(3\frac{L_{ij}L_{ik}}{L_{jk}}+\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ij}L_{jk}}{L_{ik}}\right)\right], \\
\left(\mathbb{I}_{ijk}^{(0)}\right)_{10} &= \mathfrak{q}L_{ij}\left[2-\mathfrak{q}\left(3\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ij}L_{ik}}{L_{jk}}+\frac{L_{ij}L_{jk}}{L_{ik}}\right)\right], \\
\left(\mathbb{I}_{ijk}^{(0)}\right)_{11} &= \mathfrak{q}L_{ik}\left[1-\mathfrak{q}\left(3\frac{L_{ij}L_{jk}}{L_{ik}}+\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ik}L_{ij}}{L_{jk}}\right)\right].
\end{aligned} \tag{5.8}$$

and

$$\begin{aligned}
\left(\mathbb{I}_{ijk}^{(1)}\right)_{00} &= \mathfrak{q}L_{ik}\left[-2+\mathfrak{q}\left(3\frac{L_{ij}L_{jk}}{L_{ik}}+\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ik}L_{ij}}{L_{jk}}\right)\right], \\
\left(\mathbb{I}_{ijk}^{(1)}\right)_{01} &= \mathfrak{q}L_{ij}\left[-1+\mathfrak{q}\left(3\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ij}L_{ik}}{L_{jk}}+\frac{L_{ij}L_{jk}}{L_{ik}}\right)\right], \\
\left(\mathbb{I}_{ijk}^{(1)}\right)_{10} &= \mathfrak{q}L_{jk}\left[-1+\mathfrak{q}\left(3\frac{L_{ij}L_{ik}}{L_{jk}}+\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ij}L_{jk}}{L_{ik}}\right)\right], \\
\left(\mathbb{I}_{ijk}^{(1)}\right)_{11} &= \frac{\mathfrak{q}^2}{4}\left(\frac{L_{ij}L_{jk}}{L_{ik}}+\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ik}L_{ij}}{L_{jk}}\right)^2-\frac{\mathfrak{q}}{4}\left(\frac{L_{ij}L_{jk}}{L_{ik}}+\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ik}L_{ij}}{L_{jk}}\right)-\frac{1}{4} \\
&\quad +\mathfrak{q}^2\left(L_{ij}^2+L_{jk}^2+L_{ik}^2\right).
\end{aligned} \tag{5.9}$$

Combining these component matrices with the common exponential factor gives the $\mathcal{N}=4$ length kernel,

$$I_3^{\mathcal{N}=4}(L_{ij},L_{jk},L_{ik}) = \frac{\exp\left[-\mathfrak{q}\left(\frac{L_{ij}L_{jk}}{L_{ik}}+\frac{L_{jk}L_{ik}}{L_{ij}}+\frac{L_{ik}L_{ij}}{L_{jk}}\right)\right]}{2\mathfrak{q}L_{ik}}\left(\mathbb{I}_{ijk}^{(0)},\mathbb{I}_{ijk}^{(1)}\right). \tag{5.10}$$

Normalization, and three-point functions. The first place the kernel appears is the composition law. Schematically, after quotienting the common symmetry and passing to length variables,

$$\int d\mu_{\text{phys}}(j)\, P_{ij}^{\text{BPS}} P_{jk}^{\text{BPS}} \;\longrightarrow\; I_3^{\mathcal{N}=4}(L_{ij},L_{jk},L_{ik})\, P_{ik}^{\text{BPS}}.$$

Since the BPS propagator has two fermionic structures, this single composition statement splits into two radial projections. With the normalization in (5.10), these projections are

$$\int_0^\infty \frac{dL_{ij}}{L_{ij}}\frac{dL_{jk}}{L_{jk}}\frac{e^{-\mathfrak{q}(\cdots)}}{2\mathfrak{q}L_{ik}}\Big((2\mathfrak{q}L_{ij})^2K_0(2\mathfrak{q}L_{ij})\,(2\mathfrak{q}L_{ij})^2K_1(2\mathfrak{q}L_{ij})\Big)\mathbb{I}^{(0)}_{ijk}\begin{pmatrix}(2\mathfrak{q}L_{jk})^2K_0(2\mathfrak{q}L_{jk})\\(2\mathfrak{q}L_{jk})^2K_1(2\mathfrak{q}L_{jk})\end{pmatrix}=-2\mathfrak{q}L_{ik}K_0(2\mathfrak{q}L_{ik}), \tag{5.11}$$

$$\int_0^\infty \frac{dL_{ij}}{L_{ij}}\frac{dL_{jk}}{L_{jk}}\frac{e^{-\mathfrak{q}(\cdots)}}{2\mathfrak{q}L_{ik}}\Big((2\mathfrak{q}L_{ij})^2K_0(2\mathfrak{q}L_{ij})\,(2\mathfrak{q}L_{ij})^2K_1(2\mathfrak{q}L_{ij})\Big)\mathbb{I}^{(1)}_{ijk}\begin{pmatrix}(2\mathfrak{q}L_{jk})^2K_0(2\mathfrak{q}L_{jk})\\(2\mathfrak{q}L_{jk})^2K_1(2\mathfrak{q}L_{jk})\end{pmatrix}=2\mathfrak{q}L_{ik}K_1(2\mathfrak{q}L_{ik}). \tag{5.12}$$

Equations (5.11) and (5.12) are the practical check of the normalization.

We now turn from the composition law to correlators. We first focus on BPS three-point functions. For a scalar three-point function, the matter input is the SCFT three-point structure. Before reducing to length variables, the corresponding gluing formula has the same form as the four-point formula below; one should first quotient by the common diagonal orbit, or equivalently choose a representative and include the quotient Jacobian in the reduced length measure:

$$\mathcal{A}_3^{\Delta_1,\Delta_2,\Delta_3}=\int\frac{\prod_{a=1}^3 d\mu_{\rm phys}(a)}{\mathrm{Vol}(G)}\,P_{12}^{\rm BPS}P_{23}^{\rm BPS}P_{31}^{\rm BPS}\,\mathcal{O}_{\Delta_1,\Delta_2,\Delta_3}(1,2,3). \tag{5.13}$$

Equivalently, one can fix one endpoint and include the quotient Jacobian in the reduced length measure. The object $\mathcal{O}_{\Delta_1,\Delta_2,\Delta_3}(1,2,3)$ is the matter three-point structure, which should be read from the boundary SCFT. For three scalar dimensions $\Delta_1,\Delta_2,\Delta_3$, set $\alpha_{12}=\Delta_1+\Delta_2-\Delta_3$, $\alpha_{23}=\Delta_2+\Delta_3-\Delta_1$, and $\alpha_{31}=\Delta_3+\Delta_1-\Delta_2$. For the scalar component considered here, using the scalar boundary data of Section 4.3, this SCFT structure becomes a length-space monomial,

$$\mathcal{O}_{\Delta_1,\Delta_2,\Delta_3}(1,2,3)=C_{123}\,\frac{\Pi_1^{\Delta_1}\Pi_2^{\Delta_2}\Pi_3^{\Delta_3}}{\widetilde{X}_{12}^{\alpha_{12}}\widetilde{X}_{23}^{\alpha_{23}}\widetilde{X}_{31}^{\alpha_{31}}}=C_{123}L_{12}^{\alpha_{12}}L_{23}^{\alpha_{23}}L_{31}^{\alpha_{31}}. \tag{5.14}$$

The integrated three-point function is then obtained by multiplying this length-space matter structure by one composition kernel and integrating over the three lengths. For the $\mathcal{N}=2$ zero-energy example above, the integral is

$$\mathcal{A}_3^{\mathcal{N}=2}=C_{123}\int\frac{dL_{12}}{L_{12}}\frac{dL_{23}}{L_{23}}\frac{dL_{31}}{L_{31}}\,\Psi_0^{\mathcal{N}=2}(L_{12})\Psi_0^{\mathcal{N}=2}(L_{23})\Psi_0^{\mathcal{N}=2}(L_{31})\,I_3^{\mathcal{N}=2}(L_{12},L_{23},L_{31})\,L_{12}^{\alpha_{12}}L_{23}^{\alpha_{23}}L_{31}^{\alpha_{31}}. \tag{5.15}$$

The same Bessel and Gamma-integral techniques reviewed in Appendix D evaluate this integral directly.[16] The answer is

$$\mathcal{A}_3^{\mathcal{N}=2}=C_{123}\,\frac{\sqrt{\pi}}{2^{2(\Delta_1+\Delta_2+\Delta_3)+1}}\,\mathfrak{q}^{-(\Delta_1+\Delta_2+\Delta_3)}\,\frac{\Gamma(2\Delta_1+1)\Gamma(2\Delta_2+1)\Gamma(2\Delta_3+1)}{\Gamma(\Delta_1+\Delta_2+\Delta_3+1)}. \tag{5.16}$$

[16] The $K_{1/2}$ wavefunction is special: $K_{1/2}(z)=\sqrt{\pi/(2z)}e^{-z}$. This turns the intermediate length integrals into a particularly simple Gamma integral.

For the $\mathcal{N}=4$ BPS sector, the same statement keeps the two K_0/K_1 components:

$$\mathcal{A}_3^{\mathcal{N}=4} = C_{123}\int \frac{dL_{12}}{L_{12}}\frac{dL_{23}}{L_{23}}\frac{dL_{31}}{L_{31}}\left[\vec{\Psi}_{s_1}(L_{12})^T\otimes\vec{\Psi}_{s_2}(L_{23})^T\otimes\vec{\Psi}_{s_3}(L_{31})^T\right]\cdot I_3^{\mathcal{N}=4}(L_{12},L_{23},L_{31})\,L_{12}^{\alpha_{12}}L_{23}^{\alpha_{23}}L_{31}^{\alpha_{31}}. \tag{5.17}$$

The dot denotes the finite component contraction fixed by the BPS wavefunctions and by the two entries of $I_3^{\mathcal{N}=4}$. Using the same Bessel and Gamma-integral techniques as in Appendix D, each term in (5.17) reduces to a six-pair Gamma integral. After the final spectral integral the answer is a finite sum of balanced ${}_4F_3(1)$ functions,

$$\mathcal{A}_3^{\mathcal{N}=4} = C_{123}\sum_{\ell=1}^{24}\mathcal{C}_\ell(\alpha_{12},\alpha_{23},\alpha_{31})\,{}_4F_3\Big(\vec{a}_\ell(\alpha_{12},\alpha_{23},\alpha_{31});\vec{b}_\ell(\alpha_{12},\alpha_{23},\alpha_{31});1\Big), \tag{5.18}$$

where $\vec{a}_\ell$, $\vec{b}_\ell$, and $\mathcal{C}_\ell$ are explicit linear and Gamma-function expressions in $(\alpha_{12},\alpha_{23},\alpha_{31})$. The full expression is recorded in the Mathematica supplement.

5.2 The scalar four-point function

We now spell out the four-point calculation in a form which will be useful for comparison between $\mathcal{N}=0$, $\mathcal{N}=2$, and $\mathcal{N}=4$. For equal scaling dimensions, the time-ordered four-point correlator is

$$\mathcal{A}^{\Delta}_{4,\mathrm{TOC}} = \int \frac{\prod_{a=1}^4 d\mu_{\mathrm{phys}}(a)}{\mathrm{Vol}(G)}\,P_{12}P_{23}P_{34}P_{41}\,\mathcal{O}_\Delta(1,2)\mathcal{O}_\Delta(3,4). \tag{5.19}$$

Since the two bilocals do not cross, this is just the product of the two corresponding two-point functions. The hard part is the crossed pairing, which gives the OTOC. The OTOC expression is

$$\mathcal{A}^{\Delta}_{4,\mathrm{OTOC}} = \int \frac{\prod_{a=1}^4 d\mu_{\mathrm{phys}}(a)}{\mathrm{Vol}(G)}\,P_{12}P_{23}P_{34}P_{41}\,\mathcal{O}_\Delta(1,3)\mathcal{O}_\Delta(2,4). \tag{5.20}$$

From here on we focus on the OTOC; later in this subsection we specialize to the equal-scaling-dimension case. We use hyperbolic geometry to determine the two diagonal lengths L_{13} and L_{24}:

$$\frac{1}{L_{13}L_{24}} = \frac{1}{L_{12}L_{34}} + \frac{1}{L_{23}L_{41}}, \qquad L_{13}L_{24} = \frac{L_{12}L_{23}L_{34}L_{41}}{L_{12}L_{34}+L_{23}L_{41}}. \tag{5.21}$$

These diagonal lengths appear when we write the two scalar operators in the length basis. For two equal scalar dimensions,

$$\mathcal{O}_\Delta(1,3)\mathcal{O}_\Delta(2,4) = (L_{13}L_{24})^{2\Delta} = \left(\frac{L_{12}L_{23}L_{34}L_{41}}{L_{12}L_{34}+L_{23}L_{41}}\right)^{2\Delta}, \tag{5.22}$$

The useful feature of the equal-dimension scalar case is that the product of scalar bilocals is independent of the gluing length L_{13}.

$\mathcal{N}=0$ **detour.** In ordinary JT gravity the three-length kernel is the scalar kernel $I_3^{\mathcal{N}=0}$ in (5.4). In dimensionless lengths, gluing two copies along L_{13} gives

$$\begin{aligned} I_4(L_{12},L_{23},L_{34},L_{41}) &= \int_0^\infty \frac{dL_{13}}{L_{13}}\, I_3^{\mathcal{N}=0}(L_{12},L_{23},L_{13}) I_3^{\mathcal{N}=0}(L_{34},L_{41},L_{13}) \\ &= \frac{2}{\pi^2} K_0(2\mathcal{R}_{1234})\,, \\ \mathcal{R}_{1234} &= \sqrt{\frac{(L_{12}L_{23}+L_{34}L_{41})(L_{12}L_{34}+L_{23}L_{41})(L_{12}L_{41}+L_{23}L_{34})}{L_{12}L_{23}L_{34}L_{41}}}. \end{aligned} \tag{5.23}$$

so that

$$\begin{aligned} \mathcal{A}_4^{\mathcal{N}=0} = \int \frac{dL_{12}}{L_{12}}\frac{dL_{23}}{L_{23}}\frac{dL_{34}}{L_{34}}\frac{dL_{41}}{L_{41}}\, \Psi_{s_1}(L_{12})\Psi_{s_2}(L_{23})\Psi_{s_3}(L_{34})\Psi_{s_4}(L_{41}) \\ \times \mathcal{O}_\Delta(1,3)\mathcal{O}_\Delta(2,4)\, I_4(L_{12},L_{23},L_{34},L_{41}). \end{aligned} \tag{5.24}$$

The details of the length calculation are given in Appendix D. After the length integrals and some standard manipulations, this reproduces the JT OTOC kernel with a constant prefactor,

$$\mathcal{A}_4^{\mathcal{N}=0} \sim \mathcal{A}_4^{\mathrm{JT}}, \tag{5.25}$$

where $\mathcal{A}_4^{\mathrm{JT}}$ is the usual Schwarzian OTOC [36, 37, 50]. The supersymmetric cases use the same length geometry, with the scalar bosonic kernel replaced by the K_0/K_1 matrix structure produced by the supercharges.

$\mathcal{N}=2$ **analogue.** The zero-energy $\mathcal{N}=2$ OTOC is given by

$$\begin{aligned} \mathcal{A}_4^{\mathcal{N}=2} = \int \frac{dL_{12}}{L_{12}}\frac{dL_{23}}{L_{23}}\frac{dL_{34}}{L_{34}}\frac{dL_{41}}{L_{41}}\, \mathcal{O}_\Delta(1,3)\mathcal{O}_\Delta(2,4) \prod_{(ab)=(12),(23),(34),(41)} \frac{2L_{ab}}{\pi} K_{1/2}(2L_{ab}) \\ \times \int_0^\infty \frac{dL_{13}}{L_{13}}\, I_3^{\mathcal{N}=2}(L_{12},L_{23},L_{13})\, I_3^{\mathcal{N}=2}(L_{34},L_{41},L_{13}). \end{aligned} \tag{5.26}$$

Compared with the bosonic case, the rational factor in $I_3^{\mathcal{N}=2}$ given in (5.7) produces powers of the diagonal length L_{13} and of the adjacent lengths $L_{12}, L_{23}, L_{34}, L_{41}$. To cast the result in a form familiar from the bosonic calculation, we perform an IBP reduction which is explained in Appendix C. The result is a finite numerator multiplying the same scalar-bilocal product and the same K_0 four-length kernel:

$$\begin{aligned} \mathcal{A}_4^{\mathcal{N}=2} = \int \frac{dL_{12}}{L_{12}}\frac{dL_{23}}{L_{23}}\frac{dL_{34}}{L_{34}}\frac{dL_{41}}{L_{41}}\, (T_1+T_2+T_3+T_4+T_5)\, \mathcal{O}_\Delta(1,3)\mathcal{O}_\Delta(2,4) \\ \times K_0(2\mathcal{R}_{1234}) \prod_{(ab)=(12),(23),(34),(41)} \frac{2L_{ab}}{\pi} K_{1/2}(2L_{ab}). \end{aligned} \tag{5.27}$$

Here $\mathcal{R}_{1234}$ is the length combination in (5.23), and

$$T_1 = 4\Delta(2\Delta+1)\frac{L_{12}L_{23}L_{34}L_{41}}{(L_{12}L_{34}+L_{23}L_{41})^2}, \tag{5.28}$$

$$T_2 = 4\Delta(2\Delta+1)\frac{L_{23}L_{34}+L_{12}L_{41}}{L_{12}L_{34}+L_{23}L_{41}}, \tag{5.29}$$

$$T_3 = 2\Delta(2\Delta+1), \tag{5.30}$$

$$T_4 = -\frac{2(2\Delta+1)}{L_{12}L_{34}+L_{23}L_{41}}\left[L_{12}L_{23}L_{34}+L_{23}L_{34}L_{41}+L_{12}(L_{23}+L_{34})L_{41}\right], \tag{5.31}$$

$$T_5 = \frac{2}{L_{12}L_{34}+L_{23}L_{41}}\left[L_{12}L_{34}(L_{12}+L_{34})+L_{23}L_{41}(L_{23}+L_{41})\right]. \tag{5.32}$$

The resulting length integral can be evaluated completely. The steps are explained in Appendix D; here we only note that the $K_{1/2}$ structure simplifies the calculation substantially. Writing

$$\mathcal{A}_4^{\mathcal{N}=2} = \sum_{r=1}^{5} A_r(\Delta), \tag{5.33}$$

one finds

$$\pi\, 2^{4\Delta} A_1(\Delta) = \frac{\Delta\,\Gamma(1+2\Delta)^4}{4\,\Gamma(2+4\Delta)}\,{}_2F_1\left(\begin{matrix}2+2\Delta,\ 2+2\Delta\\ 3+4\Delta\end{matrix};-1\right), \tag{5.34}$$

$$\pi\, 2^{4\Delta} A_2(\Delta) = \frac{\Gamma(1+2\Delta)^3\Gamma(2+2\Delta)}{4\,\Gamma(2+4\Delta)}\,{}_2F_1\left(\begin{matrix}1+2\Delta,\ 1+2\Delta\\ 3+4\Delta\end{matrix};-1\right), \tag{5.35}$$

$$\pi\, 2^{4\Delta} A_3(\Delta) = \frac{\Delta(1+2\Delta)\Gamma(1+2\Delta)^4\Gamma(1+4\Delta)}{2\,\Gamma(2+4\Delta)^2}\,{}_3F_2\left(\begin{matrix}1+2\Delta,\ 1+2\Delta,\ 1+4\Delta\\ 2+4\Delta,\ 2+4\Delta\end{matrix};-1\right), \tag{5.36}$$

$$\pi\, 2^{4\Delta} A_4(\Delta) = -\frac{(1+2\Delta)\Gamma(1+2\Delta)^2\Gamma(2+2\Delta)^2\Gamma(2+4\Delta)}{4\,\Gamma(3+4\Delta)^2}\,{}_3F_2\left(\begin{matrix}1+2\Delta,\ 1+2\Delta,\ 2+4\Delta\\ 3+4\Delta,\ 3+4\Delta\end{matrix};-1\right), \tag{5.37}$$

$$\pi\, 2^{4\Delta} A_5(\Delta) = \frac{\Gamma(1+2\Delta)^2\Gamma(2+2\Delta)^2\Gamma(1+4\Delta)}{4\,\Gamma(2+4\Delta)\Gamma(3+4\Delta)}\,{}_3F_2\left(\begin{matrix}1+2\Delta,\ 1+2\Delta,\ 1+4\Delta\\ 2+4\Delta,\ 3+4\Delta\end{matrix};-1\right). \tag{5.38}$$

This exact expression can be used as a benchmark: when an analytic integral as precise as this is hard to obtain, it tells us how large the deviation is under a given approximation. A useful approximation is to study a large-Δ saddle with equal lengths. The dominant configuration is the symmetric one,

$$L_{12} = L_{23} = L_{34} = L_{41} = \lambda.$$

The leading saddle sits at $\lambda_* = (2-\sqrt{2})\Delta$, and the Gaussian fluctuations around this symmetric configuration give the large-Δ estimate

$$\mathcal{A}^{\mathcal{N}=2}_{4,1-\text{loop}} \sim 2^{-1/4}\sqrt{\pi}\,\Delta^{3/2}\left(\frac{(2-\sqrt{2})\Delta}{\sqrt{2}\,e}\right)^{4\Delta}. \tag{5.39}$$

Figure 1 compares the logarithm of the exact answer (5.33) with the logarithm of this saddle approx-

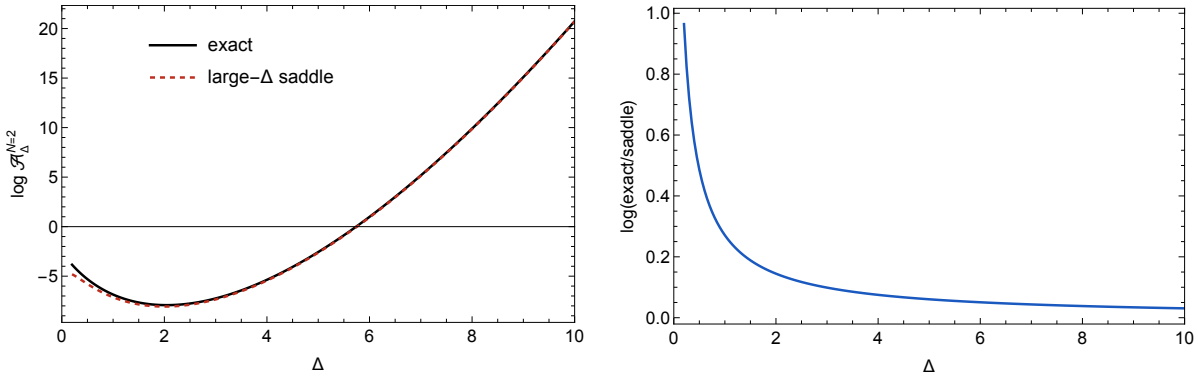


Figure 1: The zero-energy $\mathcal{N}=2$ OTOC as a function of Δ, plotted on a logarithmic scale. Left: $\log \mathcal{A}_4^{\mathcal{N}=2}$, where the solid curve is the exact finite sum $A_1+\cdots+A_5$, and the dashed curve is the logarithm of the large-Δ saddle estimate in (5.39). Right: the logarithmic residual $\log \mathcal{A}_4^{\mathcal{N}=2} - \log \mathcal{A}_{4,1-\text{loop}}^{\mathcal{N}=2}$, which is small on the scale of the left panel but visible after zooming in.

imation. The right panel resolves the logarithmic residual $\log \mathcal{A}_4^{\mathcal{N}=2} - \log \mathcal{A}_{4,1-\text{loop}}^{\mathcal{N}=2}$. Thus the two curves can nearly overlap in the left panel while the right panel still shows a visible residual: the left panel has the much larger vertical range of the logarithmic amplitude, whereas the right panel zooms in on the log-ratio itself. Near $\Delta=1$, the one-loop saddle estimate is only about 30% away from the exact answer which shows the large Δ approximation, with one-loop corrections, is actually not a bad approximation, and the agreement improves quickly as Δ is increased.

$\mathcal{N}=4$ BPS length integral and saddle. We now return to the $\mathcal{N}=4$ BPS sector. The direct analogue of (5.26) keeps the two components of the $I_3^{\mathcal{N}=4}$ kernel. Suppressing the external K_0/K_1 indices of the BPS wavefunctions, the pre-IBP length integral is

$$\begin{aligned}\mathcal{A}_4^{\mathcal{N}=4} = \int \frac{dL_{12}}{L_{12}}\frac{dL_{23}}{L_{23}}\frac{dL_{34}}{L_{34}}\frac{dL_{41}}{L_{41}} \Big[\vec{\Psi}_{s_1}(L_{12})^T \otimes \vec{\Psi}_{s_2}(L_{23})^T \otimes \vec{\Psi}_{s_3}(L_{34})^T \otimes \vec{\Psi}_{s_4}(L_{41})^T\Big]\cdot \\ \times \mathcal{O}_\Delta(1,3)\mathcal{O}_\Delta(2,4)\int_0^\infty \frac{dL_{13}}{L_{13}}\, I_3^{\mathcal{N}=4}(L_{12},L_{23},L_{13})^T\, I_3^{\mathcal{N}=4}(L_{34},L_{41},L_{13}).\end{aligned} \tag{5.40}$$

Here $I_3^{\mathcal{N}=4}$ is the two-component kernel in (5.10).

The expression (5.40) is not yet a useful final form. As in the $\mathcal{N}=2$ calculation, one can simplify it by integration by parts; the needed identities are collected in Appendix C. A useful check is obtained by setting the scalar-bilocal product to one. Then the IBP identities reduce the two $I_3^{\mathcal{N}=4}$ kernels to the composition law of the BPS propagator, with the K_0 and K_1 pieces treated separately and with the boundary terms accounted for as in Appendix C. For the OTOC, the same IBP operation also differentiates the scalar-bilocal product $(L_{12}L_{23}L_{34}L_{41}/(L_{12}L_{34}+L_{23}L_{41}))^{2\Delta}$. The result is a finite sum of ordinary length integrals with the same exponential factors and Bessel functions, but with shifted powers of the lengths and with a finite K_0/K_1 combination structure. From this form one can either

try to do the full analytic length integral term by term,[17] or extract the large-Δ saddle. Below we take the second route.

The full length integral has not yet been reduced to a closed energy-space crossing kernel, but the large-Δ saddle can already be extracted from this length representation. The dominant configuration is the same symmetric one,

$$L_{12} = L_{23} = L_{34} = L_{41} = \lambda, \qquad \lambda_* = (2-\sqrt{2})\Delta.$$

Thus the exponential part of the answer is unchanged from the $\mathcal{N}=2$ calculation. The difference between the $\mathcal{N}=4$ and $\mathcal{N}=2$ integrands is then contained in the finite polynomial in the lengths multiplying the same saddle exponent. Evaluating this polynomial at the symmetric saddle, and including the Gaussian factors, gives the one-loop saddle estimate

$$\mathcal{A}^{\mathcal{N}=4}_{4,1-\text{loop}} \sim \frac{2^{3/4}\pi^{9/2}}{16}(111-78\sqrt{2})\,\Delta^{15/2}\left(\frac{(2-\sqrt{2})\Delta}{\sqrt{2}\,e}\right)^{4\Delta}. \tag{5.41}$$

This gives a useful comparison with the zero-energy $\mathcal{N}=2$ answer,

$$\frac{\mathcal{A}^{\mathcal{N}=4}_{4,1-\text{loop}}}{\mathcal{A}^{\mathcal{N}=2}_{4,1-\text{loop}}} \sim \frac{\pi^4}{8}(111-78\sqrt{2})\,\Delta^6. \tag{5.42}$$

This Δ^6 enhancement is an interesting fact, and we believe it comes from a shorter effective wormhole length in the higher supersymmetry case.

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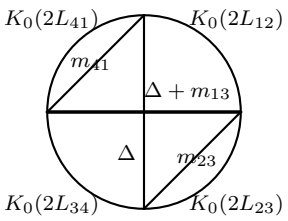
[17] A typical term after the IBP has the schematic form

$$\int_0^\infty \frac{dL_{13}}{L_{13}}\,E_{123}E_{341}\,L_{13}^{m_{13}}\prod_{a=12,23,34,41} L_a^{m_a}K_0(2L_a),$$

with E_{ijk} defined in (C.1); other terms replace some external K_0's by K_1's or shift the integer powers m_a. After the L_{13}-integral this gives ordinary JT Witten-diagram type integrals, schematically

$$\prod_{a=12,23,34,41} L_a^{m_a}K_0(2L_a)\,K_0(2\mathcal{R}_{1234})$$

or the corresponding $\mathcal{R}_{1234}^{-1}K_1(2\mathcal{R}_{1234})$ form. In this sense the Witten diagram is of the following crossed shape,



Following the Appendix D route opens these integrals into Mellin–Barnes variables, but for the generic $\mathcal{N}=4$ terms we do not find a collapse to a single-fold Mellin–Barnes integral. Therefore the term-by-term answer is not simply a single ${}_pF_q$. A better IBP basis may still combine the terms more efficiently, but we have not found such a basis yet.

6 Discussion and future directions

An OPE/representation-theoretic way to derive the OTOC.

We now describe an alternative representation-theoretic route to the four-point function, following the discussion of Wilson functions and polynomials in [47, 48]. The reason to bring this up is that our formalism supplies one of the main ingredients, which is the three-point functions. This suggests that a closed-form expression for the super-JT OTOC may be within reach.

The time-ordered channel is the cleanest place to see what data are needed. We write an OPE representation of the time-ordered four-point function: in the 12|34 channel it takes the form

$$\mathcal{A}^{12|34}_{4,\mathrm{TOC}} = \sum_{n=0}^{\infty} \mathcal{A}^{12}_{3,n}(s_1, s_2, s_3)\, \eta_n \left(\Gamma^{h_n}_{13}\right)^{-1} \mathcal{A}^{34}_{3,n}(s_1, s_4, s_3), \tag{6.1}$$

Equation (6.1) should be read as requiring three pieces of information.

1. *The channel sum.* One must know which exchanged representations appear in the OPE. For identical bosonic scalars this is

$$D^+_\Delta \otimes D^+_\Delta = \bigoplus_{n=0}^{\infty} D^+_{2\Delta+n}, \qquad h_n = 2\Delta + n. \tag{6.2}$$

   In the notation of [47, 48], this is the same n-sum appearing in the Wilson-polynomial identity resolution $\sum_n P_n^{\Delta,\Delta}(s_2; s_1, s_3) P_n^{\Delta,\Delta}(s_4; s_1, s_3)$.

2. *The gluing kernel.* In our raw $\mathcal{A}_3$ normalization, the object sitting between the two three-point vertices is

$$\eta_n \left(\Gamma^{h_n}_{13}\right)^{-1}, \qquad \Gamma^h_{13} := \frac{\Gamma(h \pm i s_1 \pm i s_3)}{\Gamma(2h)}.$$

   Here Γ^h_{13} is the Schwarzian two-point leg of the exchanged operator. The scalar part of the gluing kernel is

$$\eta_n = \frac{\Gamma(h_n)^2}{n!\,(4\Delta + n - 1)_n} = \frac{\Gamma(2\Delta + n)^2}{n!\,(4\Delta + n - 1)_n}. \tag{6.3}$$

   The denominator is the norm of the ordered double-trace primary $[O_\Delta O_\Delta]_n$[18], while the two powers of $\Gamma(h_n)$ match the normalization of the raw length-integral three-point vertex. In [47, 48] this gluing kernel is hidden by the normalized Wilson-polynomial basis: their P_n's include the Wilson norm, so the TOC identity resolution has no visible $\eta_n(\Gamma^{h_n}_{13})^{-1}$ between the two factors.

3. *The three-point function.* In our notation the three-point block in (6.1) is the fixed-energy three-

[18]
$$[O_\Delta O_\Delta]^{12}_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{(2\Delta + n - k)_k}{(2\Delta)_k} \partial^k O_1\, \partial^{n-k} O_2.$$

point function for unit-normalized operators

$$\mathcal{A}^{12}_{3,n}(s_1,s_2,s_3) = \int \frac{dL_{12}}{L_{12}}\frac{dL_{23}}{L_{23}}\frac{dL_{31}}{L_{31}}\,\Psi_{s_1}(L_{12})\Psi_{s_2}(L_{23})\Psi_{s_3}(L_{31}) \\ \times I_3^{\mathcal{N}=0}(L_{12},L_{23},L_{31})\,V_n(L_{12},L_{23},L_{31}), \tag{6.4}$$

with $V_n = L_{12}^{-n}L_{23}^{2\Delta+n}L_{31}^{2\Delta+n}$. This is the analog, with a different normalization, of the Wilson-polynomial $P_n^{\Delta,\Delta}(s_2;s_1,s_3)$ in the notation of [47, 48].

The OTOC requires one more piece of data, namely the exchange action on the same channel space:

$$\mathcal{A}^{12|34}_{4,\mathrm{OTOC}} = \sum_{n=0}^{\infty} \mathcal{A}^{12}_{3,n}(s_1,s_2,s_3)\,\eta_n\,\lambda_n\,\left(\Gamma^{h_n}_{13}\right)^{-1}\mathcal{A}^{34}_{3,n}(s_1,s_4,s_3). \tag{6.5}$$

For identical bosonic scalars this exchange coefficient is

$$\lambda_n = (-1)^n. \tag{6.6}$$

This comes from $[O_\Delta O_\Delta]^{12}_n = (-1)^n[O_\Delta O_\Delta]^{21}_n$.

Now let us study the supersymmetric analogue using the same three ingredients. The external insertions are components of graded matter multiplets, so a channel label is no longer just a number n. A convenient notation for small $\mathcal{N}=4$ multiplets labels the superprimary by $[2J]_h$, where h is its $SL(2)$ weight and J is its highest $SU(2)_{\mathfrak{R}}$ spin; we write $A[2J]_h$ for short multiplets and $L[2J]_h$ for long multiplets, following [61]. The analogue of (6.1) has the schematic form

$$\mathcal{A}^{12|34}_{4,\mathrm{TOC}} = \sum_{\Xi} \mathcal{A}^{12}_{3,\Xi,a}\,\mathcal{G}^{ab}_{\Xi}\,\mathcal{A}^{34}_{3,\Xi,b}, \tag{6.7}$$

where repeated tensor-structure indices are summed. This formula should again be read as requiring three pieces of information.

1. *The channel sum.* One must know which exchanged supermultiplets Ξ appear in the OPE of the two external multiplets, including possible multiplicity spaces. The representation-theory literature, in particular [61], supplies this part. For example,

$$A[1]_{1/2}\otimes A[1]_{1/2} = A[2]_1 \oplus \bigoplus_{m=0}^{\infty} L[0]_{1+m}.$$

   This example is multiplicity-free, but in general there can be multiplicities from different tensor structures.

2. *The gluing kernel.* The factor $\mathcal{G}^{ab}_{\Xi}$ is the supersymmetric analogue of $\eta_n(\Gamma^{h_n}_{13})^{-1}$. It contains the inverse two-point function of the exchanged multiplet in the fixed-energy basis, together with an

energy-independent normalization factor. In the supersymmetric case this factor can become a matrix on the tensor-structure space.

3. *The three-point function.* The objects $\mathcal{A}^{12}_{3,\Xi,a}$ and $\mathcal{A}^{34}_{3,\Xi,b}$ are the fixed-energy three-point functions obtained from the length-space propagator construction in this paper. For genuine supermultiplets the three-point function is more complicated than a product of powers of lengths, so the composition kernel is more complicated than the scalar $I_3^{\mathcal{N}=2,4}$ in Section 5.1. The logic of obtaining it from propagators, however, is the same.

The OTOC needs one further finite piece of data, the graded exchange map on the same channel space:

$$\mathcal{A}^{12|34}_{4,\mathrm{OTOC}} = \sum_{\Xi} \mathcal{A}^{12}_{3,\Xi,a}\, \mathcal{G}^{ac}_{\Xi}\, \Lambda_{\Xi,c}{}^{b}\, \mathcal{A}^{34}_{3,\Xi,b}. \tag{6.8}$$

Here Λ_Ξ is a finite exchange matrix. For component operators of grading $\#_i = 0, 1$, it includes the ordinary grading sign $(-1)^{\#_1\#_2}$.

To summarize, the problem of getting OTOC from this method is very concrete: combine the two three-point functions computed by the present propagator method with the supermultiplet decomposition and exchange data from [61], keeping careful track of grading and normalization. That data gives the supersymmetric TOC through (6.7) and the supersymmetric OTOC through (6.8).

D1–D5–P Berry curvature. Another direction is to study the Berry curvature of BPS black hole states. Recently, Berry curvature was proposed as a diagnostic of chaos in degenerate BPS sectors [42]; see also related work [62]. These analyses are mostly formulated from the CFT point of view; here we study the same physics from gravity. As mentioned in the introduction, our approach is to follow the flow from the UV region to the edge of the near-horizon throat, and then use dimensional reduction to map the calculation to a two-dimensional JT problem.

First recall the definition. Let $\mathcal{H}_{\mathrm{BPS}}$ be the degenerate BPS Hilbert space at fixed charges, with projector $\Pi_{\mathrm{BPS}} = \sum_a |a\rangle\langle a|$. The basic definition is that the non-Abelian Berry curvature is the curvature of this projector,

$$(F_{IJ})_{ab} = \langle a|[\partial_I \Pi_{\mathrm{BPS}}, \partial_J \Pi_{\mathrm{BPS}}]|b\rangle. \tag{6.9}$$

Equivalently, after inserting a complete set of states outside $\mathcal{H}_{\mathrm{BPS}}$, the same curvature is

$$(F_{IJ})_{ab} = \sum_{n\notin\mathcal{H}_{\mathrm{BPS}}} \frac{\langle a|V_I|n\rangle\langle n|V_J|b\rangle - \langle a|V_J|n\rangle\langle n|V_I|b\rangle}{(E_n - E_0)^2}. \tag{6.10}$$

Here E_0 is the common BPS energy and V_I is the equal-time perturbing operator. This is the form we use below; the equivalent Euclidean conformal-perturbation derivation of [63] gives the same expression after spectral decomposition, with the two Euclidean time integrals producing the factor $(E_n - E_0)^{-2}$.

The system we have in mind is D1–D5–P. Without the momentum charge, the bulk dual is $\mathrm{AdS}_3 \times$

$S^3 \times M_4$, and the Berry curvature is a highly protected quantity determined by symmetry and the chiral-ring data [64, 65]. With large momentum charge, the bulk dual is expected to be an extremal BTZ black hole times the same transverse dimensions [66]. In this case there is less symmetry, and the Berry curvature is not expected to be uniquely fixed. We provide a gravity approach to computing it. Since the low-energy spectrum of a (near-)BPS black hole is captured by super-JT gravity, we use the techniques developed in this paper to study the corresponding Berry curvature. The index I labels a direction in the D1–D5 conformal manifold, which has dimension 20 for $M_4 = T^4$ and 84 for $M_4 = K3$ [67, 68]. More concretely, the conformal-manifold deformation is represented by an exactly marginal operator

$$\mathcal{O}_I^{\text{marg}} \sim \left(\mathsf{F}^{(L)}_{-1/2}\mathsf{F}^{(R)}_{-1/2}\Phi_I\right), \qquad (h,\bar{h}) = (1,1), \tag{6.11}$$

where the superconformal primary obeys

$$\Phi_I : \qquad (h,\bar{h}) = \left(\frac{1}{2},\frac{1}{2}\right), \qquad (j,\bar{j}) = \left(\frac{1}{2},\frac{1}{2}\right). \tag{6.12}$$

At strong coupling $\mathcal{O}_I^{\text{marg}}$ is dual to the neutral massless scalar t_I [68, 69].

$$\delta S_{\text{CFT}} = \sum_I \lambda^I \int d^2z\, \mathcal{O}_I^{\text{marg}}(z,\bar{z}), \qquad V_I = \int d\sigma\, \mathcal{O}_I^{\text{marg}}(0,\sigma). \tag{6.13}$$

We then propagate this boundary source to the edge of the AdS_2 region. This amounts to solving the massless scalar equation in the BTZ background

$$ds_3^2 = -N^2dt^2 + N^{-2}dr^2 + r^2(d\varphi + N^\varphi dt)^2, \qquad N^2 = \frac{(r^2-r_+^2)(r^2-r_-^2)}{r^2},$$

with $N^\varphi = -r_+r_-/(r^2)$. The source is a static zero mode, so we take $t_I(t,r,\varphi) = \lambda_I R(r)$, with

$$R(r) = c_1 + \frac{c_2}{2(r_+^2-r_-^2)}\log\left(\frac{r^2-r_+^2}{r^2-r_-^2}\right)$$

which solves[19]

$$\partial_r\left[\frac{(r^2-r_+^2)(r^2-r_-^2)}{r}\partial_r R\right] = 0. \tag{6.14}$$

Regularity at the horizon removes the nonconstant branch, so the propagation to the throat is trivial.

We can now compute the moments of Berry curvature from $\mathcal{N}=4$ super-JT by a simple symmetry argument: with our convention the D1–D5–P momentum charge is carried by the left-moving sector, while the extremal BTZ geometry preserves the $PSU(1,1|2)$ superisometry in the right-moving sector.

In the JT description we should not expect to compute a prescribed matrix element of F_{IJ} in a microscopic basis of $\mathcal{H}_{\text{BPS}}$. Constructing such a basis for individual black-hole microstates is difficult,

[19] In the extremal limit the nonconstant solution becomes $(r^2-r_0^2)^{-1}$.

despite substantial progress from cohomological constructions [70–78]. The natural gravity observables are instead statistical and basis-independent quantities, such as traces of moments, and even the spectral statistics of the matrix $(F_{IJ})_{ab}$ as in [42]. Here we fix two independent deformation directions I, J and use a, b to label BPS states.[20] The simplest such basis-independent observable is

$$\mathcal{V}_{IJ;IJ} = \frac{1}{D_{\rm BPS}} \overline{\mathrm{Tr}_{\rm BPS}\left(F_{IJ}F_{IJ}^{\dagger}\right)}, \qquad D_{\rm BPS} = \dim \mathcal{H}_{\rm BPS}, \tag{6.15}$$

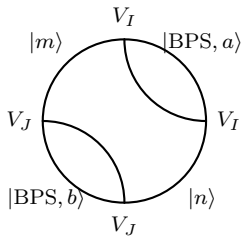
where the overline denotes the statistical average that the JT calculation is meant to approximate.

The next step is to translate this observable into explicit JT correlators. Substituting (6.10) into (6.15), we obtain an average over four V matrix elements, which translates into JT four-point functions.[21] Concretely, the two different orderings of I, J, namely I, J, J, I and I, J, I, J, give

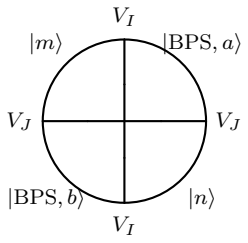
$$\begin{aligned}\mathcal{V}_{IJ;IJ} = \frac{2}{D_{\rm BPS}} \sum_{m,n\notin\mathcal{H}_{\rm BPS}} \sum_{a,b\in\mathcal{H}_{\rm BPS}} \frac{1}{(E_n-E_0)^2(E_m-E_0)^2} \\ \times \left[\overline{\langle a|V_I|n\rangle\langle n|V_J|b\rangle\langle b|V_J|m\rangle\langle m|V_I|a\rangle} - \overline{\langle a|V_I|n\rangle\langle n|V_J|b\rangle\langle b|V_I|m\rangle\langle m|V_J|a\rangle}\right].\end{aligned} \tag{6.16}$$

Since V_I and V_J correspond to two different deformation directions, there is no Wick contraction between them. The two averages should therefore be translated into the following two JT gravity correlators:

$\overline{\langle a|V_I|n\rangle\langle n|V_J|b\rangle\langle b|V_J|m\rangle\langle m|V_I|a\rangle}$ $\longleftrightarrow$



$\overline{\langle a|V_I|n\rangle\langle n|V_J|b\rangle\langle b|V_I|m\rangle\langle m|V_J|a\rangle}$ $\longleftrightarrow$



where the lines in the disk, denoted by $\mathcal{O}_I^{\rm marg}(i,j)$, are the Wilson-line two-point insertions for V_I and V_J. Following the propagator gluing method of Section 5.2, we obtain the corresponding four-point

[20] In the specific D1–D5–P Berry curvature problem all directions in the moduli space are equivalent, so we can pick any two independent deformation directions.

[21] See especially Eq. (2.41) of [19], where a normalized trace with four energy projectors is identified with a product of four energy-basis matrix elements and the JT four-point function.

expressions

$$\mathcal{A}_{4,\mathrm{TOC}}(E_n,E_m)=\int\frac{\prod_{a=1}^4 d\mu_{\mathrm{phys}}(a)}{\mathrm{Vol}(G)}\,P_{12}^{\mathrm{BPS}}P_{23}^{\text{non-BPS}}(E_n)P_{34}^{\mathrm{BPS}}P_{41}^{\text{non-BPS}}(E_m)\big[\mathcal{O}_I^{\mathrm{marg}}(1,2)\mathcal{O}_J^{\mathrm{marg}}(3,4)+(I\leftrightarrow J)\big] \tag{6.17}$$

$$\mathcal{A}_{4,\mathrm{OTOC}}(E_n,E_m)=\int\frac{\prod_{a=1}^4 d\mu_{\mathrm{phys}}(a)}{\mathrm{Vol}(G)}\,P_{12}^{\mathrm{BPS}}P_{23}^{\text{non-BPS}}(E_n)P_{34}^{\mathrm{BPS}}P_{41}^{\text{non-BPS}}(E_m)\big[\mathcal{O}_I^{\mathrm{marg}}(1,3)\mathcal{O}_J^{\mathrm{marg}}(2,4)+(I\leftrightarrow J)\big] \tag{6.18}$$

Here $P^{\text{non-BPS}}(E)$ is the fixed-energy propagator for non-BPS states.

We will comment on the bilocal operator $\mathcal{O}_I^{\mathrm{marg}}$ below. With the spectral normalization of (6.10), the second moment is

$$\mathcal{V}_{IJ;IJ}=\int dE_n\,dE_m\,\frac{\rho_{\text{non-BPS}}(E_n)\rho_{\text{non-BPS}}(E_m)}{D_{\mathrm{BPS}}(E_n-E_0)^2(E_m-E_0)^2}\left[\mathcal{A}_{4,\mathrm{TOC}}(E_n,E_m)-\mathcal{A}_{4,\mathrm{OTOC}}(E_n,E_m)\right]. \tag{6.19}$$

The two BPS propagators in $\mathcal{A}_4$ include the sums over the two BPS labels a,b.[22]

We now turn to the operator $\mathcal{O}^{\mathrm{marg}}$ and the supersymmetric selection rule. The $\mathcal{O}_I^{\mathrm{marg}}$'s in (6.17) and (6.18) are the AdS_2 boundary-anchored two-point functions for the deformation V_I. As discussed in Section 4.3, for a primary component of weight h and $SU(2)_{\mathfrak{R}}$ spin j, the invariant dependence is $L_{ij}^{2h}D^{(j)}(U_{ij})$. The marginal operator is instead a supercharge descendant of the spin-1/2, dimension-1/2 primary, so its invariant expression is more complicated: the physical boundary supercharge includes the compensating transformation needed by the Hamiltonian reduction. In the normalization of (4.52), this means $\mathcal{O}_I^{\mathrm{marg}}(i,j)=Q_iQ_j\big[L_{ij}D^{(1/2)}(U_{ij})\big]=L_{ij}^2+iL_{ij}\eta_j^aW_a{}^b(i,j)\eta_{i,b}$, where Q_i is the physical boundary supercharge, not the naive bare F_- action.

This also tells us which non-BPS sector to use: there is a supersymmetry plus angular-momentum selection rule. In any vertex $\langle a|V_I|n\rangle$, angular momentum conservation forces n to have zero angular momentum. There are two non-BPS $\mathcal{N}=4$ super-Poincare worldline multiplets in Section 4.2 which contain such scalar states: one has maximum angular momentum $J=1/2$ and contains the Ψ,χ states, while the other has maximum angular momentum $J=1$ and contains the L state. Since V_I is a superdescendant of a spin-1/2 superprimary, the L state is ruled out, and we are left with

$$\Pi_{\mathrm{marg}}=\Pi_{\Psi,j=0}^{J=\frac{1}{2}}+\Pi_{\chi,j=0}^{J=\frac{1}{2}}. \tag{6.20}$$

To summarize, the ingredients needed for the second moment are already present in this paper. The BPS and $J=1/2$ supermultiplet propagators were derived in Section 4.2, while the marginal bilocal operator was discussed in Section 4.3. Combining these inputs with the four-point gluing formula (6.19) gives a concrete JT protocol for computing the second moment of the Berry curvature. We emphasize that this is not limited to the D1–D5–P Berry curvature. In principle the same procedure applies to any BPS black hole with a moduli space; D1–D5–P is just the concrete example for which we assemble

[22] The explicit factor $D_{\mathrm{BPS}}=e^{S_0}$ in the denominator and the factor of e^{S_0} hidden in P combine to make $\mathcal{V}_{IJ;IJ}\sim e^{S_0}$.

the ingredients in this paper.

A more ambitious goal is indeed more chalenging, that is to quantify the exact chaotic behavior of F_{IJ}. A natural diagnostic is the statistical distribution of its eigenvalues, but determining that distribution requires essentially all moments. In the present formalism the n-th moment is a higher $2n$-point function of the same marginal bilocal. It is not yet clear whether an all-n formula is practical; if the length-integral representation admits a useful recursion relation, then there may be a systematic way forward. This would provide a direct bridge between super-JT correlators and statistical diagnostics of chaos in the degenerate BPS Hilbert space.

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A Notations and conventions

This appendix collects notation used in the main text. Bulk component fields are ordinary symbols such as $G_{\mu\nu}$, $E_\mu{}^a$, and Ψ_μ. Bulk superfields are bold, for example $\mathbf{E}^A$ and $\mathbf{\Phi}$. BF fields are calligraphic, for example $\mathcal{A}$ and $\mathcal{X}$, and the full supergroup element is $\mathcal{G}$.

A.1 (Super) Lie algebra and bilinear form

Algebra generators are sans-serif:

$$\mathsf{L}_-, \quad \mathsf{L}_0, \quad \mathsf{L}_+, \qquad \mathsf{R}, \quad \mathsf{R}_i, \qquad \mathsf{F}_{-,p}, \quad \bar{\mathsf{F}}_-^p, \quad \mathsf{F}_{+,p}, \quad \bar{\mathsf{F}}_+^p.$$

Here R is the $U(1)_{\mathfrak{R}}$ generator of the $\mathcal{N}=2$ subalgebra. For the $SU(2)_{\mathfrak{R}}$ part of the $\mathcal{N}=4$ algebra, $p,q=1,2$ are fundamental indices and $i,j,k=1,2,3$ are adjoint indices.

$SL(2)$ basis and bracket. Throughout the paper, the bracket $\langle\,,\,\rangle$ is always the gravity-normalized invariant bilinear form. For the bosonic $SL(2)$ block we fix the basis by

$$[\mathsf{L}_0,\mathsf{L}_\pm]=\pm\mathsf{L}_\pm, \qquad [\mathsf{L}_+,\mathsf{L}_-]=2\mathsf{L}_0. \tag{A.1}$$

The bosonic invariant form is normalized by

$$\langle \mathsf{L}_0, \mathsf{L}_0 \rangle = 1, \qquad \langle \mathsf{L}_+, \mathsf{L}_- \rangle = 2, \tag{A.2}$$

with all other independent pairings vanishing. In first-order JT variables we also use

$$\mathsf{P}_{\hat{u}} = \frac{1}{2}(\mathsf{L}_- + \mathsf{L}_+), \qquad \mathsf{P}_{\hat{r}} = \mathsf{L}_0, \qquad \mathsf{J} = \frac{1}{2}(\mathsf{L}_- - \mathsf{L}_+), \tag{A.3}$$

which gives

$$\langle \mathsf{P}_a, \mathsf{P}_b \rangle = \delta_{ab}, \qquad \langle \mathsf{J}, \mathsf{J} \rangle = -1. \tag{A.4}$$

This convention is twice the common representation-theory normalization of the $SL(2)$ Casimir; it is the convention used everywhere below.

$\mathcal{N} = 2$ algebra and pairing. The $\mathcal{N} = 2$ subalgebra is generated by

$$\mathsf{L}_-, \quad \mathsf{L}_0, \quad \mathsf{L}_+, \qquad \mathsf{R}, \qquad \mathsf{F}_-, \quad \bar{\mathsf{F}}_-, \quad \mathsf{F}_+, \quad \bar{\mathsf{F}}_+.$$

The nonzero brackets and anticommutators are

$$\begin{gathered}
[\mathsf{L}_0, \mathsf{L}_\pm] = \pm \mathsf{L}_\pm, \qquad [\mathsf{L}_+, \mathsf{L}_-] = 2\mathsf{L}_0, \qquad [\mathsf{R}, \mathsf{L}_0] = [\mathsf{R}, \mathsf{L}_\pm] = 0, \\
[\mathsf{L}_0, \mathsf{F}_\pm] = \pm\frac{1}{2}\mathsf{F}_\pm, \qquad [\mathsf{L}_0, \bar{\mathsf{F}}_\pm] = \pm\frac{1}{2}\bar{\mathsf{F}}_\pm, \qquad [\mathsf{R}, \mathsf{F}_\pm] = \frac{1}{2}\mathsf{F}_\pm, \qquad [\mathsf{R}, \bar{\mathsf{F}}_\pm] = -\frac{1}{2}\bar{\mathsf{F}}_\pm, \\
[\mathsf{L}_\pm, \mathsf{F}_\pm] = [\mathsf{L}_\pm, \bar{\mathsf{F}}_\pm] = 0, \qquad [\mathsf{L}_\pm, \mathsf{F}_\mp] = -\mathsf{F}_\pm, \qquad [\mathsf{L}_\pm, \bar{\mathsf{F}}_\mp] = \bar{\mathsf{F}}_\pm, \\
\{\mathsf{F}_\pm, \bar{\mathsf{F}}_\pm\} = \mathsf{L}_\pm, \qquad \{\mathsf{F}_+, \bar{\mathsf{F}}_-\} = \mathsf{R} - \mathsf{L}_0, \qquad \{\mathsf{F}_-, \bar{\mathsf{F}}_+\} = \mathsf{R} + \mathsf{L}_0, \\
\{\mathsf{F}_\epsilon, \mathsf{F}_{\epsilon'}\} = \{\bar{\mathsf{F}}_\epsilon, \bar{\mathsf{F}}_{\epsilon'}\} = 0, \qquad \epsilon, \epsilon' = \pm.
\end{gathered} \tag{A.5}$$

The invariant pairing is the gravity-normalized Casimir pairing. In these conventions the nonzero pairings are

$$\begin{gathered}
\langle \mathsf{L}_+, \mathsf{L}_- \rangle = 2, \qquad \langle \mathsf{L}_0, \mathsf{L}_0 \rangle = 1, \qquad \langle \mathsf{R}, \mathsf{R} \rangle = -1, \\
\langle \mathsf{F}_+, \bar{\mathsf{F}}_- \rangle = -2, \qquad \langle \bar{\mathsf{F}}_-, \mathsf{F}_+ \rangle = 2, \qquad \langle \mathsf{F}_-, \bar{\mathsf{F}}_+ \rangle = -2, \qquad \langle \bar{\mathsf{F}}_+, \mathsf{F}_- \rangle = 2.
\end{gathered} \tag{A.6}$$

The signs in the odd pairings are the graded signs of the invariant form. All other independent pairings vanish.

$\mathcal{N} = 4$ algebra and pairing. For the full $\mathcal{N} = 4$ particle we use the $PSU(1,1|2)$ algebra in an $SU(2)_{\mathfrak{R}}$-covariant basis. The even generators are

$$\mathsf{L}_-, \quad \mathsf{L}_0, \quad \mathsf{L}_+, \qquad \mathsf{R}_i \quad (i = 1, 2, 3),$$

and the odd generators are

$$\mathsf{F}_{\pm,p}, \qquad \bar{\mathsf{F}}^p_{\pm}, \qquad p=1,2.$$

The $SL(2)$ brackets are still (A.1). The remaining nonzero brackets and anticommutators are

$$\begin{gathered}
[\mathsf{R}_i,\mathsf{R}_j]=i\epsilon_{ijk}\mathsf{R}_k, \qquad [\mathsf{R}_i,\mathsf{L}_0]=[\mathsf{R}_i,\mathsf{L}_\pm]=0,\\
[\mathsf{L}_0,\mathsf{F}_{\pm,p}]=\pm\frac{1}{2}\mathsf{F}_{\pm,p}, \qquad [\mathsf{L}_0,\bar{\mathsf{F}}^p_{\pm}]=\pm\frac{1}{2}\bar{\mathsf{F}}^p_{\pm}, \qquad [\mathsf{R}_i,\mathsf{F}_{\pm,p}]=\frac{1}{2}(\sigma_i)^q{}_p\mathsf{F}_{\pm,q}, \qquad [\mathsf{R}_i,\bar{\mathsf{F}}^p_{\pm}]=-\frac{1}{2}(\sigma_i)^p{}_q\bar{\mathsf{F}}^q_{\pm},\\
[\mathsf{L}_\pm,\mathsf{F}_{\pm,p}]=[\mathsf{L}_\pm,\bar{\mathsf{F}}^p_{\pm}]=0, \qquad [\mathsf{L}_\pm,\mathsf{F}_{\mp,p}]=-\mathsf{F}_{\pm,p}, \qquad [\mathsf{L}_\pm,\bar{\mathsf{F}}^p_{\mp}]=\bar{\mathsf{F}}^p_{\pm}.\\
\{\mathsf{F}_{\pm,p},\bar{\mathsf{F}}^q_{\pm}\}=\delta_p{}^q\,\mathsf{L}_\pm, \qquad \{\mathsf{F}_{+,p},\bar{\mathsf{F}}^q_{-}\}=-\delta_p{}^q\,\mathsf{L}_0+(\sigma^i)^q{}_p\mathsf{R}_i, \qquad \{\mathsf{F}_{-,p},\bar{\mathsf{F}}^q_{+}\}=\delta_p{}^q\,\mathsf{L}_0+(\sigma^i)^q{}_p\mathsf{R}_i,\\
\{\mathsf{F}_{\epsilon,p},\mathsf{F}_{\epsilon',q}\}=\{\bar{\mathsf{F}}^p_{\epsilon},\bar{\mathsf{F}}^q_{\epsilon'}\}=0, \qquad \epsilon,\epsilon'=\pm.
\end{gathered} \tag{A.7}$$

Here σ_i are the standard Pauli matrices. The invariant bilinear form is the gravity-normalized Casimir pairing of the projective algebra.[23] In these conventions the nonzero pairings are

$$\begin{gathered}
\langle\mathsf{L}_+,\mathsf{L}_-\rangle=2, \qquad \langle\mathsf{L}_0,\mathsf{L}_0\rangle=1, \qquad \langle\mathsf{R}_i,\mathsf{R}_j\rangle=-\delta_{ij}, \qquad \langle\mathsf{F}_{+,p},\bar{\mathsf{F}}^q_{-}\rangle=-2\delta_p{}^q, \qquad \langle\bar{\mathsf{F}}^q_{-},\mathsf{F}_{+,p}\rangle=2\delta_p{}^q,\\
\langle\mathsf{F}_{-,p},\bar{\mathsf{F}}^q_{+}\rangle=-2\delta_p{}^q, \qquad \langle\bar{\mathsf{F}}^q_{+},\mathsf{F}_{-,p}\rangle=2\delta_p{}^q.
\end{gathered} \tag{A.8}$$

All other independent pairings vanish.

Ordered coordinates and left/right derivatives. For the $PSU(1,1|2)$ group element we use the ordered coordinates

$$\mathcal{G}=e^{x\mathsf{L}_-}e^{\theta^p_-\mathsf{F}_{-,p}+\bar\theta_{-,p}\bar{\mathsf{F}}^p_-}e^{\rho\mathsf{L}_0}Ue^{\theta^p_+\mathsf{F}_{+,p}+\bar\theta_{+,p}\bar{\mathsf{F}}^p_+}e^{y\mathsf{L}_+}. \tag{A.9}$$

For a function $f(\mathcal{G})$, the left and right vector fields are defined by

$$\mathcal{D}^L_T f(\mathcal{G})=\frac{d}{d\epsilon}f(e^{\epsilon T}\mathcal{G})\Big|_{\epsilon=0}, \qquad \mathcal{D}^R_T f(\mathcal{G})=\frac{d}{d\epsilon}f(\mathcal{G}e^{\epsilon T})\Big|_{\epsilon=0}. \tag{A.10}$$

It is useful to separate the fermionic number operators

$$N_-=\theta^p_-\partial_{\theta^p_-}+\bar\theta_{-,p}\partial_{\bar\theta_{-,p}}, \qquad N_+=\theta^p_+\partial_{\theta^p_+}+\bar\theta_{+,p}\partial_{\bar\theta_{+,p}}.$$

[23] A useful warning is that one can easily write a 3×3, or $2|1$, matrix realization for the $\mathfrak{psu}(1,1|1)$ subalgebra, but one should not expect an analogous honest 4×4, or $2|2$, matrix realization for $\mathfrak{psu}(1,1|2)$. This is why the small $\mathcal{N}=2$ check works, while the natural full $\mathcal{N}=4$ check is only projective. After complexification the former is essentially $\mathfrak{sl}(2|1)$, whereas the latter is $\mathfrak{psl}(2|2)=\mathfrak{sl}(2|2)/\mathbb{C}\,\mathbf{1}$. In the $2|1$ case, $\mathrm{Str}\,\mathbf{1}_{2|1}=2-1\neq0$, so the identity is not an element of $\mathfrak{sl}(2|1)$. The scalar-looking singlet in the odd anticommutator can therefore be absorbed into the genuine $U(1)_{\mathfrak{R}}$ generator, giving literal identities such as $\{\mathsf{F}_+,\bar{\mathsf{F}}_-\}=\mathsf{R}-\mathsf{L}_0$ and $\{\mathsf{F}_-,\bar{\mathsf{F}}_+\}=\mathsf{R}+\mathsf{L}_0$. In the $2|2$ case, $\mathrm{Str}\,\mathbf{1}_{2|2}=2-2=0$, so the identity lies inside $\mathfrak{sl}(2|2)$ and is central. The simple algebra is the projective quotient, and the elementary $2|2$ matrices close only up to this central term: $\{\mathsf{F}_{+,p},\bar{\mathsf{F}}^q_-\}=-\delta_p{}^q\mathsf{L}_0+(\sigma^i)^q{}_p\mathsf{R}_i+\frac{1}{2}\delta_p{}^q\mathbf{1}_{2|2}$. The last term vanishes in $\mathfrak{psl}(2|2)$, but not as an ordinary 4×4 matrix. Thus the obstruction is the special $m=n$ feature of $\mathfrak{sl}(m|n)$, not a bad basis choice. Accordingly the pairing below is defined from the projective Casimir, not from an ordinary 4×4 trace.

The simple left derivatives are

$$\mathcal{D}^L_{\mathsf{L}_-} = \partial_x, \quad \mathcal{D}^L_{\mathsf{L}_0} = -x\partial_x + \partial_\rho - \frac{1}{2}N_-, \quad \mathcal{D}^L_{\mathsf{F}_{-,p}} = \partial_{\theta^p_-} - \frac{1}{2}\bar{\theta}_{-,p}\partial_x, \quad \mathcal{D}^L_{\bar{\mathsf{F}}^p_-} = \partial_{\bar{\theta}_{-,p}} - \frac{1}{2}\theta^p_-\partial_x, \qquad p = 1, 2. \tag{A.11}$$

The simple right derivatives are

$$\mathcal{D}^R_{\mathsf{L}_+} = \partial_y, \quad \mathcal{D}^R_{\mathsf{L}_0} = -y\partial_y + \partial_\rho - \frac{1}{2}N_+, \quad \mathcal{D}^R_{\mathsf{F}_{+,p}} = \partial_{\theta^p_+} + \frac{1}{2}\bar{\theta}_{+,p}\partial_y, \quad \mathcal{D}^R_{\bar{\mathsf{F}}^p_+} = \partial_{\bar{\theta}_{+,p}} + \frac{1}{2}\theta^p_+\partial_y, \qquad p = 1, 2. \tag{A.12}$$

The remaining left-plus and right-minus vector fields are longer and will be provided in the supplemental Mathematica code.

A.2 Superspace, superfield, and super-Schwarzian

Bulk and boundary supercoordinates. Bulk spacetime indices are $\mu, \nu = r, u$, tangent-space indices are $a, b = \hat{r}, \hat{u}$, and spinor indices are α, β. For superspace bookkeeping, the bulk and boundary supercoordinates are

$$\mathbf{Z}^M = (X^\mu = (r, u); \Theta^{p\alpha}, \bar{\Theta}_p{}^\alpha), \qquad z^m = (u, \vartheta^p, \bar{\vartheta}_p).$$

Here $\mathbf{Z}^M$ are the bulk supercoordinates, while z^m are the boundary superline coordinates.

On the flat boundary superline we use the basis one-forms

$$\Delta u = du + i\, d\vartheta^p \bar{\vartheta}_p + i\, d\bar{\vartheta}_p\, \vartheta^p, \qquad d\vartheta^p, \qquad d\bar{\vartheta}_p.$$

The corresponding flat superspace derivatives are

$$D_p = \partial_{\vartheta^p} - i\bar{\vartheta}_p\, \partial_u, \qquad \bar{D}^p = \partial_{\bar{\vartheta}_p} - i\vartheta^p\, \partial_u, \qquad \{D_p, \bar{D}^q\} = -2i\delta_p{}^q \partial_u.$$

For any boundary superfield $F(u, \vartheta, \bar{\vartheta})$,

$$dF = \Delta u\, \partial_u F + d\vartheta^p D_p F + d\bar{\vartheta}_p \bar{D}^p F.$$

Supergravity multiplets. The bulk component fields are grouped into a supergravity multiplet and a dilaton multiplet. We use the component diamonds

$$\mathfrak{G}_{\mathcal{N}} = \begin{array}{ccc} & E_\mu{}^a & \\ \Psi_\mu{}^p & & \bar{\Psi}_{\mu,p} \\ & \omega_\mu,\ A_\mu{}^I & \end{array} \qquad \mathfrak{D}_{\mathcal{N}} = \begin{array}{ccc} & \Phi & \\ \zeta_\alpha{}^p & & \bar{\zeta}_{\alpha,p} \\ & X^a,\ b^I & \end{array} .$$

The index ranges are

$$p = \begin{cases} 1, & \mathcal{N}=2, \\ 1,2, & \mathcal{N}=4, \end{cases} \qquad I = \begin{cases} \mathfrak{R}, & \mathcal{N}=2, \quad \mathfrak{R}=U(1)_{\mathfrak{R}}, \\ i=1,2,3, & \mathcal{N}=4, \quad \mathfrak{R}=SU(2)_{\mathfrak{R}}. \end{cases}$$

The fields X^a and b^I are the bosonic auxiliary, or multiplier, components of the superdilaton multiplet: X^a imposes the supertorsion constraint, while b^I is conjugate to the $\mathfrak{R}$-symmetry field strength. For $\mathcal{N}=2$, $b^I = b$ is the $U(1)_{\mathfrak{R}}$ multiplier; for $\mathcal{N}=4$, $b^I = b^i$ is an $SU(2)_{\mathfrak{R}}$ triplet [79, 80].

These component fields are encoded in the superfields as follows. In Wess–Zumino gauge, the bulk supervielbein begins schematically as

$$\mathbf{E}^a = dX^\mu \big[\underbrace{E_\mu{}^a}_{\substack{\text{frame}\\ \text{field}}} + i\,\bar\Theta_p \gamma^a \underbrace{\Psi^p_\mu}_{\text{gravitino}} + i \underbrace{\bar\Psi_{\mu,p}}_{\overline{\text{gravitino}}} \gamma^a \Theta^p + O(\Theta^2) \big] + i\, d\bar\Theta_p\, \gamma^a \Theta^p + i\, \bar\Theta_p \gamma^a d\Theta^p + O(\Theta^2 d\Theta),$$

and the odd components begin as

$$\mathbf{E}^{p\alpha} = d\Theta^{p\alpha} + dX^\mu \big[\underbrace{\Psi^{p\alpha}_\mu}_{\text{gravitino}} + O(\Theta) \big], \qquad \bar{\mathbf{E}}_p{}^\alpha = d\bar\Theta_p{}^\alpha + dX^\mu \big[\underbrace{\bar\Psi_{\mu,p}{}^\alpha}_{\overline{\text{gravitino}}} + O(\Theta) \big].$$

The bulk superdilaton has component expansion

$$\mathbf{\Phi}(\mathbf{Z}) = \underbrace{\Phi(X)}_{\text{dilaton}} + \Theta^{p\alpha} \underbrace{\bar\zeta_{p\alpha}(X)}_{\overline{\text{dilatino}}} + \bar\Theta_p{}^\alpha \underbrace{\zeta^p{}_\alpha(X)}_{\text{dilatino}} + O(\Theta^2).$$

First-order fields. The first-order fields are packaged into a $\mathfrak{psu}(1,1|2)$-valued connection

$$\mathcal{A} = \mathcal{A}^- \mathsf{L}_- + \mathcal{A}^0 \mathsf{L}_0 + \mathcal{A}^+ \mathsf{L}_+ + \underbrace{\mathsf{A}^i}_{\substack{\text{SU}(2)_{\mathfrak{R}}\\ \text{connection}}} \mathsf{R}_i + \underbrace{\Psi^p_-}_{\text{gravitino}} \mathsf{F}_{-,p} + \underbrace{\bar\Psi_{-,p}}_{\overline{\text{gravitino}}} \bar{\mathsf{F}}^p_- + \underbrace{\Psi^p_+}_{\text{gravitino}} \mathsf{F}_{+,p} + \underbrace{\bar\Psi_{+,p}}_{\overline{\text{gravitino}}} \bar{\mathsf{F}}^p_+.$$

The adjoint scalar, equivalently the first-order dilaton multiplet, is

$$\mathcal{X} = X_- \mathsf{L}_- + X_0 \mathsf{L}_0 + X_+ \mathsf{L}_+ + \underbrace{X^i_{\mathcal{R}}}_{\substack{\text{SU}(2)_{\mathfrak{R}}\\ \text{multiplier}}} \mathsf{R}_i + \underbrace{\zeta^p_-}_{\text{dilatino}} \mathsf{F}_{-,p} + \underbrace{\bar\zeta_{-,p}}_{\overline{\text{dilatino}}} \bar{\mathsf{F}}^p_- + \underbrace{\zeta^p_+}_{\text{dilatino}} \mathsf{F}_{+,p} + \underbrace{\bar\zeta_{+,p}}_{\overline{\text{dilatino}}} \bar{\mathsf{F}}^p_+.$$

The radial asymptotic form of the boundary connection is

$$\mathcal{A}_u = e^r \underbrace{e^{(0)}_u}_{\substack{\text{metric}\\ \text{source}}} \mathsf{L}_- + e^{r/2} \underbrace{v^{(0)p}_u}_{\substack{\text{gravitino}\\ \text{source}}} \mathsf{F}_{-,p} + e^{r/2} \underbrace{\bar v^{(0)}_{u,p}}_{\substack{\overline{\text{gravitino}}\\ \overline{\text{source}}}} \bar{\mathsf{F}}^p_- + \underbrace{a^{(0)i}_u}_{\substack{\text{SU}(2)_{\mathfrak{R}}\\ \text{source}}} \mathsf{R}_i$$

$$+ e^{-r/2} \underbrace{\mathcal{S}^p}_{\text{supercurrent}} \mathsf{F}_{+,p} + e^{-r/2} \underbrace{\bar{\mathcal{S}}_p}_{\overline{\text{supercurrent}}} \bar{\mathsf{F}}^p_+ + \underbrace{\mathcal{R}^i}_{\substack{\text{SU}(2)_{\mathfrak{R}}\\ \text{current}}} \mathsf{R}_i + e^{-r} \underbrace{\mathcal{L}}_{\substack{\text{stress}\\ \text{tensor}}} \mathsf{L}_+ + \cdots .$$

This identifies the boundary super-Schwarzian multiplet as

$$\big(\mathcal{L}, \mathcal{S}^p, \bar{\mathcal{S}}_p, \mathcal{R}^i\big).$$

Super-Schwarzian. In the $\mathcal{N}=2$ truncation one may repackage the component vev multiplet into the standard superspace super-Schwarzian. We use the flat boundary superline coordinates

$$z=(u,\vartheta,\bar{\vartheta}), \qquad D=\partial_\vartheta - i\bar{\vartheta}\,\partial_u, \qquad \bar{D}=\partial_{\bar{\vartheta}} - i\vartheta\,\partial_u, \qquad \{D,\bar{D}\}=-2i\partial_u.$$

A super-reparametrization is described by target supercoordinates

$$Z(z)=\big(X(z),\Theta(z),\bar{\Theta}(z)\big)$$

obeying the $\mathcal{N}=2$ superconformal constraints

$$D\bar{\Theta}=0, \qquad \bar{D}\Theta=0, \qquad DX+i\bar{\Theta}D\Theta=0, \qquad \bar{D}X+i\Theta\bar{D}\bar{\Theta}=0. \tag{A.13}$$

In these conventions the $\mathcal{N}=2$ super-Schwarzian used in [7, 58] is written as

$$\mathcal{S}_{\mathcal{N}=2}(Z;z)=\frac{\partial_u(\bar{D}\bar{\Theta})}{\bar{D}\bar{\Theta}}-\frac{\partial_u(D\Theta)}{D\Theta}+2i\,\frac{\partial_u\Theta\,\partial_u\bar{\Theta}}{(D\Theta)(\bar{D}\bar{\Theta})}. \tag{A.14}$$

The factor of i in the last term is a convention effect: it converts the more common convention $\{D,\bar{D}\}=2\partial_\tau$ to our convention $\{D,\bar{D}\}=-2i\partial_u$. The corresponding superspace action is therefore

$$I_{\mathcal{N}=2}=-C\int du\,d\vartheta\,d\bar{\vartheta}\,\mathcal{S}_{\mathcal{N}=2}(Z;z), \tag{A.15}$$

up to the overall normalization C. The component variables used in the particle-on-a-group calculation are the lowest components of the super-reparametrization,

$$X|=x, \qquad \Theta|=\theta_-, \qquad \bar{\Theta}|=\bar{\theta}_-, \qquad (D\Theta)(\bar{D}\bar{\Theta})|=\Pi, \qquad \left.\frac{D\Theta}{\bar{D}\bar{\Theta}}\right|=e^{-2a}, \tag{A.16}$$

where

$$\Pi=\dot{x}+\bar{\theta}_-\dot{\theta}_-+\theta_-\dot{\bar{\theta}}_-.$$

With this identification, the $\vartheta\bar{\vartheta}$ component of $\mathcal{S}_{\mathcal{N}=2}$ reproduces the component super-Schwarzian multiplet $(\mathcal{L},\mathcal{S},\bar{\mathcal{S}},\mathcal{R})$ obtained from the particle on a group, with $\mathcal{L}$ understood as the improved stress-tensor component. Thus the component calculation in the main text is equivalently the all-orders superspace $\mathcal{N}=2$ super-Schwarzian written in the derivative convention of this appendix.

The $\mathcal{N}=4$ case is parallel, except that the spinor derivative is an $SU(2)_{\mathfrak{R}}$ doublet. We use

$$z=(u,\vartheta^p,\bar\vartheta_p),\qquad D_p=\partial_{\vartheta^p}-i\bar\vartheta_p\partial_u,\qquad \bar D^p=\partial_{\bar\vartheta_p}-i\vartheta^p\partial_u,\qquad \{D_p,\bar D^q\}=-2i\delta_p{}^q\partial_u.$$

A super-reparametrization

$$Z(z)=\big(X(z),\Theta^p(z),\bar\Theta_p(z)\big)$$

obeys

$$\bar D^p\Theta^q=0,\qquad D_p\bar\Theta_q=0,\qquad D_pX+i\bar\Theta_qD_p\Theta^q=0,\qquad \bar D^pX+i\Theta^q\bar D^p\bar\Theta_q=0. \tag{A.17}$$

In the old $\mathcal{N}=4$ superspace construction [59, 60], the super-Schwarzian object is the logarithm of the spinor Jacobian,

$$\Sigma_{\mathcal{N}=4}(Z;z)=\log\det\big(D_p\Theta^q\big). \tag{A.18}$$

The constraints imply the equivalent barred and bosonic Jacobian forms, but we will not need them explicitly. The corresponding superspace action is

$$I_{\mathcal{N}=4}=-C_4\int du\,d^2\vartheta\,d^2\bar\vartheta\,\Sigma_{\mathcal{N}=4}(Z;z), \tag{A.19}$$

up to the overall normalization C_4, with $d^2\vartheta\,d^2\bar\vartheta=d\vartheta^1d\vartheta^2d\bar\vartheta_1d\bar\vartheta_2$. The component variables used in the particle-on-a-group calculation are obtained from the lowest components of the super-reparametrization and of the spinor Jacobian,

$$X|=x,\qquad \Theta^p|=\theta^p_-,\qquad \bar\Theta_p|=\bar\theta_{-,p},\qquad (D_p\Theta^q)|=\Pi^{1/2}U^q{}_p,\qquad (\bar D^p\bar\Theta_q)|=\Pi^{1/2}(U^{-1})^p{}_q, \tag{A.20}$$

where $U(u)\in SU(2)_{\mathfrak{R}}$ and

$$\Pi=\dot x+\bar\theta_{-,p}\dot\theta^p_-+\theta^p_-\dot{\bar\theta}_{-,p}.$$

Thus $\det(D_p\Theta^q)|=\Pi$, in direct analogy with $(D\Theta)(\bar D\bar\Theta)|=\Pi$ in (A.16). The same $\mathcal{N}=4$ data can also be packaged as an $SU(2)_{\mathfrak{R}}$ triplet superfield [7],

$$\begin{aligned}\mathcal{J}^i_{\rm Sch}&\sim -2D_p(\sigma^i)^p{}_q\bar D^q\,\Sigma_{\mathcal{N}=4}\\&=2\mathcal{R}^i+\vartheta^p(\sigma^i)_p{}^q\bar{\mathcal{S}}_q+\mathcal{S}^p(\sigma^i)_p{}^q\bar\vartheta_q-\vartheta^p(\sigma^i)_p{}^q\bar\vartheta_q\,\mathcal{L}+\cdots,\end{aligned} \tag{A.21}$$

up to an overall convention-dependent normalization.

B $\mathcal{N}=0$ JT, BF Theory, and Particle on a Group

This appendix gives expansion of Section 2. It is supposed to be a careful and hopefully self-contained and pedegogical derivation of the bosonic particle-on-a-group description of JT gravity.

B.1 From metric JT gravity to first-order BF variables

In this subsection we review the translation from metric formalism to the BF formalism of JT gravity, and the reason is that from the latter one can easily pass to the particle on a group description.

Metric variables. We begin with the metric formalism, or say the second-order description of Euclidean JT gravity. The fundamental fields are a two-dimensional metric $G_{\mu\nu}$ and a dilaton Φ. The spacetime M is regulated by an asymptotic curve ∂M, with coordinate u and induced line element $ds_\partial = \sqrt{h_{uu}}\, du$. The nearly-$AdS_2$ Dirichlet boundary condition fixes the leading proper length of this curve together with the leading dilaton:[24]

$$h_{uu}\big|_{\partial M} = \left(\frac{q}{\epsilon}\right)^2, \qquad \Phi\big|_{\partial M} = \frac{\phi_r^{(0)}(u)}{\epsilon}, \qquad \epsilon \to 0. \tag{B.1}$$

We take the Euclidean action to be

$$I[G,\Phi] = -\frac{1}{2}\int_M d^2X\,\sqrt{G}\,\Phi\,(R[G]+2) - \int_{\partial M} ds\,\Phi_\partial (K-1). \tag{B.2}$$

The variational problem is the standard gravitational Dirichlet problem for the regulated data in (B.1). The dilaton variation imposes the constant curvature condition

$$R + 2 = 0. \tag{B.3}$$

while the metric variation gives the dilaton equation[25]

$$\nabla_\mu\nabla_\nu\Phi - G_{\mu\nu}\nabla^2\Phi + G_{\mu\nu}\Phi = 0. \tag{B.4}$$

First-order variables. We now rewrite the same theory in first-order form in order to expose the BF structure. The independent fields are the zweibein, the Euclidean spin connection, two auxiliary zero-forms, and the physical dilaton. It is convenient to use the plus/minus frame basis

$$E^\pm = E^{\hat{u}} \pm iE^{\hat{r}}, \qquad X^\pm = X_{\hat{u}} \pm iX_{\hat{r}}, \qquad \omega = \omega_\mu dX^\mu, \qquad \Phi.$$

[24] Many JT conventions set the boundary einbein to 1 and put the physical thermal length in the range of u, for example $u \sim u + \beta$. We keep the bosonic source q of Section 2 explicit because the same number becomes the parabolic charge of [40] in the particle reduction.

[25] Tracing the equation displayed below in two dimensions gives $\nabla^2\Phi = 2\Phi$. Substitution then gives the stronger Hessian equation $\nabla_\mu\nabla_\nu\Phi = G_{\mu\nu}\Phi$. This equation has an immediate Killing-vector interpretation: if $k_\mu = \epsilon_\mu{}^\nu\nabla_\nu\Phi$, then $\nabla_{(\mu}k_{\nu)} = 0$, because the Hessian of Φ is proportional to the metric. Conversely, on a constant-curvature AdS_2 background the three Killing vectors can be generated this way from three independent solutions of the Hessian equation. The existence of these three modes is special to the maximally symmetric background: commuting derivatives on $\nabla_\rho\Phi$ in the Hessian equation requires the curvature tensor of unit Euclidean AdS_2, equivalently $R = -2$ in our convention. The three independent Hessian modes may be expanded with three coefficients, and these coefficients furnish the adjoint representation of $\mathfrak{sl}(2,\mathbb{R})$.

The metric is recovered from

$$G_{\mu\nu} = E_\mu{}^a E_\nu{}^b \eta_{ab} = \frac{1}{2}\left(E_\mu{}^+ E_\nu{}^- + E_\mu{}^- E_\nu{}^+\right). \tag{B.5}$$

We choose $\epsilon_{\hat{r}\hat{u}} = +1$.[26]

In these variables the bulk action equivalent to (B.2) is

$$I = -\int_M \left[\frac{1}{2}X^-\left(dE^+ + i\omega\wedge E^+\right) + \frac{1}{2}X^+\left(dE^- - i\omega\wedge E^-\right) + \Phi\left(d\omega + \frac{1}{2i}E^+\wedge E^-\right)\right] + I_\partial. \tag{B.6}$$

and we comment on I_∂ later.

A nice way to repackage the first order variables is to use the BF theory language, which is a gauge theory of the group $SL(2,\mathbb{R})$. We introduce two gauge fields: a connection one form which encodes the einbein [27]

$$\mathcal{A} = \left[\frac{1}{4}(E^+ + E^-) + \frac{1}{2}\omega\right]\mathsf{L}_- + \frac{1}{2i}(E^+ - E^-)\,\mathsf{L}_0 + \left[\frac{1}{4}(E^+ + E^-) - \frac{1}{2}\omega\right]\mathsf{L}_+. \tag{B.7}$$

and an adjoint zero-form which encodes the dilaton

$$\mathcal{X} = \left(\frac{\Phi}{2} + \frac{X^+ + X^-}{4}\right)\mathsf{L}_- + \frac{X^+ - X^-}{2i}\,\mathsf{L}_0 + \left(-\frac{\Phi}{2} + \frac{X^+ + X^-}{4}\right)\mathsf{L}_+. \tag{B.8}$$

Defining $\mathcal{F}[\mathcal{A}] = d\mathcal{A} + \mathcal{A}\wedge\mathcal{A}$, the bulk first-order action can be written as

$$I_{\text{BF,bulk}}[\mathcal{A},\mathcal{X}] = -\int_M \langle\mathcal{X},\mathcal{F}[\mathcal{A}]\rangle\,. \tag{B.9}$$

At this stage the equivalence is only a bulk equivalence. And the equations of motion provide the first check of the equivalence. Varying $X^\pm$ imposes following two equations which in the gravity context is the torsion-free condition

$$dE^+ + i\omega\wedge E^+ = 0, \qquad dE^- - i\omega\wedge E^- = 0. \tag{B.10}$$

The variation of Φ gives

$$d\omega + \frac{1}{2i}E^+\wedge E^- = 0. \tag{B.11}$$

After combining with torsionlessness, this equation is precisely $R + 2 = 0$.

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[26] The minimal dictionary is $T^+ = dE^+ + i\omega\wedge E^+$, $T^- = dE^- - i\omega\wedge E^-$, $R^a{}_b = d\omega^a{}_b + \omega^a{}_c\wedge\omega^c{}_b$, and $d\omega = \frac{R}{4i}E^+\wedge E^-$.

[27] This is the $\mathsf{L}_\pm,\mathsf{L}_0$ version of the standard first-order parametrization $\mathcal{A} = E^{\hat{u}}\mathsf{P}_{\hat{u}} + E^{\hat{r}}\mathsf{P}_{\hat{r}} + \omega\mathsf{J}$ and $\mathcal{X} = X_{\hat{u}}\mathsf{P}_{\hat{u}} + X_{\hat{r}}\mathsf{P}_{\hat{r}} + \Phi\mathsf{J}$. The P,J basis is defined in (A.3); we also use $E^\pm = E^{\hat{u}} \pm iE^{\hat{r}}$ and $X^\pm = X_{\hat{u}} \pm iX_{\hat{r}}$.

The variations of ω and $E^{\pm}$ give the remaining first-order equations,

$$d\Phi + \frac{1}{2i}\left(X^+E^- - X^-E^+\right) = 0, \tag{B.12}$$

and

$$dX^+ + i\omega X^+ + i\Phi E^+ = 0, \qquad dX^- - i\omega X^- - i\Phi E^- = 0. \tag{B.13}$$

The equation obtained from ω shows that $X^{\pm}$ are NOT new local degrees of freedom. They are just the plus/minus frame components of the dilaton gradient, $X^- = 2iE_+{}^{\mu}\partial_\mu\Phi$ and $X^+ = -2iE_-{}^{\mu}\partial_\mu\Phi$. Substituting these expressions into (B.13) reproduces the metric equation (B.4).[28] This verifies that the first-order system carries the same local content as second-order JT.

Gibbons–Hawking term and the path to the BF boundary action. We now discuss the boundary condition and boundary term. The metric problem fixes the induced line element and the leading physical dilaton, while the BF variation uses the connection $\mathcal{A}$ and the full adjoint zero-form $\mathcal{X}$. In a boundary-adapted orthonormal frame, the JT boundary data take the form

$$E^{\hat{u}}\big|_{\partial M} = ds = E_u{}^{\hat{u}}du \sim \frac{e_u^{(0)}(u)}{\epsilon}\,du, \qquad E^{\hat{r}}\big|_{\partial M} = 0, \qquad \Phi_\partial \sim \frac{\phi_r^{(0)}(u)}{\epsilon}.$$

These data fix the leading ratio $\Phi_\partial/E_u{}^{\hat{u}}$. In Fefferman–Graham gauge the same boundary-adapted frame also fixes the boundary connection. Using (B.7) and $E_u^+ = E_u^- = E_u{}^{\hat{u}}$, the coefficients of $\mathcal{A}_u$ along L_-, L_0, and L_+ are $(E_u{}^{\hat{u}}+\omega_u)/2$, 0, and $(E_u{}^{\hat{u}}-\omega_u)/2$, respectively.[29] For the zero-form, however, the metric condition fixes only the dilaton component Φ_∂. It does not by itself specify the boundary values of $X^{\pm}$, which are auxiliary fields imposing torsion in the bulk rather than additional metric JT sources. This is the missing boundary input that the BF boundary action must supply.

We can get some clue by looking at the boundary action. The JT boundary term is the Gibbons–Hawking–York term, and in first-order variables it becomes one part of the BF boundary term,

$$-\int_{\partial M} ds\,\Phi K = -\int_{\partial M}\Phi\,\omega = -\int du\,\Phi\,\omega_u. \tag{B.14}$$

The full BF boundary term is suggested by the bulk variation

$$\delta I_{\mathrm{BF,bulk}} = \text{bulk EoM} - \int_{\partial M} du\,\langle\mathcal{X}, \delta\mathcal{A}_u\rangle\,, \tag{B.15}$$

[28] Using torsionlessness, $E^{\pm}$ are covariantly constant, and the two $E^{\pm}$ equations become the frame components of $\nabla_\mu\nabla_\nu\Phi = G_{\mu\nu}\Phi$. Taking the trace gives $\nabla^2\Phi = 2\Phi$, hence (B.4).

[29] On the cutoff curve the spin connection pulls back as $\omega|_{\partial M} = K\,ds$, so $\omega_u = KE_u{}^{\hat{u}}$.

Since (B.15) contains $-\int_{\partial M}\langle \mathcal{X}, \delta\mathcal{A}_u\rangle$, the natural BF boundary term is

$$\int_{\partial M}\langle \mathcal{X}, \mathcal{A}_u\rangle = \int_{\partial M}\left(\frac{1}{2}X^- E_u^+ + \frac{1}{2}X^+ E_u^- - \Phi\,\omega_u\right). \tag{B.16}$$

It contains the Gibbons–Hawking–York term in (B.14), but also the auxiliary pieces $X^- E_u^+$ and $X^+ E_u^-$; as emphasized above, the metric boundary condition does not fix $X^\pm$ as independent sources. Moreover, although this term cancels the $\delta\mathcal{A}_u$ variation from the bulk, its own variation leaves $\int_{\partial M}\langle \mathcal{A}_u, \delta\mathcal{X}\rangle$. This means $I_\partial = \int_{\partial M}\langle \mathcal{A}_u, \delta\mathcal{X}\rangle$ can not be the end of the story, because the $\delta\mathcal{X} = \prime$ condition is just too strong.[30]

To avoid this, one can add an additional boundary term which only contains $\mathcal{X}$ like $H(\langle \mathcal{X}, \mathcal{X}\rangle)$. This is the remaining input for BF theory that we mentioned above. Therefore the boundary action is

$$I_\partial = \int_{\partial M} du\ \langle \mathcal{X}, \mathcal{A}_u\rangle + \int_{\partial M} du\, H(\langle \mathcal{X}, \mathcal{X}\rangle)\,. \tag{B.17}$$

At this stage H is not fixed by the Gibbons–Hawking–York term alone. It is the additional boundary input which tells us how $\mathcal{X}$, including $X^\pm$, is tied to the boundary connection. Different BF boundary theories are specified by different boundary Hamiltonians. Below we derive the JT choice, which is just $\frac{1}{2\phi_r}\langle \mathcal{X}, \mathcal{X}\rangle$, from the mixed boundary condition, obtaining (B.25), and then use it to obtain the particle-on-a-group action.

B.2 From BF theory to the reduced particle on a group

To see what BF boundary term that we need to introduce, we want to dive into the boundary dynamics. Here we follow the the standard AdS/CFT boundary paradigm: the coefficient of the growing piece, which is usually called source, is fixed, while the decaying piece, which is usually called vev, is allowed to fluctuate. To determine them, we need to solve the bulk equations of motion and find the radial solutions.

Bulk equations and radial solutions. The BF equations take the compact form

$$\mathcal{F}[\mathcal{A}] = 0, \qquad D_{\mathcal{A}}\mathcal{X} = 0. \tag{B.18}$$

Here $D_{\mathcal{A}}\mathcal{X} = d\mathcal{X} + [\mathcal{A}, \mathcal{X}]$. We impose radial gauge $\mathcal{A}_r = \mathsf{L}_0$, which is weaker than Fefferman–Graham gauge. The ru component of the flatness condition, $\partial_r\mathcal{A}_u + [\mathsf{L}_0, \mathcal{A}_u] = 0$, then fixes the radial weights without fixing the boundary representative. In the simple constant-source representative used below this gives

$$\mathcal{A} = e^r e_u^{(0)}(u)\,\mathsf{L}_-\,du + \mathsf{L}_0\,dr + e^{-r}t(u)\,\mathsf{L}_+\,du. \tag{B.19}$$

[30] If one literally solve $\delta\mathcal{X} = \prime$, the out come is the same as $\partial_u x, u = 0$ where x, u is the Schwarzian derivative.

Using (B.7), the same solution translates back to the first-order gravity fields as

$$\omega = \left(e^r e_u^{(0)} - e^{-r}t\right) du, \qquad \frac{E^+ + E^-}{2} = \left(e^r e_u^{(0)} + e^{-r}t\right) du, \qquad \frac{E^+ - E^-}{2} = i\, dr. \tag{B.20}$$

The radial equation for the zero-form, obtained from $D_{\mathcal{A}}\mathcal{X} = 0$, fixes the same $e^r, 1, e^{-r}$ hierarchy for $\mathcal{X}$. This hierarchy is the source/vev split: the leading $e^r \mathsf{L}_-$ coefficient is fixed source data, while the $e^{-r}\mathsf{L}_+$ coefficient is a normalizable vev mode. In the constant-source JT ensemble we set $e_u^{(0)} = q$ and fix $\chi_-/e_u^{(0)} = \phi_r$, equivalently $\chi_- = q\phi_r$. If we then use only the leading and finite parts of the u-component of $D_{\mathcal{A}}\mathcal{X} = 0$, the same representative gives $\chi_0 = 0$ and $\chi_+ = \phi_r t(u)$. The off-shell asymptotic fields are therefore

$$\mathcal{A}_u(r,u) = e^r q\, \mathsf{L}_- + e^{-r}t(u)\, \mathsf{L}_+, \qquad \mathcal{X}(r,u) = e^r q\phi_r\, \mathsf{L}_- + e^{-r}\phi_r t(u)\, \mathsf{L}_+. \tag{B.21}$$

We mention that Equation (B.21) represents off-shell asymptotic data, not a full solution of all BF equations. The leading and finite parts of $D_u\mathcal{X} = 0$ identify the fixed source and its conjugate vev data, while the remaining decaying L_+ component gives the equation of motion for $t(u)$, which is actually the Schwarzian equation of motion $\dot{t}(u) = \partial_u\{x,u\} = 0$. We do not impose this last equation in the off-shell Dirichlet problem; the vev coefficient $t(u)$ is path integrated over and its equation of motion is generated by the particle-on-a-group action.[31]

JT mixed boundary condition and BF Hamiltonian. We can now determine the boundary condition for $\mathcal{X}$. In the constant-source representative it is tied to the boundary connection by

$$\mathcal{X}(r,u)|_{\partial M} = \phi_r\ \mathcal{A}_u(r,u)|_{\partial M}\,, \qquad \delta\mathcal{X}(r,u)|_{\partial M} = \phi_r\ \delta\mathcal{A}_u(r,u)|_{\partial M}\,. \tag{B.22}$$

The corresponding first-order BF boundary action should keep $\mathcal{X}$ as the boundary momentum:

$$I[\mathcal{A},\mathcal{X}] = -\int_M \langle \mathcal{X}, \mathcal{F}[\mathcal{A}]\rangle + \int_{\partial M} du\, \left[\langle \mathcal{X}, \mathcal{A}_u\rangle - \frac{1}{2\phi_r}\langle \mathcal{X},\mathcal{X}\rangle\right]. \tag{B.23}$$

The coefficient of the quadratic term is fixed by the variational principle.

$$\delta I|_{\partial M} = \int_{\partial M} du\, \left\langle \mathcal{A}_u - \frac{1}{\phi_r}\mathcal{X}, \delta\mathcal{X}\right\rangle = 0. \tag{B.24}$$

This vanishes precisely on the mixed JT boundary condition $\mathcal{X} = \phi_r \mathcal{A}_u$. Equivalently,

$$H(\langle \mathcal{X},\mathcal{X}\rangle) = -\frac{1}{2\phi_r}\langle \mathcal{X},\mathcal{X}\rangle \tag{B.25}$$

[31] In the shifted radial representative $\mathcal{A} = e^r e_u^{(0)}\mathsf{L}_- du + \mathsf{L}_0(dr + s\, du) + e^{-r}t\mathsf{L}_+ du$, the leading and finite parts of $D_u\mathcal{X} = 0$ determine χ_0, χ_+. For $e_u^{(0)} = q$ and $\chi_- = q\phi_r$, the result is $\chi_0 = \phi_r s$ and $\chi_+ = \phi_r t + (\phi_r/2q)\dot{s}$.

up to an additive constant.

This form also clarifies the role of the auxiliary fields in the path integral. The fields $X^{\pm}$ imposed torsion in the first-order gravity action; in the BF boundary action they remain auxiliary variables should be integrated over, not fixed as extra sources. Since $\mathcal{X}$ appears without derivatives and only quadratically, it is a Gaussian integral but with non-trivial peak centered at $\mathcal{X} = \phi_r \mathcal{A}_u$. Integrating it out gives the equivalent $\mathcal{A}$-only boundary term

$$I_\partial = \frac{\phi_r}{2} \int_{\partial M} du \, \langle \mathcal{A}_u, \mathcal{A}_u \rangle \,. \tag{B.26}$$

Particle on a group I (path integral): action and constraints from Lagrange multiplier. We now pass from the BF formulasion to the particle on a group formalism. The bulk equation $\mathcal{F}[\mathcal{A}] = 0$ is solved locally by writing the connection as a pure gauge. If one also imposed the zero-form equation completely, the solution would take the form

$$\mathcal{A} = \mathcal{G}^{-1} d\mathcal{G}, \qquad \mathcal{X} = \mathcal{G}^{-1} \mathcal{X}_0 \mathcal{G}, \qquad d\mathcal{X}_0 = 0. \tag{B.27}$$

Here $\mathcal{G}$ is a group valued field on AdS_2 which will become the particle-on-a-group variable later. The above is true for any flat connection, but it does not yet define the JT boundary theory.[32] The JT boundary theory requires the additional asymptotic structure, meaning $\mathcal{A}$ can not be any flat conection, so that $\mathcal{G}$ can not be any group-valued funciton.

Now let us make it a bit more precise. For the off-shell path integral we therefore impose flatness and the JT source condition on $\mathcal{A}$, but not the final decaying component of $D_{\mathcal{A}} \mathcal{X} = 0$. Thus flatness allows us to write the boundary connection as $\mathcal{A}_u = \mathcal{G}^{-1} \partial_u \mathcal{G}$.[33] Substituting this into the first-order boundary action gives

$$I_{\mathrm{pG}}[\mathcal{G}, \mathcal{X}] = \int du \, \left[\left\langle \mathcal{X}, \mathcal{G}^{-1} \partial_u \mathcal{G} \right\rangle - \frac{1}{2\phi_r} \langle \mathcal{X}, \mathcal{X} \rangle \right]. \tag{B.28}$$

The algebraic equation for $\mathcal{X}$ is $\mathcal{X} = \phi_r \mathcal{G}^{-1} \partial_u \mathcal{G}$. Equivalently, in terms of the normalized particle momentum $\mathcal{P} \equiv \frac{\mathcal{X}}{\phi_r}$, the Schwarzian coupling ϕ_r multiplies the whole first-order particle action,

$$I_{\mathrm{pG}}[\mathcal{G}, \mathcal{P}] = \phi_r \int du \, \left[\left\langle \mathcal{P}, \mathcal{G}^{-1} \partial_u \mathcal{G} \right\rangle - \frac{1}{2} \langle \mathcal{P}, \mathcal{P} \rangle \right]. \tag{B.29}$$

Integrating out $\mathcal{X}$, or equivalently $\mathcal{P}$, gives the quadratic current action

$$I_{\mathrm{pG}}[\mathcal{G}] = \frac{\phi_r}{2} \int du \, \left\langle \mathcal{G}^{-1} \dot{\mathcal{G}}, \mathcal{G}^{-1} \dot{\mathcal{G}} \right\rangle. \tag{B.30}$$

[32] The constant adjoint element $\mathcal{X}_0$ has three components. A rigid $SL(2, \mathbb{R})$ conjugation removes two of them on a generic orbit, leaving only the invariant Casimir $\langle \mathcal{X}_0, \mathcal{X}_0 \rangle$. This is why the on-shell phase space can be parametrized by the JT scale ϕ_r and its conjugate boundary variable, rather than by three independent zero-form sources.

[33] The group current $\mathcal{G}^{-1} d\mathcal{G}$ is standard in particle actions on group manifolds and related coadjoint-orbit constructions; see [81, 82].

Here there is no constraints on the particle yet, and the constraint comes from to impose the same asymptotic source condition on the group current. In the constant-source FG representative, after removing the radial weights, the current is constrained to have the form $\mathcal{G}^{-1}\partial_u\mathcal{G} = q\mathsf{L}_- + t(u)\mathsf{L}_+$, with fixed source q and fluctuating vev coefficient $t(u)$. In a more general gauge one writes $\mathcal{G}^{-1}\partial_u\mathcal{G} = e_u^{(0)}\mathsf{L}_- + s\mathsf{L}_0 + t\mathsf{L}_+$. The constant-source FG representative sets $e_u^{(0)} = q$ and $s = 0$. The scalar source fixes $\chi_- = q\phi_r$, while $t(u)$ remains off-shell vev data. In invariant language, fixing q is a genuine charge constraint, while $s = 0$ is a choice of gauge slice rather than an additional boundary charge.

It is useful to check that (B.30) contains the usual Schwarzian action. Use the Gauss parameterization

$$\mathcal{G}(u) = e^{x(u)\mathsf{L}_-}e^{\rho(u)\mathsf{L}_0}e^{\gamma(u)\mathsf{L}_+}. \tag{B.31}$$

The worldline group current is

$$\mathcal{G}^{-1}\dot{\mathcal{G}} = e^{\rho}\dot{x}\,\mathsf{L}_- + (\dot{\rho} - 2\gamma e^{\rho}\dot{x})\,\mathsf{L}_0 + \left(\dot{\gamma} + \gamma\dot{\rho} - \gamma^2 e^{\rho}\dot{x}\right)\mathsf{L}_+. \tag{B.32}$$

Now impose the constant-source FG form of the group current reading from (B.21),

$$\mathscr{J}(u) \equiv \mathcal{G}^{-1}\dot{\mathcal{G}} = q\mathsf{L}_- + t(u)\mathsf{L}_+. \tag{B.33}$$

Comparing the three components of (B.32) gives

$$e^{\rho} = \frac{q}{\dot{x}}, \qquad \gamma = \frac{\dot{\rho}}{2q} = -\frac{\ddot{x}}{2q\dot{x}}, \qquad t(u) = -\frac{1}{2q}\left[\frac{\dddot{x}}{\dot{x}} - \frac{3}{2}\left(\frac{\ddot{x}}{\dot{x}}\right)^2\right] = -\frac{1}{2q}\{x,u\}. \tag{B.34}$$

Here $\{x,u\}$ is the Schwarzian derivative, and the sign in $e^{\rho} = q/\dot{x}$ follows from the orientation convention in (B.31). Since

$$\langle q\mathsf{L}_- + t\mathsf{L}_+, q\mathsf{L}_- + t\mathsf{L}_+\rangle = 4qt,$$

the quadratic particle action becomes

$$I_{\mathrm{pG}} = -\phi_r\int du\,\{x,u\}, \tag{B.35}$$

with the Euclidean sign convention inherited from (B.23).

Note that the above is at action level, now we quantize the theory. For the path integral, and later for canonical quantization, we use the Lagrange multiplier to impose teh following constraint

$$\langle\mathcal{P},\mathsf{L}_+\rangle = q, \qquad \text{or equivalently} \qquad \langle\mathcal{X},\mathsf{L}_+\rangle = \phi_r q. \tag{B.36}$$

We introduce a Lagrange multiplier, $\lambda(u)$ to the action and integrate over all $\lambda(u)$

$$I[\mathcal{G},\mathcal{P},\lambda] = \phi_r\int du\,\left[\left\langle\mathcal{P},\mathcal{G}^{-1}\partial_u\mathcal{G} - \lambda\mathsf{L}_+\right\rangle - \frac{1}{2}\langle\mathcal{P},\mathcal{P}\rangle + q\lambda\right]. \tag{B.37}$$

Interestingly, the same Lagrange multiplier also keeps track of the endpoint character of the reduced wavefunction. Here $\mathcal{G}$ is still the unreduced $SL(2,\mathbb{R})$ group variable. The subgroup being divided out is the right $N_+ = \exp(\mathbb{R}\mathsf{L}_+)$ direction; the reduced configuration space is the right quotient $SL(2,\mathbb{R})/N_+$, not a separate quotient group. Local right N_+ transformations which vanish at the endpoints are gauge redundancies of (B.37). If the right transformation has nonzero endpoint values, the action changes by a boundary term, which is precisely the character carried by the endpoint states. Explicitly,

$$\mathcal{G} \to \mathcal{G}e^{\alpha\mathsf{L}_+}, \qquad \mathcal{P} \to e^{-\alpha\mathsf{L}_+}\mathcal{P}e^{\alpha\mathsf{L}_+}, \qquad \lambda \to \lambda + \dot{\alpha},$$

[34] shifts the action only by the fixed charge term:

$$I[\mathcal{G}e^{\alpha\mathsf{L}_+}, e^{-\alpha\mathsf{L}_+}\mathcal{P}e^{\alpha\mathsf{L}_+}, \lambda + \dot{\alpha}] = I[\mathcal{G}, \mathcal{P}, \lambda] + \phi_r q\big[\alpha_f - \alpha_i\big]. \tag{B.38}$$

Equivalently, the bracketed normalized action shifts by $q[\alpha_f - \alpha_i]$; the full action is obtained by multiplying this normalized action by ϕ_r. This shows that the path integral produces an N_+-equivariant wavefunction on the original $SL(2,\mathbb{R})$ variable, and we will see later this leads to a quotient by right N_+ imposed in the measure and inner product.

Just from the equalvariance property, the propagator should obey

$$K_q(\mathcal{G}_f e^{\alpha_f\mathsf{L}_+}; \mathcal{G}_i e^{\alpha_i\mathsf{L}_+}) = e^{-q\alpha_f}\, K_q(\mathcal{G}_f; \mathcal{G}_i)\, e^{q\alpha_i}. \tag{B.39}$$

Equivalently, a final-state wavefunction transforms as

$$\Psi(\mathcal{G}e^{\alpha\mathsf{L}_+}) = e^{-q\alpha}\Psi(\mathcal{G}),$$

This is the Lagrangian origin of the constraint $L_+^R\Psi = -q\Psi$ later appearing in the Hamiltonian formalism.

Particle on a group II (canonical quantization): reduced Hamiltonian and left invariant right equalvarient states We now give the Hamiltonian version of the same reduction, following the conventions of [40]. We keep the Gauss parameterization (B.31) and the group current (B.32). Substituting (B.32) into the canonical one-form in (B.29), we obtain

$$\begin{aligned} p_x &= e^{\rho}\left[\langle\mathcal{P}, \mathsf{L}_-\rangle - 2\gamma\langle\mathcal{P}, \mathsf{L}_0\rangle - \gamma^2\langle\mathcal{P}, \mathsf{L}_+\rangle\right], \\ p_\rho &= \langle\mathcal{P}, \mathsf{L}_0\rangle + \gamma\langle\mathcal{P}, \mathsf{L}_+\rangle, \qquad p_\gamma = \langle\mathcal{P}, \mathsf{L}_+\rangle. \end{aligned} \tag{B.40}$$

[34] Right and left multiplication are different here. For right multiplication $\delta\mathcal{G} = \mathcal{G}T$, so $\mathcal{G}^{-1}\delta\mathcal{G} = T$ and the normalized canonical pairing sees $\langle\mathcal{P}, T\rangle$. For left multiplication $\delta\mathcal{G} = T\mathcal{G}$, so $\mathcal{G}^{-1}\delta\mathcal{G} = \mathcal{G}^{-1}T\mathcal{G}$ and the corresponding normalized charge is $\langle\mathcal{G}\mathcal{P}\mathcal{G}^{-1}, T\rangle$. The parabolic reduction of [40] uses the right action generated by L_+.

Solving these equations gives the right dual momenta in terms of the canonical coordinates:

$$\langle \mathcal{P}, \mathsf{L}_+ \rangle = p_\gamma, \qquad \langle \mathcal{P}, \mathsf{L}_0 \rangle = p_\rho - \gamma p_\gamma, \qquad \langle \mathcal{P}, \mathsf{L}_- \rangle = e^{-\rho} p_x + 2\gamma p_\rho - \gamma^2 p_\gamma. \tag{B.41}$$

This derivation uses the right covariant derivatives. Meaning

$$p_\gamma \mathcal{G} = \mathcal{G} \mathsf{L}_+ \tag{B.42}$$

why the right not the left we will explain below when we talk about detailed definition of states.

The Hamiltonian is obtained by an ordinary Legendre transform. The first term in the normalized particle Lagrangian, $\langle \mathcal{P}, \mathcal{G}^{-1}\dot{\mathcal{G}} \rangle$, is precisely the canonical one-form $p_x \dot{x} + p_\rho \dot{\rho} + p_\gamma \dot{\gamma}$. Therefore

$$H = p_x \dot{x} + p_\rho \dot{\rho} + p_\gamma \dot{\gamma} - L = \frac{1}{2} \langle \mathcal{P}, \mathcal{P} \rangle , \tag{B.43}$$

and in canonical quantization, we promte everything to operators. Just to comment, the actual JT Hamiltonian carries the overall Schwarzian coupling, $\widehat{H}_{\rm JT} = (2\phi_r)^{-1} \langle \widehat{\mathcal{P}}, \widehat{\mathcal{P}} \rangle$, with the ordering chosen so that the quadratic operator is the $SL(2,\mathbb{R})$ Casimir. For the present purpose we only need the ordered quadratic operator in the Gauss patch,

$$\widehat{C} = L_0^2 - \frac{1}{2} (L_+ L_- + L_- L_+) = \partial_\rho^2 + \partial_\rho - e^{-\rho} \partial_x \partial_\gamma. \tag{B.44}$$

Up to the overall factor $1/(2\phi_r)$, and the usual possible additive constant, this is the Hamiltonian of the unreduced particle on $SL(2,\mathbb{R})$.

We now impose the constraint, which is $p_\gamma = \langle \mathcal{P}, \mathsf{L}_+ \rangle$. In the Euclidean conventions of [40], this is

$$L_+^R \Psi = -q \Psi, \qquad \Psi(x, \rho, \gamma) = e^{-q\gamma} F(x, \rho). \tag{B.45}$$

Substitution of $\partial_\gamma \to -q$ in (B.44) then gives the reduced operator

$$\widehat{C}_q = \partial_\rho^2 + \partial_\rho + q e^{-\rho} \partial_x. \tag{B.46}$$

The physical reduced JT Hamiltonian is $\widehat{H}_{{\rm JT},q} = (2\phi_r)^{-1} \widehat{C}_q$. This reproduces the operator of [40] from the Lagrangian construction, with ϕ_r appearing only as the overall Schwarzian coupling.

We next fix the convention for states, since the same right action appears both in the wavefunction constraint and in the transformation of boundary propagators. Choose a reference state, or a set of reference states $|i\rangle$, and define

$$|\mathcal{G}, i\rangle = U(\mathcal{G}) |i\rangle.$$

This definition does not rely on any quotient or reduction; it is simply how the group-labeled states are

introduced. Left multiplication acts directly on the group label,

$$U(a)|\mathcal{G}, i\rangle = |a\mathcal{G}, i\rangle.$$

Right multiplication is different because the factors multiply inside the same representation. If h acts on the chosen reference states by $\rho(h)$ (namely ρ a representation), then

$$|\mathcal{G}h, i\rangle = U(\mathcal{G})U(h)|i\rangle = |\mathcal{G}, j\rangle\, \rho(h)^j{}_i.$$

Thus right multiplication should be intentionally separated from a left action such as $U(h)|\mathcal{G}, i\rangle = |\mathcal{G}h, i\rangle$. It acts on the reference-state labels instead.

The state transformation is reflected in wavefunction. For example, take $h_\alpha = e^{\alpha \mathsf{L}_+}$

$$U(h_\alpha)|q\rangle = e^{q\alpha}|q\rangle, \qquad \langle q|U(h_\alpha)^{-1} = e^{-q\alpha}\langle q|.$$

With $\langle \mathcal{G}; q| = \langle q|U(\mathcal{G})^{-1}$, the wavefunction obeys[35]

$$\Psi_q(\mathcal{G}e^{\alpha \mathsf{L}_+}) = e^{-q\alpha}\Psi_q(\mathcal{G}), \qquad \Psi_q(\mathcal{G}) = \langle \mathcal{G}; q|\Psi\rangle, \tag{B.47}$$

which is the finite form of $L^R_+\Psi_q = -q\Psi_q$ so that

$$\Psi_q(x, \rho, \gamma) = e^{-q\gamma}F(x, \rho). \tag{B.48}$$

These are the reduced wavefunctions for particle on a group.

From the propagator, we can now understand the difference between left and right action better, and the slogan is that the propagator is left invariant and right equalvariant. The global $SL(2, \mathbb{R})$ symmetry acts from the left on these states, $U(a)|\mathcal{G}; q\rangle = |a\mathcal{G}; q\rangle$.[36] The Euclidean propagator is

$$K_q(\mathcal{G}_f, \mathcal{G}_i; T) = \langle \mathcal{G}_f; q|e^{-T\widehat{H}}|\mathcal{G}_i; q\rangle = \langle q|U(\mathcal{G}_f)^{-1}e^{-T\widehat{H}}U(\mathcal{G}_i)|q\rangle.$$

Because the Hamiltonian is the Casimir, it commutes with the left action:

$$[\widehat{H}, U(a)] = 0, \qquad U(a)^{-1}e^{-T\widehat{H}}U(a) = e^{-T\widehat{H}}.$$

[35] In general, wavefunctions transform as

$$\Psi_i(\mathcal{G}h) = \big(\rho(h)^{-1}\big)_i{}^j\Psi_j(\mathcal{G}).$$

This statement is more general than the JT example. For example, reference states carrying $SU(2)_{\mathfrak{R}}$ indices transform this way under right $SU(2)_{\mathfrak{R}}$ multiplication even when $SU(2)_{\mathfrak{R}}$ is not being reduced.

[36] Infinitesimal left multiplication in these coordinates gives $L_- = \partial_x$, $L_0 = -x\partial_x + \partial_\rho$, and $L_+ = x^2\partial_x - 2x\partial_\rho + e^{-\rho}\partial_\gamma$, up to the overall sign convention.

Therefore simultaneous left multiplication gives

$$K_q(a\mathcal{G}_f, a\mathcal{G}_i; T) = \langle q|U(\mathcal{G}_f)^{-1}U(a)^{-1}e^{-T\widehat{H}}U(a)U(\mathcal{G}_i)|q\rangle = K_q(\mathcal{G}_f, \mathcal{G}_i; T). \tag{B.49}$$

On the other hand, right multiplication is instead read through the transformation of the reference states. For a multiplet of reference states the propagator is matrix-valued,

$$K_{ij}(\mathcal{G}_f, \mathcal{G}_i; T) = \langle \mathcal{G}_f, i|e^{-T\widehat{H}}|\mathcal{G}_i, j\rangle,$$

and right multiplication inserts the corresponding ρ matrices at the endpoints,

$$K_{ij}(\mathcal{G}_f h_f, \mathcal{G}_i h_i; T) = \rho(h_f^{-1})_i{}^k\, K_{kl}(\mathcal{G}_f, \mathcal{G}_i; T)\, \rho(h_i)^l{}_j.$$

For the one-dimensional N_+ character this becomes

$$\begin{aligned} K_q(\mathcal{G}_f e^{\alpha_f \mathsf{L}_+}, \mathcal{G}_i e^{\alpha_i \mathsf{L}_+}; T) &= \langle q|U(e^{\alpha_f \mathsf{L}_+})^{-1}U(\mathcal{G}_f)^{-1}e^{-T\widehat{H}}U(\mathcal{G}_i)U(e^{\alpha_i \mathsf{L}_+})|q\rangle \\ &= e^{-q\alpha_f}\, K_q(\mathcal{G}_f, \mathcal{G}_i; T)\, e^{q\alpha_i}. \end{aligned} \tag{B.50}$$

This is the Hilbert-space form of the endpoint character derived in (B.39).

Inner product and reduced measure. It remains to explain which measure should be used for these equivariant wavefunctions, so that we can define inner product and Hilbert space.

We claim that the physical inner product must divide by the right N_+ gauge volume. This is most transparent in the path-integral derivation. After the Gaussian integral over the normalized momentum, the boundary path integral is over group trajectories and the Lagrange multiplier:

$$\mathcal{D}\mu_{\text{naive}} = \prod_u d\mathcal{G}(u)\, d\lambda(u), \qquad d\mathcal{G} = e^{\rho} dx\, d\rho\, d\gamma, \tag{B.51}$$

Here $d\mathcal{G}$ is the Haar measure on $SL(2,\mathbb{R})$ in the Gauss patch. This is not yet the physical measure: it still contains the local right N_+ redundancy. That action is invariant under the local right transformation

$$\mathcal{G}(u) \to \mathcal{G}(u)e^{\alpha(u)\mathsf{L}_+}, \qquad \gamma(u) \to \gamma(u) + \alpha(u), \qquad \lambda(u) \to \lambda(u) + \dot{\alpha}(u).$$

Thus γ and λ should not be counted as two independent physical functions: the quotient removes one arbitrary function $\alpha(u)$. The group current changes nontrivially under the same right shift:

$$\mathcal{G}^{-1}\partial_u\mathcal{G} \longrightarrow e^{-\alpha\mathsf{L}_+}\left(\mathcal{G}^{-1}\partial_u\mathcal{G}\right)e^{\alpha\mathsf{L}_+} + \dot{\alpha}\,\mathsf{L}_+.$$

This is the redundancy that must be removed from the path integral.[37] The quotient form of the history measure is therefore

$$\mathcal{D}\mu_{\rm phys} = \frac{\prod_u d\mathcal{G}(u)\, d\lambda(u)}{\mathrm{Vol}(\mathcal{H}_+)}, \qquad \mathcal{H}_+ = \{e^{\alpha(u)\mathsf{L}_+}\}. \tag{B.52}$$

Equivalently, one may choose a gauge slice for this quotient. Let us consider a more general gauge for the current $\mathscr{J}(u) = \mathcal{G}^{-1}\dot{\mathcal{G}} = A\mathsf{L}_- + s\mathsf{L}_0 + t\mathsf{L}_+$, the right transformation gives $A \to A$ and $s \to s - 2A\alpha$. On the charge surface $A = q$, the condition $s(u) = 0$ is a good gauge slice in the sense that whatever other gauge can always be brought back to this gauge. Indeed, in Gauss coordinates $s = \dot{\rho} - 2\gamma e^{\rho}\dot{x} = \dot{\rho} - 2A\gamma$, with $A = e^{\rho}\dot{x}$, so $s = 0$ fixes $\gamma = \dot{\rho}/(2A)$, or $\gamma = \dot{\rho}/(2q)$ after imposing $A = q$.[38]

$$\frac{\mathcal{D}\mathcal{G}\,\mathcal{D}\lambda}{\mathrm{Vol}(\mathcal{H}_+)} = \mathcal{D}\mathcal{G}\,\mathcal{D}\lambda\,\delta[s]\,\Delta_{\rm FP}, \qquad \Delta_{\rm FP} = \det\left(\frac{\delta s^{\alpha}}{\delta\alpha}\right) = \det(-2A). \tag{B.53}$$

On the support of the charge constraint $A = q$, this determinant is the constant $\det(-2q)$, which is why it is invisible in the final bosonic Schwarzian action. The two common orders of integration are equivalent: gauge fixing supplies $\delta[s]\Delta_{\rm FP}$, while the λ-integral supplies $\delta[A-q]$, since L_+ is null and λ appears linearly after $\mathcal{P}$ is eliminated. In either order one reaches

$$\int \mathcal{D}\mathcal{G}\,\delta[A-q]\,\delta[s]\,\Delta_{\rm FP}\,e^{-I_{\rm pG}[\mathcal{G}]}. \tag{B.54}$$

Thus $\delta[A-q]$ comes from the Lagrange multiplier, whereas $\delta[s]\Delta_{\rm FP}$ is the gauge-fixed form of the quotient by $\mathcal{H}_+$.

For the Hilbert-space inner product, one can directly obtain it from from the reduced path integral measure at a time slice.

$$d\mu_{\rm phys} = \frac{d\mathcal{G}}{d\gamma} = e^{\rho}\,dx\,d\rho. \tag{B.55}$$

The corresponding position-space resolution of the identity is

$$\mathbf{1}_q = \int dx\,d\rho\,e^{\rho}\,|x,\rho;q\rangle\langle x,\rho;q|, \tag{B.56}$$

and the physical inner product for reduced wavefunctions $F(x,\rho)$ is

$$\langle F_1, F_2\rangle_{\rm phys} = \int dx\,d\rho\,e^{\rho}\,\overline{F_1(x,\rho)}F_2(x,\rho). \tag{B.57}$$

[37] For a boundary interval with specified endpoints, the pure gauge transformations are the shifts with α vanishing at the endpoints. The endpoint values themselves are kept as external transformations on the endpoint states and give the characters in (B.39). On a closed boundary, all time-dependent shifts are divided out.

[38] We can also think about this from the constraint point of view. $s = 0$ is not a second momentum-map constraint; it is the gauge-fixed representation of the quotient by $\mathcal{H}_+$. We are dealing with the first class constraint, and usually a first class constraint removes two degrees of freedom in the phase space. The $A = q$ condition freeze A, and the $s = 0$ gauge choice freezes the other.

In unreduced notation the same statement can be written as

$$\langle \Psi_1, \Psi_2 \rangle_{\text{phys}} \sim \frac{1}{\text{Vol}(N_+)} \int_G d\mathcal{G}\, \overline{\Psi_1(\mathcal{G})} \Psi_2(\mathcal{G}).$$

This completes the $\mathcal{N} = 0$ discussion of the particle-on-a-group construction.

B.3 Matter Green's functions and bilocals

We now add a matter field to the same $\mathcal{N} = 0$ BF background. A bulk Green's function is the two-point function for this matter field, so it must solve the matter equation of motion in each argument. In the BF language a matter field of dimension Δ transforms in a representation $\mathcal{R}_\Delta$, and the linear equation of motion is

$$D_{\mathcal{A}} \Phi_\Delta = d\Phi_\Delta + \mathcal{R}_\Delta(\mathcal{A}) \Phi_\Delta = 0.$$

The natural group-language candidate for the Green's function is therefore the Wilson line [37, 43, 83],

$$\mathcal{U}_\Delta(i, j) = \mathrm{P} \exp\left[- \int_j^i \mathcal{R}_\Delta(\mathcal{A}) \right]. \tag{B.58}$$

On a BF saddle the connection is flat, so locally $\mathcal{A} = \mathcal{G}^{-1} d\mathcal{G}$. Then the Wilson line can be written only in terms of the endpoint group elements $\mathcal{G}_i, \mathcal{G}_j$ as[39]

$$\mathcal{U}_\Delta(i, j) = \mathcal{R}_\Delta(\mathcal{G}_i^{-1} \mathcal{G}_j). \tag{B.59}$$

This endpoint formula has exactly the covariance expected of a matter Green's function. Under a local $SL(2, \mathbb{R})$ gauge transformation h, with $\mathcal{G} \to \mathcal{G}h$, it transforms as

$$\mathcal{U}_\Delta(i, j) \longrightarrow \mathcal{R}_\Delta(h_i)^{-1} \mathcal{U}_\Delta(i, j) \mathcal{R}_\Delta(h_j).$$

A matter state at endpoint k transforms as

$$|\Phi_{\Delta,k}\rangle \longrightarrow \mathcal{R}_\Delta(h_k)^{-1} |\Phi_{\Delta,k}\rangle.$$

Therefore $\mathcal{U}_\Delta(i, j) |\Phi_{\Delta,j}\rangle$ transforms as a matter state at endpoint i. Thus the Wilson line and the matter Green's function are naturally identified in the BF description.

We now specialize to the particle on a group after the reduction and extract the bilocal. To extract a scalar from the Wilson line, choose endpoint "polarizations" $\langle v_i|$ and $|v_j\rangle$, which are vectors in

[39] A short derivation is as follows. Let $y(\tau)$ be a path from j to i, with $\mathcal{A}_\tau = \mathcal{A}_\mu \dot{y}^\mu$. The Wilson line $\mathcal{U}_\Delta(\tau, j)$ solves $\left(\frac{d}{d\tau} + \mathcal{R}_\Delta(\mathcal{A}_\tau)\right) \mathcal{U}_\Delta(\tau, j) = 0$, with $\mathcal{U}_\Delta(j, j) = \mathbf{1}$. For $\mathcal{A} = \mathcal{G}^{-1} d\mathcal{G}$, one has $\mathcal{A}_\tau = \mathcal{G}(\tau)^{-1} \dot{\mathcal{G}}(\tau)$. The endpoint expression obeys $\frac{d}{d\tau} \mathcal{R}_\Delta(\mathcal{G}(\tau)^{-1} \mathcal{G}_j) = -\mathcal{R}_\Delta(\mathcal{A}_\tau) \mathcal{R}_\Delta(\mathcal{G}(\tau)^{-1} \mathcal{G}_j)$, with $\mathcal{R}_\Delta(\mathcal{G}_j^{-1} \mathcal{G}_j) = \mathbf{1}$. It therefore obeys the same first-order equation and initial condition as the path-ordered exponential, so the two are equal.

the Hilbert space carrying the $\mathcal{R}_\Delta$ representations. In a natural basis, these states label which matter component is inserted at the endpoints. For the scalar primary we use $|v_0\rangle_\Delta$ and ${}_\Delta\langle v_0|$; here Δ labels the representation and 0 the primary level, while the descendants $|v_n\rangle_\Delta$ span the same $\mathcal{R}_\Delta$ representation space.[40] The scalar primary polarization obeys the L_+ annihilation condition, with the dual condition at the bra endpoint, so the $e^{\gamma \mathsf{L}_+}$ factors in the full Gauss representative act trivially in this bilocal. The primary bilocal is

$$\begin{aligned}\mathcal{W}_\Delta(i,j) &= {}_\Delta\langle v_0|\mathcal{R}_\Delta(\mathcal{G}_i^{-1}\mathcal{G}_j)|v_0\rangle_\Delta \\ &= {}_\Delta\langle v_0|\mathcal{R}_\Delta\Big(e^{-\rho_i \mathsf{L}_0} e^{(x_j-x_i)\mathsf{L}_-} e^{\rho_j \mathsf{L}_0}\Big)\,|v_0\rangle_\Delta .\end{aligned} \tag{B.60}$$

The primary assumption in this step is essential. In the full Gauss representative, $\mathcal{G}_i^{-1}\mathcal{G}_j$ contains the endpoint factors associated with the right L_+ coordinate. For the scalar primary polarization these factors drop out because $|v_0\rangle_\Delta$ is annihilated by L_+, with the corresponding dual annihilation condition at the bra endpoint. This is why the reduced Wilson-line matrix element collapses to the single invariant power in (B.62). For a descendant polarization this simplification is no longer true. If

$$|v_n\rangle_\Delta \propto (\mathsf{L}_-)^n |v_0\rangle_\Delta,$$

then $\mathsf{L}_+|v_n\rangle_\Delta \neq 0$ for $n > 0$, and the same endpoint factors must be kept in the Wilson-line matrix element

$$\mathcal{W}_{\Delta;n_i,n_j}(i,j) = {}_\Delta\langle v_{n_i}|\mathcal{R}_\Delta(\mathcal{G}_i^{-1}\mathcal{G}_j)|v_{n_j}\rangle_\Delta. \tag{B.61}$$

After imposing the JT source constraint, the positive Gauss coordinate is expressed in terms of the boundary curve; in the bosonic convention used here $\gamma = -\ddot{x}/(2q\dot{x})$. Descendant bilocals therefore contain additional $\dot{x}, \ddot{x}, \ldots$ dependence, equivalently the conformal-covariant derivatives of the primary bilocal. Thus the Wilson-line construction applies to all components of the representation, but only the primary matrix element gives the simple bilocal $(\dot{x}_i\dot{x}_j/(x_i-x_j)^2)^\Delta$.[41]

Concretely, the bilocal is a boundary-anchored restriction of this Green's function.

$$\mathcal{O}_\Delta(i,j) = \left[\frac{\dot{x}_i\dot{x}_j}{(x_i-x_j)^2}\right]^\Delta . \tag{B.62}$$

The JT source constraint in (B.34) gives $\dot{x}_i = qe^{-\rho_i}$ and $\dot{x}_j = qe^{-\rho_j}$. From here, not surprisingly, we

[40] This is why the AdS/CFT dictionary is usually stated for primary operators: the primary labels the independent bulk field. In $\mathrm{AdS}_2/\mathrm{CFT}_1$, the normalizable scalar modes have $\omega_n = \Delta + n$, and $|v_n\rangle_\Delta$ is the representation-space version of the $SL(2,\mathbb{R})$ descendant $L_{-1}^n\mathcal{O}|0\rangle$ under the state–operator map.

[41] In the supersymmetric discussion the analogous statement is stronger. A superconformal primary must be annihilated not only by L_+ but also by the positive fermionic generators $\mathsf{F}_{+,p}$ and $\bar{\mathsf{F}}_+^p$. Some component fields can be ordinary $SL(2)$ primaries without being superconformal primaries. Their reduced bilocals are obtained by acting with the physical, compensated worldline supercharges described in Section 4.1, not by acting with bare bulk fermionic generators.

can write the Wilson-line operators as a function of invariants

$$\mathcal{O}_\Delta(i,j) = q^{2\Delta}\mathcal{W}_\Delta(i,j) = \left(qL_{ij}^{(0)}\right)^{2\Delta}, \qquad L_{ij}^{(0)} = \frac{e^{-\frac{1}{2}(\rho_i+\rho_j)}}{x_j - x_i}. \tag{B.63}$$

The factor of q is therefore not part of the bare matter Wilson line. It enters only after the boundary source condition converts the endpoint frame factors into the Schwarzian Jacobians $\dot{x}_i$ and $\dot{x}_j$.

C Composition kernel and IBP identities

This appendix records the integrand-level manipulations behind the composition kernel. We use the same invariant lengths as in Section 5, but set the fixed charge to one throughout this appendix. Thus all Bessel arguments are $2L$.

IBP general discussion. For a composition block $i \to j \to k$, define

$$S_{ijk} = \frac{L_{ij}L_{jk}}{L_{ik}} + \frac{L_{jk}L_{ik}}{L_{ij}} + \frac{L_{ik}L_{ij}}{L_{jk}}, \qquad E_{ijk} = e^{-S_{ijk}}. \tag{C.1}$$

The basic IBP move uses the Bessel equation

$$\mathcal{L}_{\nu,L} := L\partial_L(L\partial_L) - \left(4L^2 + \nu^2\right), \qquad \mathcal{L}_{\nu,L}K_\nu(2L) = 0. \tag{C.2}$$

With respect to the measure dL, the adjoint operator is

$$\mathcal{L}^\dagger_{\nu,L}F := \partial_L(L\partial_L(LF)) - \left(4L^2 + \nu^2\right)F. \tag{C.3}$$

The Green identity is

$$K_\nu(2L)\mathcal{L}^\dagger_{\nu,L}F - F\mathcal{L}_{\nu,L}K_\nu(2L) = \partial_L\mathcal{B}_L(K_\nu, F), \tag{C.4}$$

where

$$\mathcal{B}_L(u,F) = L^2\left(u\partial_L F - (\partial_L u)F\right) + LuF. \tag{C.5}$$

We also use

$$\frac{1}{2}\partial_L K_0(2L) = -K_1(2L), \qquad \frac{1}{2}\partial_L[2LK_1(2L)] = -2LK_0(2L). \tag{C.6}$$

These identities convert mixed K_0K_1 terms into total derivatives plus K_0K_0 or K_1K_1 terms.

$\mathcal{N} = 2$ first-order IBP and its OTOC. The $\mathcal{N} = 2$ kernel in (5.7) has a useful first-order form. For the block $1 \to 2 \to 3$, let E_{123} be the exponential in (C.1) and define

$$\mathcal{O}_{ab} := 2L_{ab} - L_{ab}\partial_{L_{ab}}. \tag{C.7}$$

Then

$$I_3^{\mathcal{N}=2}(L_{12},L_{23},L_{13}) = \left[2\left(\mathcal{O}_{12}+\mathcal{O}_{23}+\mathcal{O}_{13}\right)-1\right] E_{123}. \tag{C.8}$$

Indeed, acting on the exponential gives

$$\left[2\left(\mathcal{O}_{12}+\mathcal{O}_{23}+\mathcal{O}_{13}\right)-1\right] E_{123} = \left[-1+\frac{2}{L_{12}L_{23}L_{13}}\left(L_{12}L_{23}+L_{23}L_{13}+L_{13}L_{12}\right)^2\right] E_{123}. \tag{C.9}$$

As a check, apply this identity to the $\mathcal{N}=2$ composition integral. The zero-energy radial factor obeys

$$\left(\partial_L + 2 - \frac{1}{2L}\right)\Psi_0^{\mathcal{N}=2}(L) = 0, \qquad \Psi_0^{\mathcal{N}=2}(L) = \frac{2L}{\sqrt{\pi}} K_{1/2}(2L). \tag{C.10}$$

Equivalently, with respect to the measure dL/L, the adjoint of $\mathcal{O}_{ab}$ acts on the external wavefunction as

$$\mathcal{O}_{ab}^{\dagger}\Psi_0^{\mathcal{N}=2}(L_{ab}) = \frac{1}{2}\Psi_0^{\mathcal{N}=2}(L_{ab}). \tag{C.11}$$

Therefore the two adjacent derivatives in (C.8) can be moved onto the two external wavefunctions:

$$\begin{aligned}\int_0^\infty \frac{dL_{12}}{L_{12}}\frac{dL_{23}}{L_{23}}\,\Psi_0^{\mathcal{N}=2}(L_{12})\Psi_0^{\mathcal{N}=2}(L_{23})I_3^{\mathcal{N}=2}(L_{12},L_{23},L_{13}) \\ = (1+2\mathcal{O}_{13})\int_0^\infty \frac{dL_{12}}{L_{12}}\frac{dL_{23}}{L_{23}}\,\Psi_0^{\mathcal{N}=2}(L_{12})\Psi_0^{\mathcal{N}=2}(L_{23})E_{123}.\end{aligned} \tag{C.12}$$

After some manipulation the remaining scalar integral is identical to Appendix I of [40].

We now use the same first-order identity for the four-point function. For the two composition blocks in (5.26), set

$$F_\Delta = \left(\frac{L_{12}L_{23}L_{34}L_{41}}{L_{12}L_{34}+L_{23}L_{41}}\right)^{2\Delta}.$$

Then the product of kernels before doing the L_{13}-integral is

$$\begin{aligned}&F_\Delta\, I_3^{\mathcal{N}=2}(L_{12},L_{23},L_{13})\, I_3^{\mathcal{N}=2}(L_{34},L_{41},L_{13}) \\ &\quad = F_\Delta\,\left[2\left(\mathcal{O}_{12}+\mathcal{O}_{23}+\mathcal{O}_{13}\right)-1\right] E_{123}\left[2\left(\mathcal{O}_{34}+\mathcal{O}_{41}+\mathcal{O}_{13}\right)-1\right] E_{341},\end{aligned} \tag{C.14}$$

This is the form used for the IBP reduction of (5.26). When $\mathcal{O}_{12},\mathcal{O}_{23},\mathcal{O}_{34},\mathcal{O}_{41}$ are moved by parts, the derivatives act on F_Δ. The basic derivatives are

$$\begin{aligned}L_{12}\partial_{L_{12}}\log F_\Delta = L_{34}\partial_{L_{34}}\log F_\Delta = 2\Delta\,\frac{L_{23}L_{41}}{L_{12}L_{34}+L_{23}L_{41}}, \\ L_{23}\partial_{L_{23}}\log F_\Delta = L_{41}\partial_{L_{41}}\log F_\Delta = 2\Delta\,\frac{L_{12}L_{34}}{L_{12}L_{34}+L_{23}L_{41}}.\end{aligned}$$

The diagonal operator $\mathcal{O}_{13}$ acts only on the two exponentials, since F_Δ is independent of L_{13}. After the L_{13}-integral is performed, the three pieces in (C.14) give the five terms in (5.28)–(5.32): the adjacent-adjacent action gives the $\Delta(2\Delta+1)$ terms T_1,T_2,T_3, the mixed adjacent-diagonal action gives T_4, and

the diagonal-diagonal action gives T_5. In this way the IBP reduction turns the two $I_3^{\mathcal{N}=2}$ kernels into the four-length expression (5.27).

$\mathcal{N}=4$ IBP structure. For the $\mathcal{N}=4$ identities we use the same endpoint notation as in the main text. For one composition block $i\to j\to k$, we use the exponential E_{ijk} defined in (C.1). The two component matrices $\mathbb{I}^{(0)}_{ijk}$ and $\mathbb{I}^{(1)}_{ijk}$ in (5.8)–(5.9) give two scalar densities after they are placed between the K_0/K_1 Bessel function structures. The direct expressions are long, but the IBP form is short. The source factors are

$$\begin{aligned}
&R^{(0)}_{ijk}=-\frac{L_{ik}(L_{ij}^2+L_{jk}^2)}{L_{ij}L_{jk}}\,E_{ijk}, \quad Q^{(0)}_{ij}=\left(\frac14+\frac{L_{ij}L_{ik}}{2L_{jk}}\right)E_{ijk}, \quad Q^{(0)}_{jk}=\left(\frac14+\frac{L_{jk}L_{ik}}{2L_{ij}}\right)E_{ijk},\\
&R^{(1)}_{ijk}=0, \qquad\qquad\qquad Q^{(1)}_{ij}=\frac{L_{ij}L_{ik}}{2L_{jk}}E_{ijk}, \qquad\qquad Q^{(1)}_{jk}=\frac{L_{jk}L_{ik}}{2L_{ij}}E_{ijk}.
\end{aligned}\tag{C.16}$$

Let $I^{(0)}_{ijk}$ and $I^{(1)}_{ijk}$ denote the two resulting densities, with the Bessel function factors included and with respect to the ordinary measure $dL_{ij}\,dL_{jk}$. The pointwise IBP identities are

$$\begin{aligned}
I^{(0)}_{ijk}&=\left[R^{(0)}_{ijk}+\mathcal{L}^{\dagger}_{0,L_{ij}}Q^{(0)}_{ij}+\mathcal{L}^{\dagger}_{0,L_{jk}}Q^{(0)}_{jk}\right]K_0(2L_{ij})K_0(2L_{jk})\\
&\quad+\frac12\partial_{L_{ij}}J^{(0)}_{ij}+\frac12\partial_{L_{jk}}J^{(0)}_{jk},
\end{aligned}\tag{C.17}$$

$$\begin{aligned}
I^{(1)}_{ijk}&=-\left[\mathcal{L}^{\dagger}_{1,L_{ij}}Q^{(1)}_{ij}+\mathcal{L}^{\dagger}_{1,L_{jk}}Q^{(1)}_{jk}\right]K_1(2L_{ij})K_1(2L_{jk})\\
&\quad+\frac12\partial_{L_{ij}}J^{(1)}_{ij}+\frac12\partial_{L_{jk}}J^{(1)}_{jk}.
\end{aligned}\tag{C.18}$$

Here the currents J are explicit total-derivative terms. Their detailed form is not needed for the composition check; what matters is that they must be kept until the boundary behavior is understood. The important distinction is already visible in (C.17)–(C.18): the K_0 channel has a genuine bulk remainder $R^{(0)}_{ijk}$, while the K_1 channel is adjoint-exact in the bulk.

$\mathcal{N}=4$ composition kernel check. We now apply the preceding identities to the composition law. The result is the pair of radial projections

$$\int_0^\infty\frac{dL_{ij}}{L_{ij}}\frac{dL_{jk}}{L_{jk}}\frac{E_{ijk}}{2L_{ik}}\Big((2L_{ij})^2K_0(2L_{ij})\,(2L_{ij})^2K_1(2L_{ij})\Big)\,\mathbb{I}^{(0)}_{ijk}\begin{pmatrix}(2L_{jk})^2K_0(2L_{jk})\\(2L_{jk})^2K_1(2L_{jk})\end{pmatrix}=-2L_{ik}K_0(2L_{ik}),\tag{C.19}$$

$$\int_0^\infty\frac{dL_{ij}}{L_{ij}}\frac{dL_{jk}}{L_{jk}}\frac{E_{ijk}}{2L_{ik}}\Big((2L_{ij})^2K_0(2L_{ij})\,(2L_{ij})^2K_1(2L_{ij})\Big)\,\mathbb{I}^{(1)}_{ijk}\begin{pmatrix}(2L_{jk})^2K_0(2L_{jk})\\(2L_{jk})^2K_1(2L_{jk})\end{pmatrix}=2L_{ik}K_1(2L_{ik}).\tag{C.20}$$

Let us indicate how the IBP identities give these two lines. For the K_0 component, the adjoint Bessel operators in (C.17) can be moved onto $K_0(2L_{ij})$ and $K_0(2L_{jk})$. The bulk part of the Bessel operator action vanishes by the Bessel equation, and the regulated boundary terms vanish; the possible corner $L_{ij},L_{jk}\sim\epsilon$ does not leave a finite contribution because $K_0(z)\sim-\log z$. Thus only the simple

bulk remainder $R^{(0)}_{ijk}$ term survives.

$$\int_0^\infty dL_{ij}\, dL_{jk}\, I^{(0)}_{ijk} = -\int_0^\infty dL_{ij}\, dL_{jk}\, \frac{L_{ik}(L_{ij}^2+L_{jk}^2)}{L_{ij}L_{jk}}\, E_{ijk} K_0(2L_{ij})K_0(2L_{jk}) = -2L_{ik}K_0(2L_{ik}). \tag{C.21}$$

For the K_1 component, the bulk is adjoint-exact and there is no analogue of $R^{(0)}_{ijk}$. The whole contribution comes from the regulated corner. More explicitly, regulate the quadrant by

$$D_\epsilon = \{(L_{ij}, L_{jk}) : L_{ij} \geq \epsilon,\ L_{jk} \geq \epsilon\}.$$

The boundary terms at infinity vanish exponentially, and the only finite contribution comes from the two small-length edges near the corner $L_{ij}, L_{jk} \sim \epsilon$. The boundary one-form which appears here is not a new input. It is the boundary term obtained from the two Green identities in (C.18), together with the explicit total derivatives in the same equation. Here $\mathcal{B}_L(u,F) = L^2(u\partial_L F - (\partial_L u)F) + LuF$.

$$dI_{1,\partial} = \left[-K_1(2L_{jk})\mathcal{B}_{L_{ij}}\left(K_1(2L_{ij}), Q^{(1)}_{ij}\right) + \frac{1}{2}J^{(1)}_{ij}\right] dL_{jk} - \left[-K_1(2L_{ij})\mathcal{B}_{L_{jk}}\left(K_1(2L_{jk}), Q^{(1)}_{jk}\right) + \frac{1}{2}J^{(1)}_{jk}\right] dL_{ij}. \tag{C.22}$$

Thus $dI_{1,\partial}$ is just the boundary form of the adjoint-exact K_1 density. Parametrizing the small corner by $L_{ij} = \epsilon,\ L_{jk} = \epsilon t$, and adding the exchanged edge, gives

$$\begin{aligned}\int_0^\infty dL_{ij}\, dL_{jk}\, I^{(1)}_{ijk} = \lim_{\epsilon\to 0}\int_{\partial D_\epsilon} dI_{1,\partial} &= \int_1^\infty L_{ik}\left(1+\frac{1}{t^2}\right)e^{-L_{ik}(t+1/t)}dt \\ &= L_{ik}\int_0^\infty t^{-2}e^{-L_{ik}(t+1/t)}dt = 2L_{ik}K_1(2L_{ik}).\end{aligned} \tag{C.23}$$

$\mathcal{N}=4$ **IBP for correlation integrands.** For correlation functions, the same block is multiplied by the remaining length dependence before the L_{ij}, L_{jk} integrals are done. Let $T(L_{ij}, L_{jk})$ denote this extra factor, with L_{ik} held fixed while the IBP is performed on the two lengths L_{ij} and L_{jk}. Define

$$B^{ijk}_\alpha[T] := \int_0^\infty dL_{ij}\, dL_{jk}\, I^{(\alpha)}_{ijk}\, T(L_{ij}, L_{jk}). \tag{C.24}$$

Substituting (C.17)–(C.18) gives

$$\begin{aligned}B^{ijk}_\alpha[T] = &\int dL_{ij}\, dL_{jk}\, \widehat{I}^{(\alpha)}_{ijk}\, T - \int dL_{ij}\, dL_{jk}\left(\frac{1}{2}J^{(\alpha)}_{ij}\partial_{L_{ij}}T + \frac{1}{2}J^{(\alpha)}_{jk}\partial_{L_{jk}}T\right) \\ &+ B^{ijk}_{\alpha,\partial}[T],\end{aligned} \tag{C.25}$$

where $\widehat{I}^{(\alpha)}_{ijk}$ denotes the non-current part of (C.17) or (C.18). For $T=1$ the current terms integrate to boundary contributions. For the three-point function in (5.17), the relevant choice is the monomial

$$T_3(L_{ij}, L_{jk}; L_{ik}) = L_{ij}^{\alpha_{ij}} L_{jk}^{\alpha_{jk}} L_{ik}^{\alpha_{ik}}, \qquad \partial_{L_{ij}}T_3 = \frac{\alpha_{ij}}{L_{ij}}T_3, \qquad \partial_{L_{jk}}T_3 = \frac{\alpha_{jk}}{L_{jk}}T_3.$$

Thus the IBP reduction is still an integrand-level statement. It rewrites the $I_3^{\mathcal{N}=4}$ numerator as a finite sum of terms with shifted powers of L_{ij}, L_{jk}, and L_{ik}, multiplying the same exponential E_{ijk} and the same K_0/K_1 wavefunctions. For positive powers α_{ab} the small-length boundary terms are suppressed, so the three-point calculation is reduced to ordinary Bessel function length integrals of this finite list of shifted monomials. In this case the final bulk representative is still short:

$$\begin{aligned} B_0^{ijk}[T_3] &= \frac{1}{4}\int dL_{ij}\, dL_{jk}\, E_{ijk} K_0(2L_{ij}) K_0(2L_{jk})\, \mathcal{M}_0, \\ B_1^{ijk}[T_3] &= \frac{1}{4}\int dL_{ij}\, dL_{jk}\, E_{ijk} K_1(2L_{ij}) K_1(2L_{jk})\, \mathcal{M}_1, \end{aligned} \tag{C.26}$$

where

$$\begin{aligned} \mathcal{M}_0 = L_{ij}^{\alpha_{ij}-1} L_{jk}^{\alpha_{jk}-1} L_{ik}^{\alpha_{ik}-1} \Big[& L_{ij}^2 L_{jk}^2 (\alpha_{ij} + \alpha_{ij}^2 + \alpha_{jk} - \alpha_{ij}\alpha_{jk} + \alpha_{jk}^2) \\ & - L_{ij} L_{jk} L_{ik} (2 + \alpha_{ij} + \alpha_{jk})(\alpha_{ij} + \alpha_{ij}^2 + \alpha_{jk} + \alpha_{jk}^2) \\ & - L_{ik}^2 (2 + \alpha_{ij} + \alpha_{jk}) \left(L_{ij}^2 (1 + \alpha_{jk}) + (1 + \alpha_{ij}) L_{jk}^2 \right) \Big], \\ \mathcal{M}_1 = L_{ij}^{\alpha_{ij}-1} L_{jk}^{\alpha_{jk}-1} L_{ik}^{\alpha_{ik}-1} \Big[& - L_{ij}^2 L_{jk}^2 (\alpha_{ij}^2 - \alpha_{ij}\alpha_{jk} + \alpha_{jk}^2) \\ & + L_{ij} L_{jk} L_{ik} (2 + \alpha_{ij} + \alpha_{jk})(\alpha_{ij}^2 + \alpha_{jk}^2) \\ & + L_{ik}^2 (2 + \alpha_{ij} + \alpha_{jk}) \left(\alpha_{jk} L_{ij}^2 + \alpha_{ij} L_{jk}^2 \right) \Big]. \end{aligned} \tag{C.27}$$

This is the form used before doing the length integrals which lead to (5.18).

For the scalar OTOC the remaining factor is instead the product of the second composition block and the scalar bilocals. In that case the derivatives in (C.25) act both on the second kernel and on $\mathcal{O}_\Delta(1,3)\mathcal{O}_\Delta(2,4)$, producing the longer finite sum of four-point integrands used in Section 5.

D $\mathcal{N} = 0$ OTOC from length integrals

This appendix gives an alternative derivation of the bosonic JT gravity OTOC. The starting point is the two-I_3 length-integral representation set up in Section 5; in the $\mathcal{N} = 0$ limit it reduces to the scalar $I_3 I_3$ integral below. We then perform the length integrals in two steps: first rewriting the OTOC kernel as a two-variable Gamma integral, and then reducing that Gamma integral by Barnes identities. The result is the standard Wilson-function expression for the JT crossing kernel, equivalently the $SL(2,\mathbb{R})$ $6j$ symbol. This provides a self-contained check that the length-integral representation reproduces the known bosonic JT OTOC kernel. All Barnes contours below are vertical contours chosen in a nonempty convergence strip. For an integrand containing $\Gamma(A+z)\Gamma(B-z)$, this means that the poles $z = -A-n$ are on the left of the contour and the poles $z = B+n$ are on the right; the final expressions are understood by meromorphic continuation.

The length integral. In the bosonic normalization used in this appendix, $\Psi_s(L) = \frac{2}{\pi} K_{2is}(2qL)$ and $\rho(s) = 2s\sinh(2\pi s)$, so that

$$\int_0^\infty \frac{dL}{L}\,\Psi_s(L)\Psi_{s'}(L) = \frac{\delta(s-s')}{\rho(s)}.$$

This fixes the bosonic JT density convention used below. The composition-normalized scalar three-length kernel is

$$I_3(L_a, L_b, L_c) = \frac{1}{\pi}\exp\left[-q\left(\frac{L_aL_b}{L_c} + \frac{L_bL_c}{L_a} + \frac{L_cL_a}{L_b}\right)\right], \tag{D.1}$$

Equivalently,

$$\begin{aligned} I_3(L_a, L_b, L_c) &= \int_0^\infty ds\,\rho(s)\,\Psi_s(L_a)\Psi_s(L_b)\Psi_s(L_c) \\ &= \frac{4}{\pi^2}\int_{-\infty}^{+\infty} \frac{ds}{\Gamma(2is)\Gamma(-2is)} K_{2is}(2qL_a)K_{2is}(2qL_b)K_{2is}(2qL_c). \end{aligned} \tag{D.2}$$

with the same normalization convention. Gluing two copies of (D.1) gives

$$\begin{aligned} I_4(L_1, L_2, L_3, L_4) &= \int_0^\infty \frac{dL}{L}\, I_3(L_1, L_2, L) I_3(L_3, L_4, L) \\ &= \frac{2}{\pi^2} K_0\left(2q\sqrt{\frac{(L_1L_2 + L_3L_4)(L_1L_3 + L_2L_4)(L_1L_4 + L_2L_3)}{L_1L_2L_3L_4}}\right). \end{aligned} \tag{D.3}$$

The same kernel has the full-line Kontorovich–Lebedev representation

$$I_4(L_1, L_2, L_3, L_4) = \frac{8}{\pi^3}\int_{-\infty}^{+\infty} \frac{ds}{\Gamma(2is)\Gamma(-2is)} \prod_{j=1}^4 K_{2is}(2qL_j). \tag{D.4}$$

For equal external matter dimension Δ, the stripped bosonic length-space OTOC block can therefore be written as

$$\mathcal{A}_\Delta^{(0)} = \int \prod_{j=1}^4 \frac{dL_j}{L_j}\,\Psi_{s_j}(L_j)\left(\frac{L_1L_2L_3L_4}{L_1L_3 + L_2L_4}\right)^{2\Delta} I_4(L_1, L_2, L_3, L_4). \tag{D.5}$$

The rest of the appendix evaluates this universal kernel.

Γ integral and Barnes reduction. To turn (D.5) into a Gamma integral, use (D.4) together with the identity

$$\frac{1}{(X+Y)^{2\Delta}} = \frac{1}{(XY)^\Delta}\int_{-\infty}^{+\infty}\frac{d\kappa}{2\pi}\left(\frac{X}{Y}\right)^{i\kappa}\frac{\Gamma(\Delta + i\kappa)\Gamma(\Delta - i\kappa)}{\Gamma(2\Delta)}. \tag{D.6}$$

for $X, Y > 0$ and $\mathrm{Re}\,\Delta > 0$, with $X = L_1L_3$ and $Y = L_2L_4$. After substituting (D.6) and (D.4), the four length integrals factorize. Using the shorthand for paired Gamma factors, the needed Mellin–Bessel transform is

$$\int_0^\infty \frac{dL}{L} L^\lambda K_{2is}(2qL)K_{2is'}(2qL) = \frac{q^{-\lambda}}{8\,\Gamma(\lambda)}\Gamma\left(\frac{\lambda}{2} \pm is \pm is'\right). \tag{D.7}$$

The odd legs $1,3$ carry $\lambda = \Delta + i\kappa$, while the even legs $2,4$ carry $\lambda = \Delta - i\kappa$. The four factors of $q^{-\lambda}$ give the explicit factor $q^{-4\Delta}$ below. Thus (D.5) becomes

$$\mathcal{A}_{\Delta}^{(0)} = q^{-4\Delta} \left(\frac{2}{\pi}\right)^4 \frac{1}{2^8\pi^2\,\Gamma(2\Delta)} \mathcal{G}_{\Delta}(s_1, s_2, s_3, s_4), \tag{D.8}$$

where the two-variable Gamma integral is

$$\mathcal{G}_{\Delta} = \int_{\mathbb{R}} \frac{d\kappa}{2\pi} \int_{\mathbb{R}} \frac{ds}{2\pi} \frac{\displaystyle\prod_{\ell=1,3} \Gamma\left(\frac{\Delta + i\kappa}{2} \pm is_\ell \pm is\right) \prod_{m=2,4} \Gamma\left(\frac{\Delta - i\kappa}{2} \pm is_m \pm is\right)}{\Gamma(\Delta + i\kappa)\Gamma(\Delta - i\kappa)\Gamma(2is)\Gamma(-2is)}. \tag{D.9}$$

This is the promised Gamma integral. Starting from here, we evaluate (D.9) in several non-trivial steps, which leave a single Mellin–Barnes integral.

Set

$$z_1 = i\left(\frac{\kappa}{2} + s\right), \qquad z_2 = i\left(\frac{\kappa}{2} - s\right). \tag{D.10}$$

Then $i\kappa = z_1 + z_2$ and $2is = z_1 - z_2$, and

$$\mathcal{G}_{\Delta} = \int \frac{dz_1}{2\pi i} \frac{dz_2}{2\pi i} \frac{F(z_1)F(z_2)}{\Gamma(z_1 - z_2)\Gamma(z_2 - z_1)\Gamma(\Delta + z_1 + z_2)\Gamma(\Delta - z_1 - z_2)}, \tag{D.11}$$

where

$$F(z) = \Gamma\left(\frac{\Delta}{2} + z \pm is_1\right) \Gamma\left(\frac{\Delta}{2} + z \pm is_3\right) \Gamma\left(\frac{\Delta}{2} - z \pm is_2\right) \Gamma\left(\frac{\Delta}{2} - z \pm is_4\right). \tag{D.12}$$

The motivation for the next two steps is to first disentangle the z_i's from the external s_i's, and then recombine some of the s_i's into the factors multiplying the final one-fold integral. The two diagonal denominators in (D.11) are opened by Barnes' first lemma:

$$\frac{\prod_{j=1}^{2} \Gamma\left(\frac{\Delta}{2} + z_j \pm is_1\right)}{\Gamma(\Delta + z_1 + z_2)} = \int_C \frac{du}{2\pi i} \Gamma(z_1 + u)\Gamma(z_2 + u)\Gamma\left(\frac{\Delta}{2} - u \pm is_1\right), \tag{D.13}$$

$$\frac{\prod_{j=1}^{2} \Gamma\left(\frac{\Delta}{2} - z_j \pm is_2\right)}{\Gamma(\Delta - z_1 - z_2)} = \int_C \frac{dv}{2\pi i} \Gamma(v - z_1)\Gamma(v - z_2)\Gamma\left(\frac{\Delta}{2} - v \pm is_2\right). \tag{D.14}$$

After (D.13) and (D.14), the z_1, z_2 integrals have exactly Gustafson's A_2 form, with

$$\boldsymbol{p} = \left(u, \frac{\Delta}{2} + is_3, \frac{\Delta}{2} - is_3\right), \qquad \boldsymbol{q} = \left(v, \frac{\Delta}{2} + is_4, \frac{\Delta}{2} - is_4\right). \tag{D.15}$$

The A_2 integral gives

$$\mathcal{G}_\Delta = 2\,\Gamma(\Delta \pm is_3 \pm is_4) \int_C \frac{du}{2\pi i} \int_C \frac{dv}{2\pi i}\, \Gamma\Big(\frac{\Delta}{2} - u \pm is_1\Big)\, \Gamma\Big(\frac{\Delta}{2} - v \pm is_2\Big) \times \frac{\Gamma(u+v)\Gamma\big(u+\frac{\Delta}{2} \pm is_4\big)\,\Gamma\big(v+\frac{\Delta}{2} \pm is_3\big)}{\Gamma(u+v+2\Delta)}. \tag{D.16}$$

The last nontrivial step is Barnes' second lemma on the v-contour. After applying it, one obtains

$$\mathcal{G}_\Delta = \frac{2\,\Gamma(\Delta \pm is_3 \pm is_4)\Gamma(\Delta \pm is_3 \pm is_2)}{\Gamma(2\Delta)} \int_C \frac{du}{2\pi i}\, \frac{\Gamma\big(\frac{\Delta}{2} - u \pm is_1\big)\,\Gamma\big(u+\frac{\Delta}{2} \pm is_2\big)\,\Gamma\big(u+\frac{\Delta}{2} \pm is_4\big)}{\Gamma\big(u+\frac{3\Delta}{2} + is_3\big)\,\Gamma\big(u+\frac{3\Delta}{2} - is_3\big)}. \tag{D.17}$$

A convenient contour has $-\frac{1}{2}\operatorname{Re}\Delta < \operatorname{Re} u < \frac{1}{2}\operatorname{Re}\Delta$, so that the poles of $\Gamma(u+\frac{\Delta}{2} \pm is_2)\Gamma(u+\frac{\Delta}{2} \pm is_4)$ lie on the left and the poles of $\Gamma(\frac{\Delta}{2} - u \pm is_1)$ lie on the right.

Residues and the Wilson-function answer. Closing the u-contour to the right gives two pole families, from $u = \frac{\Delta}{2} + i\epsilon s_1 + n$ with $\epsilon = \pm 1$. For nonresonant spectral labels this gives

$$\begin{aligned} &\int_C \frac{du}{2\pi i}\, \frac{\Gamma\big(\frac{\Delta}{2} - u \pm is_1\big)\,\Gamma\big(u+\frac{\Delta}{2} \pm is_2\big)\,\Gamma\big(u+\frac{\Delta}{2} \pm is_4\big)}{\Gamma\big(u+\frac{3\Delta}{2} + is_3\big)\,\Gamma\big(u+\frac{3\Delta}{2} - is_3\big)} \\ &\quad = \sum_{\epsilon=\pm 1} \frac{\Gamma(-2i\epsilon s_1)}{\Gamma(2\Delta + i\epsilon s_1 + is_3)\Gamma(2\Delta + i\epsilon s_1 - is_3)} \Gamma(\Delta + i\epsilon s_1 \pm is_2)\Gamma(\Delta + i\epsilon s_1 \pm is_4) \\ &\quad\quad \times {}_4F_3\left(\begin{matrix} \Delta + i\epsilon s_1 + is_2,\ \Delta + i\epsilon s_1 - is_2,\ \Delta + i\epsilon s_1 + is_4,\ \Delta + i\epsilon s_1 - is_4 \\ 2\Delta + i\epsilon s_1 + is_3,\ 2\Delta + i\epsilon s_1 - is_3,\ 1 + 2i\epsilon s_1 \end{matrix}; 1\right). \end{aligned} \tag{D.18}$$

The two ${}_4F_3(1)$'s are balanced.

Equation (D.18) is already the Wilson-function answer in a nonstandard s_3-channel. To make the connection to the usual JT OTOC kernel explicit, let $L[\cdots]$ denote Bailey's normalized L-function, equivalently the standard Wilson-function package written as a sum of two balanced ${}_4F_3(1)$'s. Then

$$\begin{aligned} &\int_C \frac{du}{2\pi i}\, \frac{\Gamma\big(\frac{\Delta}{2} - u \pm is_1\big)\,\Gamma\big(u+\frac{\Delta}{2} \pm is_2\big)\,\Gamma\big(u+\frac{\Delta}{2} \pm is_4\big)}{\Gamma\big(u+\frac{3\Delta}{2} + is_3\big)\,\Gamma\big(u+\frac{3\Delta}{2} - is_3\big)} \\ &\quad = \pi\,\Gamma(\Delta \pm is_1 \pm is_2)\Gamma(\Delta \pm is_1 \pm is_4) \\ &\quad\quad \times L\left[\begin{matrix} \Delta + i(s_1+s_4),\ \Delta + i(s_1-s_4),\ \Delta + i(s_1+s_2),\ \Delta + i(s_1-s_2) \\ 1 + 2is_1,\ 2\Delta + i(s_1-s_3),\ 2\Delta + i(s_1+s_3) \end{matrix}\right]. \end{aligned} \tag{D.19}$$

For comparison, the standard Schwarzian/JT crossing kernel is usually written in the JT channel. A one-fold Mellin–Barnes form is obtained in the Schwarzian limit of the Ponsot–Teschner kernel in [36, App. F]; the same $SL(2,\mathbb{R})$ $6j$ is the crossing coefficient for boundary-anchored JT Wilson lines in [37]. With equal external dimensions and with the momenta relabeled to the present notation, the

standard JT answer is

$$\begin{aligned}\mathcal{A}_{\Delta}^{\mathrm{JT}} = &\frac{\Gamma(\Delta+is_1\pm is_4)\Gamma(\Delta-is_3\pm is_2)\Gamma(\Delta-is_1\pm is_2)\Gamma(\Delta+is_3\pm is_4)}{\Gamma(2\Delta)^2}\\ &\times\int_{C_u}\frac{du}{2\pi i}\,\frac{\Gamma(u\pm is_4)\,\Gamma(u+i(s_1+s_3)\pm is_2)\,\Gamma(\Delta-is_1-u)\,\Gamma(\Delta-is_3-u)}{\Gamma(u+\Delta+is_1)\,\Gamma(u+\Delta+is_3)}.\end{aligned} \tag{D.20}$$

The Barnes integral in (D.20) gives another Bailey L-function,

$$L_{\mathrm{JT}}(s_1,s_2,s_3,s_4;\Delta) = L\begin{bmatrix}\Delta-is_1+is_4,\ \Delta-is_1-is_4,\ \Delta+is_3+is_2,\ \Delta+is_3-is_2\\ 1+i(s_3-s_1),\ 2\Delta,\ 2\Delta+i(s_3-s_1)\end{bmatrix}. \tag{D.21}$$

It is related to the $L[\cdots]$ in (D.19) by standard L-function transformations. Combining (D.8), (D.17), and this Bailey relation, the stripped length-space answer is

$$\mathcal{A}_{\Delta}^{(0)} = q^{-4\Delta}\left(\frac{2}{\pi}\right)^4\frac{1}{2^7\pi^2}\,\mathcal{A}_{\Delta}^{\mathrm{JT}} = \frac{q^{-4\Delta}}{8\pi^6}\,\mathcal{A}_{\Delta}^{\mathrm{JT}}. \tag{D.22}$$

Thus, up to the explicit overall factor in (D.22), the length integral gives the usual bosonic JT $6j$-symbol in the normalization displayed in (D.20).

Unequal dimensions. The old I_3 calculation also records the extension before imposing equal dimensions. One keeps the two matter factors separately in the diagonal gluing as $L^{2\Delta_1}(L')^{2\Delta_2}$. Using $LL' = L_1L_2L_3L_4/(L_1L_3+L_2L_4)$, this is

$$\left(\frac{L_1L_2L_3L_4}{L_1L_3+L_2L_4}\right)^{2\Delta_2}L^{2(\Delta_1-\Delta_2)}.$$

Thus the equal-dimension case is special: the last factor is absent, the two I_3's collapse to I_4, and one obtains the length integral evaluated above. The Gamma-integral calculation applies the same Mellin/Bessel steps to this shifted diagonal gluing: the extra factor is expanded into six Mellin–Barnes blocks, and each block is reduced by the same Mellin–Bessel transform, Barnes' first lemma, the A_2 integral, and Barnes' second lemma to a one-fold Mellin–Barnes expression. The result is the unequal-dimension JT $6j$ kernel

$$\begin{aligned}\mathcal{A}_{\Delta_1,\Delta_2}^{\mathrm{JT}} = &\frac{\Gamma(\Delta_1+is_1\pm is_4)\Gamma(\Delta_1-is_3\pm is_2)}{\Gamma(2\Delta_1)}\,\frac{\Gamma(\Delta_2-is_1\pm is_2)\Gamma(\Delta_2+is_3\pm is_4)}{\Gamma(2\Delta_2)}\\ &\times\int_{C_u}\frac{du}{2\pi i}\,\frac{\Gamma(u\pm is_4)\,\Gamma(u+i(s_1+s_3)\pm is_2)\,\Gamma(\Delta_1-is_1-u)\,\Gamma(\Delta_2-is_3-u)}{\Gamma(u+\Delta_1+is_1)\,\Gamma(u+\Delta_2+is_3)}.\end{aligned} \tag{D.23}$$

Setting $\Delta_1=\Delta_2=\Delta$ reduces this expression to (D.20), which is the case evaluated explicitly above.

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